



Degree Project in Electric Power Engineering

Second cycle, 30 credits

A Municipal-Scale Net Load Forecasting Framework for Norway: Consumer-Segment Disaggregation and BTM Solar Assessment using Open Elhub Data

A Machine Learning Study Using Open AMI Metering Data in the
National Scale of Norway

IRAKLIS BOURNAZOS

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Supervisors: Valgerður Jónsdóttir, Ilias Dimoulkas, Sebastian Haglund

Examiner: Lars Nordström

School of Electrical Engineering and Computer Science

Host company: rebase.energy

Swedish title: Ett kommunalt ramverk för nettolastprognoser i Norge:
disaggregering av konsumentsegment och bedömning av
bakom-mätaren-solenergi med öppen Elhub-data

Swedish subtitle: En maskininlärningsstudie med öppen AMI-mätardata i
nationell skala i Norge

Abstract

Accurate short-term net load forecasting is essential for distribution system operators, energy traders, and flexibility service providers, particularly as distributed rooftop solar photovoltaic installations transform the electricity demand observed at the grid boundary. This thesis develops and validates a multi-stage municipality-level net load forecasting framework for Norway, where the *Plusskunde* net-metering scheme — allowing households with installations up to 100 kW to export surplus generation — has driven rapid residential prosumer adoption, using exclusively open data sources and machine learning methods.

The framework processes Elhub open AMI metering data — comprising hourly consumption disaggregated by consumer category (private residential, commercial, and industrial), solar export records, and daily installed solar capacity snapshots — across 350 Norwegian municipalities, combined with ERA5 reanalysis weather data. The empirical programme consists of three stages: a national weather feature ablation across all municipalities, a six-fold expanding-window walk-forward evaluation on eight representative municipalities spanning the full solar penetration spectrum and multiple modelling approaches, and a three-fold national validation covering the unseen 2025 calendar year.

For private residential net load, a conservative municipality-level model assignment strategy — deploying a prosumer-aware LightGBM model in the 48 municipalities where fold-consistent improvement is demonstrated and a weather-based model in the remaining 302 — achieves a mean annual municipality MAPE of 6.671%, winning against the seasonal naive benchmark in 347 of 350 municipalities (99.1%), against the best linear regression baseline in 246 of 350 (70.3%), and never underperforming the weather-based model alone across the entire portfolio. For commercial load, the framework achieves 9.655% annual mean MAPE, winning conservatively against the naive benchmark in 313 of 350 municipalities (89.4%) and against the linear regression baseline in 304 of 350 (86.9%). Industrial net load forecasting is shown to be a structurally different problem in which weather-and-calendar features reach their generalisation limits due to the heterogeneity of facility-level production patterns across municipalities, motivating a hybrid deployment strategy. A controlled synthetic experiment reveals that the near-zero benefit of solar-enriched modelling for the residential category under current data conditions is attributable to structural behind-the-meter invisibility, low solar penetration, and prosumer registry heterogeneity.

rather than an architectural deficiency: when provided with exact, noiseless capacity information, the prosumer-aware model improves consistently over its weather-based counterpart with gains of +0.16 to +0.37 percentage points across multiple solar growth scenarios.

The complete pipeline is built on open data and open-source software under realistic operational constraints, providing a reproducible benchmark for municipality-scale net load forecasting in Nordic open-data environments.

Keywords

Net load forecasting, Behind-the-meter solar, Distributed solar PV, Smart metering, LightGBM, Gradient-boosted trees, Norwegian electricity system, Elhub, AMI data, Municipality-level forecasting, Ensemble methods, Prosumer

Sammanfattning

Noggrann korttidsprognostisering av nettolast är avgörande för distributionsnätoperatörer, energihandlare och flexibilitetstjänsteleverantörer, särskilt i takt med att distribuerade solcellsinstallationer omvandlar den elefterfrågan som observeras vid nätgränsen. Denna avhandling utvecklar och validerar ett ramverk för nettolastprognostisering på kommunnivå för Norge, där *Plusskundeordningen* — ett nätmätningssystem som ger hushåll med installationer upp till 100 kW rätt att exportera överskottsenergi — har drivit en snabb tillväxt av bostadsprosumenter, med enbart öppna datakällor och maskininlärningsmetoder.

Ramverket bearbetar öppna AMI-mätardata från Elhub — bestående av timvis förbrukning uppdelad per konsumentkategori (privathushåll, handel och industri), solexportdata samt dagliga ögonblicksbilder av installerad solkapacitet — för 350 norska kommuner, kombinerat med ERA5-reanalysdata för väder. Det empiriska programmet består av tre steg: en nationell ablation av väderprediktorer, en sexfaldig expanderande walk-forward-validering på åtta representativa kommuner och en tre-faldig nationell validering för det hittills osedda kalenderåret 2025.

För privata hushålls nettolast uppnår en konservativ modelltilldelningsstrategi ett genomsnittligt kommunalt MAPE på 6.671%, vinner mot det säsongснаiva referensvärdet i 347 av 350 kommuner (99.1%) och mot den bästa linjära regressionsbaslinjen i 246 av 350 (70.3%), och underpresterar aldrig den vädersbaserade modellen ensam. För kommersiell last uppnår ramverket 9.655% MAPE med en konservativ vinstkvot på 89.4% mot naiv och 86.9% mot regressionsbaslinje. Industriell nettolastprognostisering identifieras som ett strukturellt annorlunda problem som kräver en hybridstrategi. Ett kontrollerat syntetiskt experiment visar att solmodellens nära-noll-fördel under rådande dataförhållanden beror på strukturell osynlighet bakom mätaren och låg solpenetration snarare än en arkitektonisk brist.

Det fullständiga systemet är byggt på öppna data och öppen källkod under realistiska driftsförhållanden, och utgör ett reproducerbart riktmärke för nettolastprognostisering i nordiska öppna datamiljöer.

Nyckelord

Nettolastprognostisering, Bakom-mätaren-solenergi, Distribuerad solenergi, Smart mätning, LightGBM, Gradientförstärkta träd, Norskt elsystem, Elhub, AMI-data, Kommunal prognostisering, Ensemblemetoder, Prosument

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Finally, I would like to close with a thought that has stayed with me throughout this journey: *eupsychia* ($\epsilon\upsilon\psi\chi\acute{\iota}\alpha$) — courage and nobility of soul — is considered a prerequisite for freedom and happiness, and every important choice in life should be made with passion and wholehearted dedication.

During the writing of this thesis, I used the generative AI tool Claude (developed by Anthropic) to support the refinement of text, improve language clarity, translate content between languages, troubleshoot $\text{\LaTeX}$, assist in the styling of figures, and support code development and debugging. All AI-assisted content was critically reviewed, edited, and adapted by me to ensure its accuracy, academic integrity, and relevance to the research. The intellectual

contributions, analysis, and conclusions presented in this work are entirely my own.

Stockholm, July 2026
Iraklis Bournazos

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List of acronyms and abbreviations

AMI	Advanced Metering Infrastructure
API	Application Programming Interface
AR	Autoregressive
BTM	Behind-The-Meter
CatBoost	Categorical Boosting
DER	Distributed Energy Resource
DiVA	Digitala Vetenskapliga Arkivet
DSO	Distribution System Operator
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA5	ECMWF Reanalysis v5
GBDT	Gradient-Boosted Decision Trees
KTH	KTH Royal Institute of Technology
LightGBM	Light Gradient-Boosting Machine
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MLR	Multiple Linear Regression
NLF	Net Load Forecasting
nMAPE	Normalised Mean Absolute Percentage Error
nRMSE	Normalised Root Mean Square Error
NWP	Numerical Weather Prediction
OLS	Ordinary Least Squares
PV	Photovoltaic

RF	Random Forest
RMSE	Root Mean Square Error
SDG	Sustainable Development Goal
STLF	Short-Term Load Forecasting
TSO	Transmission System Operator
UN	United Nations
WSS	Weather Station Selection
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

The reliable operation of modern electricity systems depends critically on the ability of system operators to anticipate short-term variations in electricity demand and generation. Accurate forecasts are required for power system operational scheduling, revenue projection, energy trading, congestion management, reserve allocation, and market clearing processes [1]. Global solar deployment has reached unprecedented levels, with nearly 600 GW of new capacity installed in 2024 alone, corresponding to an average installation rate of more than 1 GW per day [2] marking a record level of expansion. As a result, global installed PV capacity has surpassed 2.2 TW, and solar power now contributes more than 10% of global electricity generation [3]. Solar is expected to dominate future renewable expansion, accounting for the majority of global capacity growth through 2030 [4]. This unprecedented deployment rate is driving a structural transformation of electricity systems at all spatial scales, from national grids to individual distribution feeders.

The accelerating deployment of distributed renewable energy resources such as rooftop solar PV systems is fundamentally altering the character of electricity demand as it is observed at the grid boundary. Households and businesses that once consumed electricity passively are increasingly becoming *prosumers*: entities that both consume grid energy and generate electricity from behind-the-meter (BTM) installations, exporting any surplus to the distribution network. This transition has profound operational consequences for distribution system operators (DSOs) and transmission system operators (TSOs), who must maintain grid balance, plan infrastructure investments, and schedule ancillary services, all of which depend critically on accurate short-term forecasts of the electricity that will actually flow through the grid.

The quantity that DSOs and TSOs must forecast is not gross load (the total

electricity consumed by end-users) but *net load*: the residual demand that must be met by the grid after subtracting the self-generation of prosumers [5].

In systems with low distributed renewable penetration, the distinction between net load and gross load is operationally negligible. However, as solar PV penetration grows, the gap widens and becomes increasingly volatile: midday periods in summer may exhibit sharp load suppression, the so-called “duck curve” phenomenon in some extreme cases, while evening ramp rates increase as solar generation falls simultaneously with rising household demand [6]. Traditionally, electricity demand forecasting relied on representative load profiles constructed from aggregated or averaged consumption data, often capturing typical patterns rather than real-time system behavior. However, the widespread deployment of advanced metering infrastructure (AMI) across modern power systems has enabled the collection of high-resolution measurements at multiple points in the grid. This transition has shifted the forecasting problems from profile-based approaches towards data-driven methodologies, which leverage real system observations to capture temporal variability and complex consumption dynamics more accurately. As a result, forecasting models can better reflect the stochastic nature of electricity demand and the increasing influence of distributed energy resources [5].

Forecasting models that fail to account for this phenomenon systematically underestimate midday load suppression, leading to suboptimal generation dispatch, inflated balancing costs, and inefficient participation in electricity markets. As behind-the-meter assets including solar PV, battery storage and smart demand response devices, continue to proliferate, the accuracy of net load forecasts becomes increasingly critical not only for system operators, but also for generation portfolio managers, energy traders, and flexibility service providers who must optimize their positions under growing uncertainty.

Norway presents a particularly interesting and underexplored context for net load forecasting research. The Norwegian electricity system is almost entirely based on renewable energy, with hydropower accounting for approximately 90% of total electricity generation. In addition, the share of modern renewables in final energy consumption is among the highest globally, reflecting the strong electrification and renewable integration of the Norwegian energy system [7]. This can promote operational flexibility but also creates a demand pattern strongly driven by factors as weather variability and temperature as electric space heating accounts for a substantial share of residential consumption.

Norwegian municipalities span a broad range of climatic, geographic, and demographic conditions. From densely populated Oslo in the south to sparsely

populated Arctic municipalities in the north, all of them display heterogeneous consumer compositions across 3 major types of load: residential (private), commercial (business), and industrial categories.

In Norway, the installed solar capacity increased from approximately 100 MW in 2020 to more than 800 MW by the end of 2025, representing an almost 700% increase within a five-year period [8]. This expansion is primarily driven by distributed rooftop installations, leading to a rapid increase in the number of prosumers connected to the grid. Despite its relatively smaller contribution at the national level compared to other countries, solar PV can reach significant penetration levels at the municipal scale, particularly during summer periods. This localized variability introduces additional complexity in forecasting accuracy and grid operations, as net load becomes increasingly sensitive to both consumption dynamics and behind-the-meter generation. Data from Elhub, the Norwegian electricity data hub operated by Statnett [9], reveal that the number of E19 metering points (prosumers with net export to the grid) grew by approximately 101.5 % nationally between 2023 and 2025. When E19 export is expressed as a fraction of *residential* (private) consumption only, the consumer category that accounts for virtually all rooftop solar adoption, the municipality of Vestby (south of Oslo) records an annual private-relative penetration of 5.9%, the highest in Norway. During the summer months of May through August, when solar irradiance is sufficient to drive meaningful generation at Norwegian latitudes, this figure rises to 22.3%. By contrast, Oslo — Norway’s largest city, dominated by multi-storey apartment buildings and many residential loads, records an annual private-relative solar penetration of lower values. At the national level, the aggregate residential penetration peaks at 3.3% in July 2025, reflecting the combined effect of growing installed capacity and seasonal solar availability.

The availability of Elhub open data since 2022, following the national rollout of Advanced Metering Infrastructure (AMI) completed in 2019, provides an unprecedented resource for this type of analysis: hourly, municipality-level, category-disaggregated metering data covering all 357 Norwegian municipalities, with both consumption and prosumer export recorded separately. This is, to the best of the author’s knowledge, the most granular open electricity dataset available for any Nordic country, and its systematic exploitation for net load forecasting research has not yet been reported in the literature.

1.1 Problem statement

Accurate short-term net load forecasting is a prerequisite for efficient operation across the entire electricity value chain. Distribution system operators (DSOs) require it to plan grid operations, congestion management interventions, and infrastructure investments at the municipality level. Transmission system operators (TSOs) rely on it for system-level balancing and reserve allocation. Energy retailers and balance responsible parties use it to optimise procurement and reduce imbalance costs in day-ahead and intraday markets. Aggregators and flexibility service providers need it to schedule demand response and battery storage assets. Independent power producers and solar PV portfolio managers depend on it to optimise generation dispatch and trading positions. In short, any market participant or system actor whose decisions are conditioned on the quantity of electricity that will flow through the grid has a direct interest in the accuracy of net load forecasts.

Existing approaches were designed for systems in which prosumer generation is negligible or not explicitly represented. As a result, they are poorly suited to contexts in which a growing share of solar PV generation remains behind the meter and is only partially visible to system operators through net metered demand and export records. As solar PV penetration increases across Norway's 357 municipalities, forecast errors become more pronounced during daylight hours, particularly in summer months when solar irradiance is highest. These errors manifest as unexpected congestion, unnecessary curtailment, and suboptimal scheduling of flexible loads and energy storage. The practical problem is therefore: how can short-term net load forecasting models be adapted, at the municipality level, to explicitly account for prosumer solar PV generation, and what improvement in forecast accuracy can these stakeholders realistically expect?

The practical problem raises a set of scientific and engineering challenges that are not fully addressed by the existing literature. First, the Norwegian context is geographically and climatically distinctive: complex topography creates significant spatial variation in solar irradiance and temperature within short distances, complicating the selection of numerical weather data. Second, the Elhub consumption data are net load by construction, since they record metered grid import that is already net of behind-the-meter self-consumption. This is not merely a data availability issue, but an observability problem: the latent relationship between E19 export records, actual BTM installed capacity, and the full distribution of prosumer behaviour at the municipality level cannot be directly observed. Third, the heterogeneity of municipalities, in terms of

size, consumer mix, and solar penetration raises the question of whether a single modelling framework can be applied uniformly or whether different model specifications are appropriate for different penetration regimes. Fourth, the disaggregation of demand into three consumption categories (private, business, industry) creates both an opportunity for more granular feature engineering and a challenge: sparse data for small municipalities and the need for category-level normalisation strategies.

The overarching research question that motivates this thesis is:

To what extent can a modular forecasting framework that explicitly incorporates prosumer characteristics and consumer-category disaggregation improve short-term net load forecasting accuracy?

The thesis addresses this question at the Norwegian municipality level and examines how the improvement in residential group varies with local distributed solar PV penetration across a heterogeneous set of municipalities.

1.2 Purpose

The project serves multiple stakeholder communities, each of whom stands to benefit in distinct ways from short-term net load forecasting at the municipality level.

For distribution system operators (DSOs) and other grid actors in Norway and comparable Nordic markets, this work provides evidence on whether and when solar aware forecasting models yield operationally meaningful accuracy improvements over simpler baseline approaches. The findings directly inform decisions about investments in data infrastructure, metering systems, and modelling sophistication, enabling DSOs to prioritize resources toward municipalities where the solar signal is already material and where forecast improvements translate into measurable reductions in imbalance costs, congestion, and curtailments.

For the broader community of energy analytics companies, balance responsible parties, energy trading operations, optimization procedures, flexibility aggregators and independent power producers, this work aims to demonstrate a replicable, open-data methodology for building municipality-level net load forecasting pipelines. The modular design of the framework, in which feature groups are added and evaluated incrementally, allows practitioners to adapt the approach to their own data availability and operational constraints.

For the energy forecasting research community, this thesis contributes empirical evidence from a Nordic, municipality-level open-data context that has been largely absent from the literature. Prior work has been dominated by studies of high-penetration markets such as California, Australia, and southern Europe, or by analyses of aggregated national and regional systems [10, 11, 5, 12]. The systematic handling of the full Norwegian penetration spectrum, provides a dataset of ablation results that can serve as a benchmark for future net load forecasting (NLF) research in comparable low-to-moderate penetration contexts.

For Norwegian energy policy and regulatory bodies, this work quantifies the degree to which the Elhub AMI data infrastructure enables actionable grid intelligence as prosumer deployment continues to grow day by day [9]. The results speak directly to the policy question of whether the investment in universal smart metering completed in Norway in 2019 is translating into forecasting capabilities that support efficient grid operation under increasing renewable integration. With implications that extend to other Nordic countries too.

From a sustainability perspective, accurate NLF directly supports the energy transition by enhancing the optimal dispatch and enabling more efficient integration of distributed renewable resources. Reduced forecast errors translate into lower balancing costs, less unnecessary curtailment of renewable generation, and more efficient use of flexible loads and storage. All of these reasons contribute to a lower-carbon and more cost-efficient electricity system. The exclusive use of open data sources in this work furthermore ensures full reproducibility and supports the scientific community's ability to build on and extend these findings without proprietary data constraints.

No significant ethical issues are anticipated in connection with this project. All data used are publicly available published by Elhub; no individual-level metering data or personally identifiable information is accessed or processed at any stage of the analysis.

1.3 Objectives

The primary goals of this thesis are to develop and evaluate a scalable municipality-level net load forecasting framework, using open data sources and machine learning methods, and to assess under which conditions solar-aware modelling and prosumer-related effects provide meaningful forecasting value in practice.

These primary goals can be addressed through the following subgoals:

- Evaluate candidate open data sources for municipality-level net load forecasting and construct a complete data acquisition and preprocessing pipeline for the selected dataset (Section 3.2).
- Ensure that the developed framework is operationally relevant and practically deployable by accounting for real-world constraints such as Elhub data publication lag, realistic feature availability, and support for incremental daily updates (Section 3.1).
- Design and implement a structured baseline and ablation framework in which forecasting models of progressively increasing complexity are trained and evaluated in order to isolate the contribution of different feature groups to forecast accuracy (Section 3.5).
- Develop separate forecasting models for the residential, commercial, and industrial consumption categories, and aggregate category-level predictions to reconstruct total municipal net load (Section 3.6).
- Investigate how behind-the-meter solar PV and prosumer-related effects influence municipal net load patterns and forecasting performance, and assess whether these effects justify enhanced model specifications (Sections 3.6, 5.2, and 5.4).
- Develop an automated procedure for selecting informative weather variables and feature subsets for each municipality and consumption category, such that the framework can scale up (Section 3.3.1).
- Quantify the relationship between local solar PV penetration and the marginal improvement in forecast accuracy attributable to enhanced model specifications, and use these results to derive empirically grounded forecasting regimes (Sections 3.3.2 and 3.7).
- Analyze and interpret the obtained results in order to draw conclusions regarding when enhanced modelling complexity is justified for municipality-level net load forecasting in practice (Chapters 6 and 7).

The deliverables of the thesis include a complete open-data forecasting pipeline, a comparative evaluation of forecasting model configurations across municipalities and consumption categories, and a set of conclusions regarding when enhanced modelling is justified at the municipality level.

1.4 Research Methodology

This thesis adopts a quantitative, data-driven research methodology. The primary analytical approach is a structured ablation study: a set of forecasting models of progressively increasing complexity is defined, trained and validated on common relevant sets and evaluated on a common held-out test set. By comparing model tiers that differ in feature groups, causal attribution of performance improvements to specific input variables is made possible. A linear regression baseline derived from the Tao Hong framework [1] is compared against LightGBM gradient boosting, allowing assessment of whether model complexity is warranted given the available data volume and the nature of the forecasting task.

The data sources and their evaluation are described in detail in chapter 3. In brief: the primary dataset is Elhub open data (net-consumption, E19 production and installed capacity in the municipality level). Exogenous weather features, accessed via the Rebase Energy API are used as high-quality historical weather datasets for model development, with the understanding that operational deployment would require substitution with a numerical weather prediction (NWP) forecast. Model performance is assessed on a held-out test set, while a k-fold validation strategy is used for hyperparameter tuning, model selection, and comparative evaluation.

1.5 Delimitations

First, the study focuses on short-term forecasting at the municipality level in Norway, with primary emphasis on the day-ahead (24-hour) horizon, which falls within the short-term category as defined in Section 2.1. Very short-term (seconds to minutes), intra-day, medium-term, and long-term forecasting are therefore outside the scope of the work.

Second, the analysis is carried out at the municipality level and reflects an aggregated distribution-level footprint. It does not address transmission-system forecasting, bidding-zone forecasting, or feeder-level network analysis. The focus is therefore on forecasting methodology under municipality-level aggregation rather than on detailed physical grid modelling.

Third, the thesis does not aim to develop an indirect or decomposed net load forecasting framework in which gross demand and behind-the-meter PV generation are forecast separately and then combined. Such an approach would require more spatially detailed information about distributed

PV installations than is available in the open datasets used here. Instead, the target is municipality-level metered net consumption, while solar-related and prosumer-related effects are incorporated indirectly through weather variables, installed-capacity information, and their impact on the aggregate signal.

Fourth, the thesis does not develop standalone solar PV forecasting models or direct household-level prosumer behaviour models, since the available data is aggregated and does not support analysis at individual level.

Finally, although different forecasting approaches are reviewed in the literature, the empirical analysis is limited to selected machine learning methods suitable for the available data and the objectives of the study. Deep learning architectures, private smart-meter data, and feeder-level measurements are outside the scope. The conclusions should therefore be interpreted in the context of municipality-level forecasting under realistic open-data constraints.

1.6 Structure of the thesis

Chapter 2 provides the theoretical background for net load forecasting, covering the behind-the-meter invisibility problem, direct versus indirect forecasting strategies, modelling approaches, evaluation metrics, and the Norwegian data infrastructure. Chapter 3 describes the methodology, including data collection, the Weather Station Selection algorithm, the three-stage experimental design, and the evaluation framework. Chapter 4 presents the exploratory data analysis of the Elhub datasets and the Weather Station Selection results. Chapter 5 reports the empirical results of all three experimental stages, including the synthetic behind-the-meter solar experiment. Chapter 6 discusses the findings in terms of framework validation, the prosumer feature bundle, industrial load heterogeneity, and open-data boundaries. Chapter 7 summarises the conclusions, limitations, and directions for future work.

Chapter 2

Background

This chapter provides the theoretical and contextual foundation required to understand the net load forecasting problem addressed in this thesis. It begins by positioning electricity load forecasting within power system operations and by introducing the distinction between gross and net load, which motivates the core problem of the study. The chapter then examines the impact of behind-the-meter (BTM) solar generation, introducing the BTM invisibility problem and its implications for forecasting, as well as the emergence of solar-driven effects such as daytime load suppression and the duck-curve phenomenon. Building on this, it presents the main forecasting strategies proposed in the literature, namely direct and indirect net load forecasting. Subsequently, the chapter reviews the main modelling approaches relevant to this thesis, including multiple linear regression baselines following the Tao Hong framework, tree-based machine learning models such as gradient-boosted trees, and, more briefly, modern deep learning approaches used in recent studies. Finally, the chapter situates the study within the Norwegian electricity system and the open-data environment provided by Elhub and identifies the key research gaps that motivate the contributions of this thesis.

2.1 Electricity Load Forecasting

Electricity load forecasting is the practice of estimating the amount of electrical energy that will be consumed over a specified future horizon. Accurate forecasts are indispensable at every level of the power system: TSOs use them for system balancing, reserve dimensioning, congestion management and reserve scheduling. DSOs rely on them for network planning and congestion management. Finally, electricity traders require them for contract

settlement and risk hedging [13]. The literature traditionally segments forecasting horizons into four categories:

1. **Very short-term:** seconds to minutes, relevant for frequency regulation and ancillary services activation.
2. **Short-term:** hours to two weeks, the scope of operational planning, reserve scheduling, and activation.
3. **Medium-term:** weeks to three years, relevant for maintenance scheduling and ancillary services capacity planning.
4. **Long-term:** beyond three years, relevant for infrastructure investment.

The core modelling assumption underlying most short-term load forecasting (STLF) research is that electricity consumption is a structured signal driven by a small set of identifiable factors:

1. **Calendar effects:** the hour of day, day of week, month of year, and public holidays, which encode habitual human activity patterns.
2. **Meteorological drivers:** primarily ambient temperature for heating and cooling loads, and solar irradiance for lighting demand and, increasingly, for its indirect suppression of net load through distributed solar generation.
3. **Secular trends:** long-term effects reflecting economic activity, demographic change, and the gradual electrification of end-uses.

The modelling challenge is to disentangle these structured components from stochastic noise and from the emerging disturbances introduced by behind-the-meter (BTM) distributed energy resources, while accounting for the fact that each consumption category exhibits distinct load patterns that vary further across municipalities.

The period from 2010 to 2019 witnessed a rapid growth in energy forecasting publications, with load forecasting papers accounting for roughly half of the entire energy forecasting literature [1]. The Global Energy Forecasting Competitions (GEFCom2012, GEFCom2014, GEFCom2017) organised by Tao Hong and collaborators played a formative role in this expansion, establishing open benchmarks and the pinball loss metric that have since become standard references [14, 1, 15]. More recent competitions, including the Post-COVID Day-Ahead Demand Forecasting Competition [16]

and the Hybrid Energy Forecasting and Trading Competition 2024 [17], have extended the scope to incorporate structural shift, demand changes and the joint treatment of renewables and load.

2.1.1 Definition and the BTM Invisibility Problem

The growing deployment of rooftop solar photovoltaic (PV) installations has introduced a fundamental distinction between *gross load*, which describes the total electricity consumed by end-users, and *net load*, which refers to the electricity that must be supplied by the central grid after accounting for behind-the-meter distributed generation. Formally:

$$L_{\text{net}}(t) = L_{\text{gross}}(t) - G_{\text{BTM}}(t) \quad (2.1)$$

where $G_{\text{BTM}}(t)$ denotes the instantaneous BTM solar generation that is consumed on-site and therefore never flows through the distribution system metering point [10, 18]. The net load is what the DSO observes through its smart meters, and it determines the physical loading of distribution feeders, substation transformers, and transmission lines. As PV penetration grows, $G_{\text{BTM}}(t)$ becomes an increasingly large and volatile term, causing net load to deviate substantially. Moreover, this introduces a form of non-stationarity, as forecasting models trained on historical data that reflect pre-solar or low-penetration conditions may fail to generalise under evolving prosumer behavior and increasing distributed generation.

The central difficulty for DSOs is what the literature refers to as the *BTM invisibility problem*. More specifically, self-consumed solar energy never registers at any metering boundary owned by the network operator. As a result, $G_{\text{BTM}}(t)$ is inherently unobservable from the operator's data infrastructure alone [5, 18]. The DSO cannot directly disaggregate gross consumption from BTM self-consumption as it can only observe their difference. This creates an asymmetry between the information available to the network operator and the physical reality of its distribution network.

Net load forecasting (NLF) emerged as a response to this asymmetry as power systems transition from the traditional load forecasting to net load forecasting. Rather than forecasting gross load, which was the historical focus of the STLF literature, NLF directly targets the grid-observable quantity, either by learning the dynamics of net load from metered data alone (the direct strategy) or by forecasting gross load and BTM generation separately and subtracting (the indirect strategy). This distinction is discussed in detail in Section 2.1.3.

In the Norwegian context considered in this thesis, the Elhub dataset provides municipality-level consumption that corresponds to *net metered consumption*, that is electricity imported from the grid after behind-the-meter self-consumption has been accounted for [9]. The municipal consumption series can therefore be interpreted as a direct measurement of net load as defined in Equation (2.1).

However, the underlying solar generation process remains unobservable. Exported surplus electricity is recorded in the production dataset, but behind-the-meter self-consumption is not directly measured. As a result, temporal variation in prosumer behaviour modulates the observed net load signal in ways that are not captured by conventional temperature and calendar features alone, motivating the use of prosumer-aware inputs in forecasting models.

2.1.2 The Duck Curve and Load Profile Distortion

The macroscopic signature of BTM solar penetration on the net load profile is commonly represented by the *duck curve*, a phenomenon first identified by the California Independent System Operator (CAISO) [19]. As rooftop PV capacity grows, the midday net load trough deepens, the afternoon ramp steepens, and the morning and evening peaks are amplified or unchanged. Plotted as a daily profile, the net load curve acquires the characteristic shape of a duck: a deep trough in the solar peak hours and an elevated head in the evening.

The operational implications of the duck curve are significant for grid management. Rapid ramping requirements on dispatchable generation, potential overgeneration in midday hours, and sharp evening peaks challenge both scheduling and reserve adequacy. Forecasting accuracy during these transition periods is particularly important because underestimation of the evening ramp can lead to inadequate reserve activation, while overestimation of the midday trough can cause premature curtailment of conventional generation.

At low and moderate penetration levels, the duck curve's absolute depth remains small or negligible. However, even nascent solar markets exhibit a related effect: a systematic suppression of net load during daylight hours. As installed PV capacity increases, successive increments of behind-the-meter generation progressively reduce daytime net imports, even if the overall load profile does not yet exhibit a pronounced duck-curve shape. This effect is further reinforced by ongoing electrification, which increases underlying demand while distributed generation offsets a growing share of

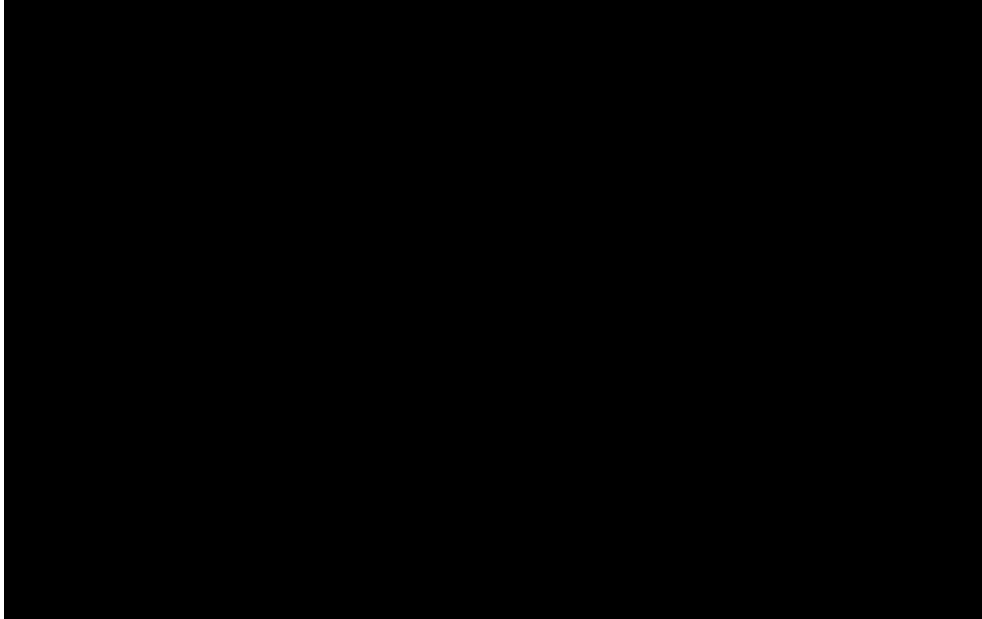


Figure 2.1: The CAISO duck curve illustrating net load changes under increasing solar generation [20]

daytime consumption, thereby dampening the observed growth in measured net load. As a result, standard calendar- and temperature-driven models tend to systematically overestimate midday net load as installed PV capacity increases. Detecting and forecasting this progressive suppression signal is therefore a key motivation for solar- and prosumer-aware feature engineering. The significance of this non-stationarity extends beyond accuracy: as discussed in [10], the forecast errors from a solar-naïve model and the BTM generation forecast errors are cointegrated, meaning that only an integrated modelling approach can fully capture the coupled dynamics of load and BTM generation.

2.1.3 Direct and Indirect Net Load Forecasting Strategies

Two broad approaches to net load forecasting are recognised in the literature [5]:

1. **Direct approach:** The direct strategy treats net load as a single forecasting target, training a model on historical metered net load together with weather and calendar covariates. The model implicitly

captures the solar suppression effect to the extent that it is reflected in the historical data and in the selected input features.

2. **Indirect approach:** The indirect strategy, also referred to as the additive approach, forecasts gross load and behind-the-meter (BTM) generation with separate models, and subsequently combines these sub-forecasts to obtain net load.

The comprehensive survey by Tziolis et al. [5], which reviewed 73 NLF studies published between 2010 and 2024, found that among studies specifying their modelling strategy, direct and indirect approaches were approximately equally represented (46.6% and 42.5%, respectively), with the remainder using hybrid or ensemble methods. The survey found that the choice of strategy was less determinative of accuracy than the choice of input features, with irradiance and other prosumer drivers consistently appearing as influential meteorological inputs after historical net load.

The trade-offs between direct and indirect strategies have been characterised empirically by Falces et al. [18], who compared four machine learning regressors. More specifically, Random Forest, XGBoost, Support Vector Regression, and Cubist rule-based regression were implemented under direct and indirect NLF architectures at a Spanish 15 kV distribution substation under three PV penetration scenarios: 0%, 17.8%, and 35.6% of installed capacity relative to peak load. At 35.6% penetration, the best indirect model achieved MAPE of 7.33% versus 7.64% for the best direct model, a margin of 0.31 percentage points. The authors concluded that the difference is “almost negligible,” suggesting that the direct strategy with appropriate solar-aware features can approach the performance ceiling of the indirect architecture without requiring a separate BTM generation sub-model. In a microgrid with 33% annual solar penetration, the direct (integrated) modelling approach outperformed the indirect (additive) alternative by 10.69% in RMSE, with a tighter uncertainty range [10].

Thepprom et al. [21] bridge the two strategies by proposing an unsupervised contrastive-based method for estimating BTM PV generation from aggregated net load without any measured BTM data. By disaggregating the net load signal into estimated pure load and estimated BTM generation using an LSTM network trained on GHI contrasts, they show that forecasting on the disaggregated series with DeepAR, TFT, and TiDE outperforms direct NLF with all three architectures, reducing RMSE by 8–13% and MAE by 15–16% on US prosumer datasets.

This thesis adopts the *direct* NLF strategy. The primary reason is practical: the Elhub dataset does not provide a direct measurement of BTM generation for the large majority of Norwegian municipalities. Without a reliable gross load series, the indirect strategy cannot be implemented. The direct framework is supplemented by proxy signals for BTM solar activity and more specifically, the metered export of prosumers to the grid which can provide directional information about solar suppression within the single-target modelling framework. The justification for this choice is further developed in Chapter 3.

2.1.4 Solar PV Penetration and the Value of Solar Features

The central empirical question of this thesis—whether and when solar-aware features improve NLF accuracy—is fundamentally a question about the penetration level at which the BTM solar signal becomes strong enough to be detected and exploited by a forecasting model. At very low penetration, BTM generation is a small perturbation and a temperature-only model fits nearly as well as a solar-aware model. As penetration grows, irradiance contributes increasing marginal information relative to what temperature features already explain.

In [10], it is demonstrated that at a microgrid with 33% annual solar penetration (UCSD), where an integrated NLF model using solar irradiance outperformed a naive additive approach by 10.69% in RMSE skill. Tziolis et al. [11] showed that a Bayesian neural network-based direct NLF model achieved $\text{nRMSE} < 5\%$ across Cyprus feeders with PV shares from 2.5% to 34.2%, with statistical post-processing reducing this further to $< 1.3\%$. Moreover, in [18], it was found that forecast error increases monotonically with penetration level under both direct and indirect architectures.

This penetration-dependent relationship motivates careful definition of the penetration metric used in this thesis. Normalising E19 metered solar export by total municipal consumption, which includes large industrial loads with negligible BTM solar exposure, would dilute even more the signal. Private residential consumption is therefore used in a try to isolate the relevant and possible penetration effect.

2.2 Modelling Approaches for Net Load Forecasting

The NLF literature has drawn on a broad range of modelling approaches. The comprehensive survey by Tziolis et al. [5] covering 73 studies published between 2010 and 2024, identifies five principal categories: persistence (naïve) methods, grey system theory models, statistical regression models, machine learning methods, and hybrid approaches. Among studies specifying their modelling strategy, machine learning and hybrid methods together account for approximately 84 % of the reviewed models, reflecting a broad consensus that data-driven approaches capture the non-linear interactions between load, weather, and behind-the-meter generation more effectively than purely parametric alternatives [5]. Across all categories, historical net load and solar irradiance consistently emerge as the most influential input variables, reinforcing a central finding of the literature: the choice of input features often determines forecast accuracy more than the choice of model class [5, 17].

A further dimension that cross-cuts all modelling approaches is the direct versus indirect strategy introduced in Section 2.1.3. Models that ignore solar dynamics have been shown to underperform integrated approaches by a non-trivial margin even at moderate penetration levels [10]. The sections that follow describe the principal modelling families in turn, with emphasis on the methods most relevant to the experiments of this thesis.

2.2.1 Statistical Regression Methods

Statistical methods can be classified into two broad categories. The first includes time-series models such as ARIMA, SARIMA, and exponential smoothing, while the second consists of linear regression based approaches. The key virtues of these methods are transparency, simplicity, computational efficiency, and well-understood statistical properties, which make them valuable both as production tools and as interpretable baselines against which more complex methods can be benchmarked.

The multiple linear regression (MLR) model proposed by Hong, commonly referred to as the *vanilla* model, is not merely a statistical formulation but a benchmark framework that has been systematically adopted in major forecasting competitions such as GEFCom and BigDEAL [14, 22].

Its role is twofold. First, it provides a transparent, and interpretable baseline grounded in well-understood statistical principles. Second, it serves as a minimum performance threshold, below which more complex models

and practices are not considered meaningful improvements. In this sense, the vanilla model functions as a sanity check for any proposed methodology [22].

It expresses hourly load as a function of calendar indicators and a third-order polynomial in dry-bulb temperature T :

$$y_t = \beta_0 + \beta_1 \text{Trend}_t + \beta_2 D_t + \beta_3 H_t + \beta_4 M_t + \beta_5 H_t D_t + f(T_t) + \varepsilon_t \quad (2.2)$$

where Trend is a linear trend variable, H is a categorical variable representing the hour of day, D is a categorical variable representing the day of week, M is a categorical variable representing the month of year and ε_t is the residual error term. The function $f(T_t)$ denotes the temperature regression terms, written as:

$$\begin{aligned} f(T_t) = & \beta_6 T_t + \beta_7 T_t^2 + \beta_8 T_t^3 \\ & + \beta_9 T_t H_t + \beta_{10} T_t^2 H_t + \beta_{11} T_t^3 H_t \\ & + \beta_{12} T_t M_t + \beta_{13} T_t^2 M_t + \beta_{14} T_t^3 M_t \end{aligned} \quad (2.3)$$

The hour-by-temperature and month-by-temperature interaction terms capture the empirical regularity that the sensitivity of electricity demand to temperature varies by time of day and by season. The cubic polynomial approximates the characteristic U-shaped temperature–load relationship without requiring a manually specified change-point. Estimated by ordinary least squares, the vanilla model achieves competitive accuracy across diverse datasets and served as a reference benchmark throughout the GEFCom competition series [14, 1].

For NLF, the vanilla model can be naturally extended by incorporating global horizontal irradiance G alongside its interactions with hour-of-day and month-of-year indicators:

$$y_t^{\text{solar}} = y_t^{\text{vanilla}} + g(G_t) \quad (2.4)$$

where $g(G_t)$ denotes irradiance terms and their interactions with hour-of-day and month-of-year effects:

$$g(G_t) = \theta_1 G_t + \theta_2 G_t H_t + \theta_3 G_t M_t \quad (2.5)$$

This extension allows the model to capture the suppressive effect of behind-the-meter (BTM) photovoltaic generation on net load, which is typically strongest around solar noon and negligible during night hours.

A structural limitation of the vanilla MLR is that it relies only on

contemporaneous temperature, thereby neglecting the thermal inertia of buildings and the delayed response of HVAC systems. This limitation is addressed in the literature by the *recency effect*, which introduces temporal memory through lagged temperature variables.

Rather than selecting these lags heuristically, the recency effect is formalised as a structured model selection problem over two hyperparameters: the number of lagged hourly temperatures (h) and the number of lagged daily moving-average temperatures (d). The optimal (d, h) pair is selected based on validation error, transforming feature engineering into a controlled model selection procedure [23].

Empirical results show that incorporating recency effects can reduce forecasting errors significantly, highlighting the importance of temporal dependencies in load and net load dynamics.

In the NLF literature, statistical regression methods account for approximately 13 % of reviewed models [5]. Their principal limitation is that non-linear interactions between load, weather, and solar generation must be explicitly specified, which becomes increasingly challenging under heterogeneous PV penetration levels. This limitation motivates the use of more flexible machine learning methods, as discussed in the following section.

2.2.2 Machine Learning Methods

Machine learning methods — particularly gradient-boosted tree ensembles — have emerged as the dominant modelling approach in recent NLF studies and energy forecasting competitions, owing to their ability to capture non-linear feature interactions automatically, handle mixed feature types natively, and generalise robustly from moderately sized datasets [5, 17]. Falces et al. [18] compared Random Forest, XGBoost, Support Vector Regression, and Cubist across three PV penetration levels at a Spanish distribution substation, finding that tree-based models consistently outperformed SVR, and that the accuracy gap between direct and indirect NLF architectures was small when solar-aware features were included — at 35.6 % penetration, the best indirect model achieved MAPE of 7.33 % versus 7.64 % for the best direct model, a margin the authors characterised as almost negligible.

The foundational building block of all gradient-boosted and ensemble tree methods is the regression tree, which recursively partitions the feature space into non-overlapping rectangular regions. Individual trees are high-variance but low-bias learners; ensemble strategies combine many trees to reduce variance while preserving low bias. All gradient-boosted models share

a common framework of stagewise additive modelling in function space: at each iteration m , a new shallow tree $h_m(\mathbf{x})$ is fitted to the negative gradient of the current loss, and the ensemble is updated as:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \quad (2.6)$$

where $\eta \in (0, 1]$ is the learning rate. For squared-error loss the negative gradient reduces to ordinary residuals, so each successive tree corrects the errors of the current ensemble. The four specific algorithms used in the thesis experiments are described below and compared in Table 2.1.

Table 2.1: Comparison of gradient-boosted tree and ensemble algorithms relevant to short-term load forecasting.

Property	Random Forest [24]	XGBoost [25]	CatBoost [26]	LightGBM [27]
Strategy	Parallel bagging	Sequential boosting	Sequential boosting	Sequential boosting
Tree growth	Level-wise	Level-wise	Symmetric	Leaf-wise
Regularisation	Random feature subsets	L1/L2 + γ	Ordered boosting	GOSS + EFB
Categoricals	Encoding required	Encoding required	Native	Native
Overfits w/ more trees	No	Yes (without tuning)	Less prone	Moderate

Random Forest [24] constructs B trees on bootstrap samples and aggregates predictions by averaging. At each node split, only a random subset of $m \approx p/3$ features is considered as candidates, decorrelating the trees and reducing ensemble variance beyond what bootstrap aggregation alone achieves. Unlike boosting, the ensemble does not overfit as B increases, and out-of-bag (OOB) error estimation provides an internal validation signal without a separate holdout set. Random Forest is valued for its robustness to noisy or redundant inputs and its minimal sensitivity to hyperparameters; in the NLF experiments of Falces et al. [18] it achieved the best performance under the indirect architecture across all penetration scenarios, owing to its superiority in PV generation sub-forecasting.

XGBoost [25] introduces a regularised objective that augments the loss with an explicit complexity penalty on leaf weights and tree size, employs a second-order Taylor expansion of the loss to derive optimal leaf weights analytically, and handles missing values natively through sparsity-aware split finding. It has dominated tabular data science competitions and serves as a well-established single-model baseline for energy forecasting tasks [28, 17].

CatBoost [26] addresses *prediction shift* — the target leakage that arises in standard gradient boosting when gradients are computed from a model

trained on the same examples — through *ordered boosting*: the gradient for each training example is computed from a model fitted only on preceding examples in a random permutation. Categorical features are encoded using ordered target statistics that avoid leakage by the same principle. Symmetric (oblivious) decision trees, where all nodes at the same depth apply the same split condition, act as structural regularisation and enable efficient inference.

LightGBM [27] introduces two algorithmic innovations for large-scale learning. Gradient-based One-Side Sampling (GOSS) retains all high-gradient instances — which carry the most information for further boosting — while subsampling low-gradient instances with a correction factor that preserves the fidelity of the gradient estimate. Exclusive Feature Bundling (EFB) identifies mutually exclusive sparse features and consolidates them into single effective features, reducing dimensionality without information loss. Leaf-wise (best-first) tree growth achieves a lower loss at a given number of leaves than level-wise alternatives, controlled against overfitting through a minimum-data-in-leaf constraint. These properties make LightGBM particularly well suited to the municipality-level NLF task in this thesis: the training sets are large enough to benefit from GOSS and leaf-wise growth, the feature space includes many potentially redundant meteorological and calendar variables that EFB can consolidate, and the fast training time supports cross-validation across multiple municipality and consumer-category combinations. LightGBM therefore constitutes the primary machine learning model in the experiments of this thesis.

2.2.3 Deep Learning Approaches

Deep learning architectures have attracted growing interest in the NLF literature for their ability to learn complex temporal dependencies directly from raw sequences. Long Short-Term Memory networks [29] introduced gated recurrent units capable of capturing long-range dependencies, and have been combined with wavelet decomposition for multi-scale NLF on European datasets [30]. The Temporal Fusion Transformer [31] combines recurrent layers for local processing with multi-head self-attention for long-range dependencies, and provides interpretable variable importance scores through learned selection networks. DeepAR [32] frames forecasting as autoregressive conditional density estimation, naturally yielding probabilistic outputs. TiDE [33] demonstrated that a simple MLP-based encoder-decoder can match or exceed Transformer-based models at substantially lower computational cost.

Thepprom et al. [21] applied TFT, DeepAR, and TiDE specifically to NLF, showing that forecasting on disaggregated BTM series reduces RMSE by 8–13 % and MAE by 15–16 % relative to direct net load forecasting with all three architectures. However, their disaggregation step requires an unsupervised LSTM trained on irradiance data matched to individual prosumer timestamps — a condition that presupposes individual-level or small-group BTM data that is not available in the municipality-aggregated open data used in this thesis. In the Norwegian context, the indirect strategy is in any case structurally infeasible because the Elhub dataset does not provide a ground-truth gross load series, making it impossible to construct the BTM sub-model that the additive architecture requires (see Section 2.1.3).

More broadly, deep learning methods require substantially more data, more demanding hyperparameter optimisation, and offer reduced interpretability compared to gradient-boosted tree methods. For a study centred on understanding which feature groups drive accuracy improvements across a heterogeneous set of municipalities, the gradient-boosted tree family provides the more appropriate balance of performance and interpretability. Deep learning extensions are identified as a direction for future work.

2.3 The Norwegian Electricity System and Elhub Data Infrastructure

2.3.1 Norwegian Grid Structure and DSO Roles

Norway occupies a singular position in European electricity systems. Approximately 90% of domestic electricity generation is hydropower [7], making Norway simultaneously one of the most capable producers of renewable electricity in the world and one of the most temperature-sensitive electricity consumers: space heating accounts for a large fraction of final energy consumption through direct electric resistance heating and heat pumps, and average household electricity consumption reaches approximately 16,000 kWh per year [34], significantly above the European average.

Norway's electricity grid operates at three levels: the transmission grid (300–420 kV) operated exclusively by Statnett SF as the certified TSO; the regional distribution grid (33–132 kV); and the local distribution grid (230 V to 22 kV). Approximately 130 distribution system operators (DSOs) manage these regional and local networks [35], most owned by municipal or county authorities, subject to incentive-based revenue cap regulation by

NVE [36]. This fragmented landscape—many operators managing networks serving small populations—creates a practical motivation for municipal-level NLF tools that can be applied uniformly without per-operator model tuning. Norway participates in the Nordic electricity market (Nord Pool) with interconnections to Sweden, Denmark, Finland, the United Kingdom, and the Netherlands; distribution network loading, however, is a purely domestic concern and it is at this level that the forecasts produced in this thesis are operationally relevant.

2.3.2 Solar PV Growth and the Plusskunde Scheme

Norwegian solar PV capacity has grown rapidly from approximately 15 MW in 2015 to over 800 MW by end of 2025, with capacity approximately doubling in 2023 alone [3, 8]. The *Plusskundeordningen* (Plus Customer scheme), effective January 2017, created a net metering framework requiring energy providers to purchase surplus generation from systems up to 100 kW [37], supported by Enova SF installation rebates. Residential systems typically fall well below this threshold, meaning that nearly all residential solar capacity is registered under this scheme and the resulting E19 prosumer export signal closely tracks the spatial and temporal distribution of BTM solar adoption across Norwegian municipalities. Norwegian E19 export grew substantially between 2023 and 2025 at the national level, making the NLF challenge studied in this thesis a moving target: the penetration-dependent accuracy will widen progressively as more municipalities enter higher penetration regimes.

2.3.3 Elhub: The Norwegian Metering Data Hub

Elhub is Norway’s centralised metering data hub, launched in February 2019, developed and operated by Statnett under mandate from NVE [9]. It manages all active metering points in Norway and serves as the central repository for all market processes, including the calculation of balance settlement data reported to eSett. Three Elhub datasets are used in this thesis.

The **consumption dataset** provides hourly net consumption per municipality, disaggregated by consumer group (private residential, business, and industry). As confirmed directly by Elhub in correspondence, the `quantityKwh` field represents net metered consumption at the grid connection point: behind-the-meter self-consumption is already subtracted, so the series directly measures municipal-level net load [9].

The **production dataset** contains hourly exported surplus for metering

point type E19 (prosumers). Elhub confirmed that E19 records net production as fed into the grid only—self-consumed generation is invisible in this series. E19 therefore aims to serve as a partial, observable proxy for BTM solar activity: high E19 export in a given hour indicates that BTM self-consumption is also elevated, making it a potentially informative input feature for direct NLF models alongside irradiance predictors. A privacy threshold applies: municipalities with fewer than 100 active prosumer metering points are suppressed, which is why the production dataset covers 276 of Norway’s 358 municipalities [9].

The **installed capacity API** provides daily snapshots of registered generation capacity (kW) per municipality. Total solar installed capacity is obtained by summing metering point types E18 and E19 for the solar production group, as confirmed by Elhub. This slowly-varying structural covariate modulates the magnitude of the irradiance signal over time and is used in the penetration metric construction described in Section 2.1.4.

2.3.4 ERA5 Reanalysis Weather Data

Operational NWP forecasts are appropriate for deployed forecasting systems but unavailable as multi-year historical archives at the spatial resolution required for a research study. This thesis uses ERA5 [38], the fifth-generation ECMWF atmospheric reanalysis, as the source of historical meteorological data for model training and evaluation. ERA5 provides hourly estimates of near-surface temperature and solar downward radiation on a 0.25° latitude–longitude grid (approximately 25–31 km spacing), covering 1940 to the present. ERA5 is constrained by actual atmospheric observations and therefore free of the forecast error that would affect NWP-based inputs, making it appropriate for research model development, though its use in place of operational NWP constitutes a deployment limitation discussed further in (Section 7.2).

2.4 Research Gaps and Positioning

The preceding review reveals four gaps that motivate the specific contributions of this thesis.

Gap 1: Absence of systematic, cross-municipality NLF studies for Nordic open-access metering data. The NLF literature is dominated by single-site or single-utility case studies, typically from Mediterranean or North American contexts. The Norwegian Elhub dataset provides multi-year,

nationally representative open data for 357 municipalities—a rare opportunity for comparative analysis across a full penetration spectrum within a single data ecosystem. No published study has exploited this resource for NLF research.

Gap 2: The penetration threshold for operationally significant solar feature benefit is unknown. The literature documents that solar features are necessary at high penetration [10] and redundant at zero penetration. At what penetration level the accuracy improvement becomes operationally meaningful has not been addressed for any European market context.

Gap 3: The E19 solar export signal as a directly measured solar proxy. The BTM invisibility problem is typically addressed by BTM disaggregation or by using irradiance as a proxy. The Elhub E19 time series provides a distinct, directly measured signal correlated with both local irradiance and installed BTM capacity. No published study has examined whether E19 adds marginal forecasting information beyond irradiance alone.

Gap 4: Consumer-segment disaggregation of the solar feature benefit. Most NLF studies model total net load without disaggregating by consumer type. The Elhub dataset distinguishes private residential, business, and industrial segments with different BTM solar exposure profiles. Norway’s *Plusskundeordningen* scheme, which has driven the residential solar adoption, provides additional motivation to examine whether the E19 signal benefit is concentrated in the residential segment.

2.5 Summary

This chapter has established the theoretical context for the short-term net load forecasting problem addressed in this thesis.

Net load is the grid-observable quantity of interest for DSOs, defined as gross consumption minus BTM self-generation and directly confirmed by Elhub to be the quantity measured by the municipal consumption aggregates. The BTM invisibility problem motivates the use of observable proxy signals (irradiance, E19 export, installed capacity) within the direct NLF strategy adopted throughout this thesis, supported by empirical evidence that the performance advantage of indirect strategies at low penetration levels is modest [18].

Section 2.2.1 provides the background for some of the statistical baselines approaches that will be tested. The gradient-boosted tree family is reviewed for context, with LightGBM selected as the primary machine learning framework constituting models. Evaluation relies on MAPE as the primary metric;

specific metric selection rationale for the NLF context is discussed in Section 3.8.

Chapter 3

Methodology

3.1 Forecasting Framework

The empirical work in this thesis is organised as a sequential four-stage pipeline, illustrated in Figure 3.1. Municipal electricity consumption is disaggregated into three load categories — residential (private), commercial (business), and industrial — each of which is modelled independently, as the categories differ in their demand patterns.

Each stage of the pipeline aims to produce a concrete decision that serves as the sole input to the subsequent stage. This design enforces a strict separation between data preparation, feature infrastructure, model selection, and national evaluation, ensuring that every modelling decision is grounded in empirical evidence and validation from the preceding stage rather than in prior assumptions.

Figure 3.1 presents an overview of the complete research pipeline. The framework consists of two prerequisite infrastructure stages — data acquisition and weather station selection — followed by three experimental stages of increasing scope. The overarching objective is not to identify the single globally optimal model, but to establish a principled, operationally grounded decision strategy for each load category and municipality. A secondary objective is to determine empirically whether publicly available solar export data and installed solar capacity can provide measurable forecasting value, given that the behind-the-meter solar suppression phenomenon described in Section 2.1.2 can influence net load signal patterns. The stages are described in detail below.

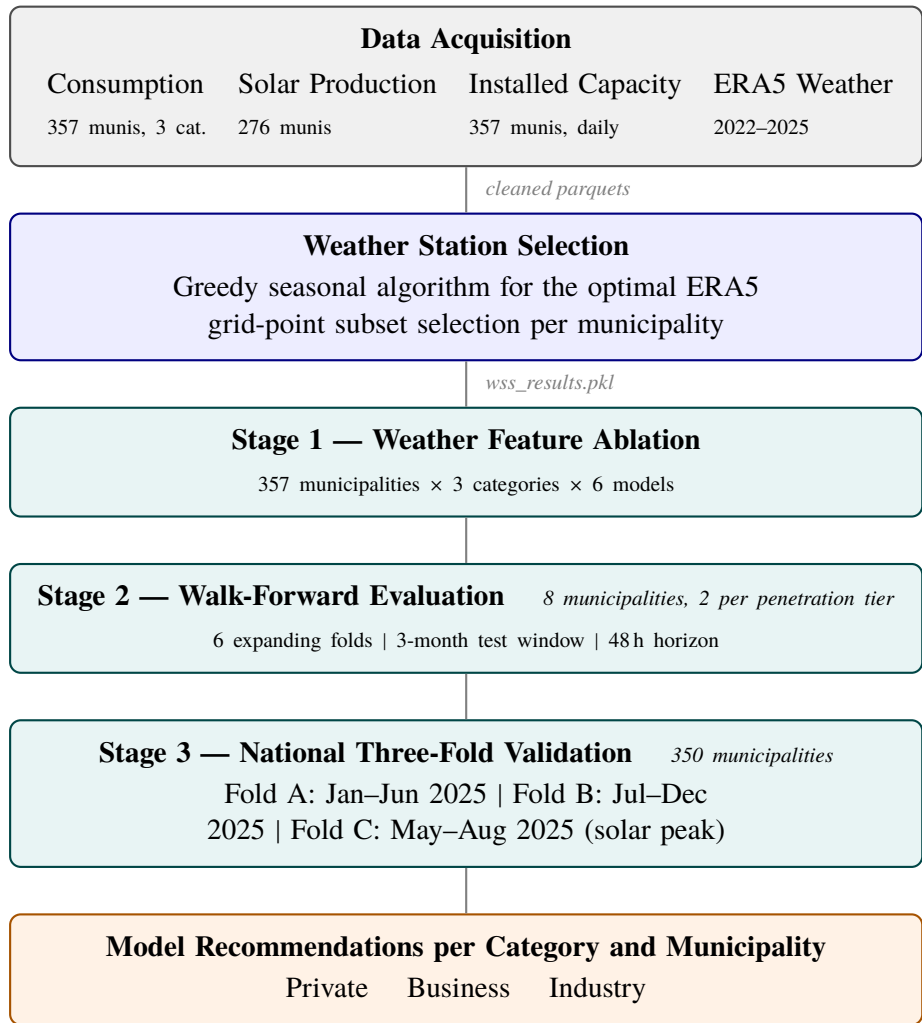


Figure 3.1: Research pipeline overview. Data Acquisition and Weather Station Selection are prerequisite infrastructure stages. Stages 1–3 are the three experimental stages, each building on the artefacts of the preceding one.

Data Acquisition

Three open datasets are assembled from the Elhub open data portal: hourly consumption records for 357 Norwegian municipalities across three consumer categories (private, business, and industry); hourly E19 solar export records available for 276 municipalities and daily informed installed solar PV capacity snapshots for all 357 municipalities. These are combined with ERA5 reanalysis weather fields for the period 2022–2025, retrieved via the Rebase Energy API. Further details are provided in Section 3.2.

Weather Station Selection

Before any model is trained, a greedy seasonal Weather Station Selection (WSS) algorithm is applied independently to each of the 357 municipalities, following and extending the approach of [39]. This stage determines the optimal combination of ERA5 grid points to represent each municipality's meteorological conditions. The WSS procedure is described in full in Section 3.3.1.

Stage 1: Weather Feature Ablation

The first experimental stage establishes the marginal forecasting value of each meteorological variable group through a cumulative exogenous weather ablation across all 357 municipalities and all three consumption categories. No autoregressive lags are included at this stage, isolating the weather signal from the load-history signal. The output is the confirmed full weather feature set and empirical evidence of weather feature importance. Section 3.5 gives full details.

Stage 2: Walk-Forward Evaluation on Representative Municipalities

The second experimental stage evaluates different modeling approaches across multiple folds on eight representative municipalities, selected to span the range of population density, solar penetration level, and geographical location. A six-fold expanding-window walk-forward protocol with a strict 24-hour rolling-prediction regime is applied. This stage aims to identify the most competitive modelling approach for national scale-up, while simultaneously providing insights on seasonal performance, generalisation capability, baseline performance and overfitting risk. Section 3.6 gives full details.

Stage 3: National Three-Fold Validation

The third experimental stage scales the promoted models to 350 municipalities using three temporally stratified evaluation folds, providing full-year performance evidence and concrete modelling recommendations and strategies for each consumer category, while highlighting the strengths and limitations of each modelling approach across load categories. Section 3.7 gives more details.

3.2 Data Acquisition

Four distinct data streams are combined to construct the modelling dataset. Table 3.1 summarizes their provenance, temporal coverage, spatial resolution, and primary role in the analysis.

Table 3.1: Summary of data sources used in this study.

Source	Period	Res.	Coverage
Elhub consumption	2023-03–2026-02	Hourly	357 municipalities
Elhub E19 production	2023-03–2026-03	Hourly	276 municipalities
Elhub solar capacity	2023-03–2026-03	Daily	357 municipalities
ERA5 via Rebase Energy	2022-01–2025-12	Hourly, 0.25°	357 / 350†

Seven municipalities (IDs: 4639, 5042, 5055, 5516, 5532, 5534, 5605) are excluded from Stages 2–3.

3.2.1 Elhub Consumption Data

Elhub publishes aggregated hourly consumption statistics at the municipality level through its open data portal. The dataset analysed in this thesis was downloaded as a set of monthly CSV files covering the period 2023-03-01 to 2026-03-01. The full consumption dataset contains approximately 28.1 million rows across 357 municipalities and three consumption categories.

Each record corresponds to one (municipality, consumption group, hour) triplet and contains the following fields:

- `START_TIME`: UTC-offset timestamp
- `MUNICIPALITY_NUMBER`: four-digit municipality code

- CONSUMPTION_GROUP: *private*, *business*, or *industry*
- QUANTITY_KWH: energy consumption in kWh
- METERING_POINT_COUNT: number of metering points
- LAST_UPDATE_TIME: timestamp of last record update

Explicit confirmation from Elhub established that QUANTITY_KWH records net grid import at the connection point: for E19-metered prosumer households, the meter records net import when household consumption exceeds generation within the hour, and net export when generation exceeds consumption. The consumption signal therefore already reflects the netted prosumer effect and constitutes a true net load signal in the DSO sense; it is the quantity that distribution planning must forecast and serves as the target variable throughout this study.

The METERING_POINT_COUNT field records the number of individual connection points whose aggregate is reported. Elhub confirmed that the standard in Norway is one metering point per consumption or production unit at the connection point. The three consumption categories are modelled separately throughout this study. The *private* (residential) category is the primary focus, as Norwegian rooftop solar installations under the Plusskunde scheme are metered as E19 residential prosumers. The *business* (commercial) category covers retail premises, offices, and small industrial units. The *industry* category covers large process and production facilities with dedicated metering arrangements.

Elhub applies one confidentiality threshold relevant to the consumption dataset: if fewer than five metering points contribute to an aggregate record, the record is not published at all. This threshold primarily affects the *industry* category in the smallest municipalities, where records with metering point counts as low as five are sometimes visible in the dataset, but records below that floor are absent. For the *private* and *business* consumption categories, which routinely aggregate thousands of metering points, this threshold is not binding. Elhub further confirmed that municipality code 0000 (name: *Ukjent tilhørighet*, meaning “unknown affiliation”) collects metering points not yet assigned to a municipality by the responsible grid owner; all records with this code are excluded from the analysis.

3.2.2 Elhub E19 Production Data

A separate section of the Elhub open data portal provides hourly E19 production records, also downloaded as monthly CSV files. Each record corresponds to one (municipality, hour) pair and contains the following fields:

- `START_TIME`: UTC-offset timestamp
- `END_TIME`: UTC-offset timestamp
- `MUNICIPALITY_NUMBER`: four-digit municipality code
- `NAME`: municipality name
- `METERING_POINT_TYPE_CODE`: always E19 in this dataset
- `QUANTITY_KWH`: net energy exported to the grid in kWh
- `LAST_UPDATE_TIME`: timestamp of last record update

The full production dataset contains approximately 7.1 million rows covering 276 of Norway's 357 municipalities, with the earliest record dated 2023-03-04 00:00 UTC. The remaining 81 municipalities have no E19 production record. As confirmed by Elhub, this is principally due to a 100-metering-point confidentiality minimum that applies specifically to the E19 production dataset: municipalities with fewer than 100 registered E19 prosumers have their production figures withheld entirely. The absence of a production record therefore indicates either that the municipality has fewer than 100 E19 meters, or that it has no registered E19 installations at all.

To understand the role of E19 within the broader Norwegian metering framework, it is necessary to distinguish the four metering point type codes defined by Elhub according to the ebIX standard [40]:

- **E17 — Consumption.** An ordinary consumption meter that records grid import only.
- **E18 — Production.** A dedicated production meter that records grid export only.
- **E19 — Combined.** A bidirectional net meter that records net import when household consumption exceeds generation within the hour, and net export when generation exceeds consumption. E19 meters are the dominant prosumer metering type for residential rooftop solar and they appear in both the consumption and production datasets depending on the hourly net balance.

- **E20 — Exchange.** A meter used at grid interconnection points between distribution system operators or across national borders. E20 meters are not relevant to this study.

To illustrate the E19 metering logic: if a prosumer household generates 4 kWh and consumes 3 kWh during a given hour, the meter records a net export of 1 kWh to the grid, which appears as `QUANTITY_KWH = 1` in the production dataset; the consumption dataset records zero import for that hour. Conversely, if the same household generates 1 kWh but consumes 3 kWh, it records a net import of 2 kWh in the consumption dataset and contributes nothing to the production dataset. In both cases, the 3 kWh consumed directly from the solar panel so self-consumption is invisible to any metering point.

The `QUANTITY_KWH` field in the production dataset thus records only the *exported surplus*, not total solar generation. All features derived from the production dataset are consequently lower-bound estimates of true solar activity; the solar penetration metrics defined in Section 3.3.2 are similarly lower bounds.

Elhub confirmed that the E19 production category represents predominantly solar generation, but may also include other small-scale distributed generation types at the prosumer level; the exact mix behind each metering point is not disclosed in order to protect confidentiality. The strong seasonal amplitude of the production signal with summer-to-winter export ratios of approximately 25:1 and near-zero values outside of May to September provides circumstantial evidence that solar generation dominates, but this cannot be asserted as exclusive. Throughout this study, E19 production features are therefore described as solar export proxies rather than solar generation proxies.

National E19 export totals 0.159 TWh in 2023, 0.240 TWh in 2024, and 0.321 TWh in 2025, corresponding to cumulative growth of approximately +100% over the two-year period. The seasonal amplitude is extreme: summer (May–August) export exceeds winter (November–February) export by a factor of approximately 25, consistent with a predominantly solar-driven signal.

3.2.3 Installed Solar Capacity Registry

Elhub provides a third dataset accessible via a dedicated API endpoint, reporting the registered installed solar PV capacity (in kW) per municipality and metering point type on a daily basis. For each calendar day, a record is published indicating the total installed capacity of all active E18 and E19 metering points in the municipality. The dataset covers the period 2023-03-01

to 2025-12-31 and contains 374,470 rows across 357 municipalities, broken down by metering point type (E18 and E19). The API enforces a rate limit that requires sequential, per-municipality data retrieval.

The dataset serves two distinct roles in this study, both restricted to E19 metering points. For E19, hourly production data is available in the open data (Section 3.2.2), making it possible to compute a capacity factor. The daily E19 capacity snapshot therefore provides the normalising denominator for the E19 capacity factor feature:

$$CF_t^{E19} = \frac{Q_t^{E19} [\text{kWh}]}{C_{d(t)}^{E19} [\text{kW}] \times 1 \text{ h}}, \quad (3.1)$$

where $C_{d(t)}^{E19}$ is the installed E19 capacity on the calendar date of hour t . This normalisation converts raw export volume, which grows monotonically with fleet expansion, into a physically bounded signal ($CF \in [0, 1]$ under ideal conditions) that is comparable across years and municipalities. The time-varying E19 capacity series additionally informs the year-over-year capacity growth feature (`cap_growth_yoy`), which captures regime changes in fleet size that affect the statistical relationship between lagged solar signals and future net load.

For E18 metering points, no hourly production data is available in the open data; the E18 capacity figures are therefore not used in the capacity factor calculation or in any forecasting feature. The sum $C^{E18} + C^{E19}$ is retained as a total installed solar proxy for the exploratory characterisation of municipalities.

3.2.4 ERA5 Weather Data via the Rebase Energy API

Meteorological inputs are sourced from the ERA5 reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) [38]. ERA5 provides hourly fields at 0.25° horizontal resolution (approximately 28 km at Norwegian latitudes). The two primary variables are two-metre air temperature (T_{2m} , in degrees Celsius) and surface solar downward radiation (SSRD, in W m^{-2}).

A second set of ERA5 variables—wind speed at 10 m (WS), wind direction decomposed into sine and cosine components (WD_sin, WD_cos), relative humidity (RH), cloud cover fraction (CC), total precipitation (TP), and snow depth (SD) is retrieved for the same grid and period and forms the *extended weather feature bundle*. These variables capture additional meteorological

drivers of electricity demand that complement the primary temperature and irradiance inputs.

ERA5 data is retrieved through the Rebase Energy API [41], which provides a pre-processed interface to the ECMWF ERA5 archive with Parquet output format and configurable grid-point selection. For each target municipality, a candidate set of ERA5 grid points is retrieved by specifying a bounding box centred on the municipality centroid. After the Weather Station Selection procedure (Section 3.3.1) selects the optimal subset, the retained grid-point series are averaged with equal weights to form the weather input to the forecasting model.

3.2.5 Data Quality and Limitations

- **ERA5 gap:** Seven municipalities lack extended ERA5 weather files (IDs: 4639, 5042, 5055, 5516, 5532, 5534, 5605). These seven municipalities are excluded from all analyses that require extended weather features (Stages 2 and 3); the remaining 350 municipalities are fully covered.
- **E19 coverage:** E19 production and installed capacity data are available for only 276 of the 357 municipalities; the remaining 81 have no registered prosumer metering points in the Elhub open dataset and are assigned to the weather-only model (M3) by default.
- **E19 aggregation:** Unlike the consumption dataset, which is disaggregated by consumer segment, the E19 production series is a municipality-level aggregate across all prosumer types. E19-derived features used in private residential models therefore incorporate export from commercial prosumers as well, introducing a signal contamination that cannot be quantified without installation-level data. Its practical impact is expected to be limited, since the Plusskunde scheme is dominated by small residential installations.
- **Temporal coverage:** All three Elhub datasets begin in March 2023, limiting the effective training window to approximately three years and restricting the capacity growth feature (`cap_growth_yoy`) to at most two complete annual cycles.

3.3 Weather Features

3.3.1 Weather Station Selection

ERA5 reanalysis data are on a 0.25° grid, meaning a municipality can span several grid points, not all of which are equally representative of local load behaviour. By averaging all available grid points introduces noise from meteorologically irrelevant locations and it is not a well validated decision based on actual results. This thesis applies a Weather Station Selection (WSS) algorithm adapted from [39] to identify the optimal subset and number of ERA5 grid points for all Norwegian municipalities.

The original [39] framework addresses two simultaneous questions: (1) *how many* weather stations should be used for a load zone, and (2) *which* stations should be used. Their greedy algorithm simultaneously determines both the optimal count k^* and the optimal station set, and was shown to outperform all fixed- k alternatives on two US utility case studies.

The algorithm proceeds in two phases:

1. **Phase 1:** Temperature grid point scoring.
2. **Phase 2:** Irradiance grid point scoring.

Both phases of the algorithm use Tao's Vanilla Benchmark Model Equation (3.2) as the predefined scoring model M . In both phases, the scoring target is the *private residential* consumption series, normalised by private metering point count (kWh/MPC h^{-1}). The WSS-selected grid point indices are subsequently reused without modification for the commercial and industrial models.

$$\begin{aligned} \text{Target} = & \beta_0 + \beta_1 \text{Trend} + \beta_2 \text{Hour} + \beta_3 \text{Weekday} + \beta_4 \text{Month} + \beta_5 \text{Hour} \cdot \text{Weekday} \\ & + \beta_6 T + \beta_7 T^2 + \beta_8 T^3 \\ & + \beta_9 T \cdot \text{Hour} + \beta_{10} T^2 \cdot \text{Hour} + \beta_{11} T^3 \cdot \text{Hour} \\ & + \beta_{12} T \cdot \text{Month} + \beta_{13} T^2 \cdot \text{Month} + \beta_{14} T^3 \cdot \text{Month} \end{aligned} \quad (3.2)$$

where *Trend* is a linear trend, *Hour* a 24-level class variable, *Weekday* a 7-level class variable, *Month* a 12-level class variable, and T the candidate temperature series. Phase 2 additionally includes irradiance terms $\beta_{15} G + \beta_{16} G \cdot \text{Hour} + \beta_{17} G \cdot \text{Month}$, with T held fixed at its Phase 1 composite.

Three adaptations extend the original framework to this study.

(i) ERA5 grid points as candidate stations. Rather than physical weather stations, the candidate pool consists of ERA5 reanalysis grid points within a dynamic bounding box centred on the municipality centroid. The bounding box half-width is scaled to municipality area:

- $< 200 \text{ km}^2$: 3×3 grid (box half-width 0.25° , 9 candidates)
- $200\text{--}2,000 \text{ km}^2$: 5×5 grid (box half-width 0.50° , 25 candidates)
- $> 2,000 \text{ km}^2$: 7×7 grid (box half-width 0.75° , 49 candidates)

This scaling ensures that small urban municipalities are not penalised by distant grid points, while large rural municipalities have adequate spatial coverage. Municipality centroids are sourced from the Kartverket API [42].

(ii) Seasonal validation criterion. Rather than a single temporal validation window, the algorithm uses season-specific scoring. Phase 1 evaluates temperature candidates on the *winter months* of the validation period, ensuring selection for heating-dominated load behaviour. Phase 2 evaluates irradiance candidates on the *summer months* of the validation period, ensuring selection for solar-production-dominated patterns. The temporal split used for WSS is: training 2023-03-01 to 2024-06-30 and validation 2024-07-01 to 2024-12-31. As a consequence of this split, only two months of each season fall within the validation window (November–December for winter, July–August for summer). The year 2025 is deliberately excluded from the WSS procedure to prevent any form of data leakage into the subsequent forecasting experiments, which use 2025 as the held-out test period.

(iii) Minimum municipality size filter. Municipalities with fewer than 500 private metering points are excluded from WSS ($n = 9$ skipped). For these municipalities, the ERA5 grid point closest to the municipality centroid is used directly as the weather representative. WSS was successfully applied to 348 of 357 municipalities.

The error measure used throughout is MAE, consistent with [39]. The maximum pool size is $K_{\max} = 25$.

The equal-weighted averages of the selected grid points constitute the WSS-composite inputs T_{WSS} and G_{WSS} used by all downstream models. Numerical outcomes of the WSS procedure are reported in Section 4.4.

Table 3.2: WSS greedy algorithm [39], applied twice per municipality: Phase 1 for temperature (T, winter scoring), Phase 2 for irradiance (G, summer scoring).

Step	Action
1	Collect all ERA5 grid points within the municipality bounding box as candidates.
2	Test each candidate point individually: fit the linear regression model on the full training period and measure its prediction error restricted to the relevant season of the validation period — winter months for Phase 1, summer months for Phase 2.
3	Rank all candidate points from best to worst based on their seasonal prediction error.
4	Starting from the best-ranked point, progressively add points one by one. After each addition, average all selected points with equal weights and record the prediction error on the validation period.
5	Stop when all candidates have been tried. Select the combination that gave the lowest validation error. This is the final weather series for the municipality.

3.3.2 Solar Penetration Metrics and Scope of Solar Features

To investigate whether the benefit of solar enriched modelling varies with the level of solar activity, each municipality is characterised by a flow-based penetration percentage (*solar_pen*), defined as the fraction of private grid consumption offset by E19 export during the peak solar window:

$$\text{solar_pen (\%)} = \frac{\sum \text{E19_kWh(Jun–Aug 2024)}}{\sum \text{private_kWh(Jun–Aug 2024)}} \times 100. \quad (3.3)$$

June–August 2024 is used as the reference window because it falls within the training set, predates the 2025 test year, and captures peak solar production.

Based on this metric, four penetration tiers are assumed: T0 (no E19 data), T1 (< 1%), T2 (1–5%), T3 (5–10%), T4 ($\geq$ 10%). These tiers are used throughout the results analysis (Chapter 5) to examine whether the benefit of the provided solar information is scaled with the level of solar penetration. The tier distribution across the 350 run municipalities is reported in Section 5.3.1.

The solar information is derived from the datasets described in Sections 3.2.1 and 3.2.3 are used exclusively for the private (residential) consumption category. This design choice rests on two arguments.

Physical argument: Norwegian rooftop solar installations are concentrated in the residential sector. Under the Norwegian *Plusskunde* scheme, the regulatory framework that governs small-scale prosumers with installed capacity below a defined threshold, households are granted the right to export surplus solar generation to the grid and are assigned a combined E19 meter. As a result, E19 meters in Norway are predominantly held by residential customers. The E19 production signal therefore can carry information about residential solar activity.

Empirical argument: As demonstrated in the Stage 1 weather feature ablation (Section 5.1), solar irradiance (G) contributes more to residential solar predictions rather than industrial and business category.

3.4 Model Baselines and Operational Protocol

A well-defined baseline is essential for quantifying the value added by each modelling step. This study uses two complementary baselines.

The *primary baseline* is the naive seasonal persistence model (M_{naive}), which predicts each hour using the observed value from exactly one week prior: $\hat{y}(t) = y(t-168)$. This model requires no training, has zero parameters, and captures the dominant weekly periodicity of electricity consumption. It is used across all stages, all categories, and all municipalities, serving as the universal lower bound against which all other models must improve.

The *statistical baseline family* consists of three linear regression models in the Tao Hong formulation [14]:

- $M0a$: Tao Vanilla — calendar features plus third-order temperature polynomial with hour and month interactions.
- $M0b$: $M0a$ plus temperature recency lags (T_{lag24h} , T_{lag48h} , moving averages).
- $M0c$: $M0b$ plus solar irradiance G with hour and month interactions and G lags.

All three linear regression models are evaluated in Stage 2 (Section 3.6) on the eight representative municipalities, allowing each incremental improvement to be measured in isolation. In Stage 3 (Section 3.7), only $M0c$ — the most competitive statistical baseline as established in Stage 2 — and the naive model are retained as final baseline candidates.

A central methodological choice affecting all model specifications is the adoption of a 24-hour rolling-prediction regime in Stages 2 and 3. This section explains the rationale, as it directly determines which autoregressive features and solar information modelling are permissible.

In the Norwegian smart metering infrastructure, Elhub consolidates hourly metering data from all DSOs. While smart meters record consumption in near-real time, the publication of finalised municipality-level aggregates involves a delay of at least one day (every day at 18:00 the data of the previous day are uploaded to Elhub) due to data consolidation and validation. This means that at the time of issuing a day-ahead forecast for day $D+1$, the most recent Elhub data available is from day $D-1$ at the earliest. The knowledge of what happened in the nearest past therefore carries a publication lag of at least 24 hours and is not reliably available at day-ahead prediction time. Using load features with lags shorter than 48 hours would constitute implicit data leakage. To guard against this, the shortest autoregressive lag in any model is 48 hours ($\text{kwh_lag_48h} = y(t-48)$).

This constraint is operationalised through a 24-hour rolling prediction regime: on each calendar day of the test period, a prediction block covering the following 24 hours is generated using feature vectors constructed strictly from data available at prediction time.

This constraint has direct implications for the solar features, making the detection of the solar suppression effect even more challenging.

3.5 Stage 1: Feature Engineering and Ablation Ladder in National Level

The features in this section are organised in a six-level cumulative ablation ladder for Stage 1, ascending from calendar-only to the full extended weather set (Table 3.3).

In Stages 2 and 3, the weather feature set confirmed by Stage 1 serves as the backbone of all LightGBM models, augmented with autoregressive load lags (kwh_lag_48h , kwh_lag_168h) and, for the private category, the prosumer feature bundle described in Section 3.6. The LightGBM configuration is kept uniform across all municipalities and consumption categories. The model is trained with an MAE objective and the following hyperparameters:

- `num_leaves = 63`

Table 3.3: Weather feature ablation ladder (Stage 1). Each level adds one group to all preceding levels. No AR lags are included in Stage 1.

Model	Group	Features added
M0_cal	Calendar	hour, dayofweek, month, Trend, is_holiday, is_holiday_tomorrow
M1_T	Temperature	T_{wss} , T^2 , T^3 + $T \times \text{hour}$, $T^2 \times \text{hour}$, $T^3 \times \text{hour}$ + $T \times \text{month}$, $T^2 \times \text{month}$, $T^3 \times \text{month}$
M2_G	Solar irradiance	G_{wss} , $G \times \text{hour}$, $G \times \text{month}$
M3_W	Wind	WS, WD_sin, WD_cos, WS $\times$ hour, WS $\times$ month
M4_RH	Humidity & cloud	RH, CC, RH $\times$ hour, RH $\times$ month, CC $\times$ hour, CC $\times$ month
M5_P	Precipitation & snow	TP, SD

- `learning_rate = 0.05`
- `feature_fraction = 0.8`
- `min_child_samples = 20`
- `n_estimators = 2000`

Early stopping is applied after 50 rounds without improvement in the validation MAE.

3.6 Stage 2: Walk-Forward Evaluation on Representative Municipalities

Stage 2 evaluates the full modelling ladder on a controlled set of eight municipalities under an expanding-window walk-forward protocol. Its purpose is to identify the most competitive modelling approach per consumption category for national scale-up, and to provide evidence on seasonal behaviour and generalisation capability.

Municipality selection: Eight municipalities are selected to span all four solar penetration tiers, with two municipalities per tier. The penetration metric π_m is defined in subsection 3.3.2. Table 3.4 also reports the annual E19-to-private-consumption ratio for each municipality for better understanding.

Table 3.5 defines the eight models evaluated in Stage 2.

Table 3.4: Representative municipalities used in Stage 2

Municipality	Tier	π_m (%)	Annual (%)
Oslo	T1	0.62	0.15
Bergen	T1	0.80	0.25
Stavanger	T2	1.30	0.38
Ås	T2	4.81	1.32
Gran	T3	8.29	2.23
Hå	T3	8.63	2.91
Hægebostad	T4	10.97	3.69
Vestby	T4	16.63	4.51

Prosumer feature bundle: The prosumer bundle added in M5_clean consists of five features derived from the E19 production and capacity datasets:

cap_e19_per_mpc : Installed E19 capacity (kW) per private metering point which is a slowly-varying proxy for local solar penetration intensity.

G_x_cap : Interaction between WSS-selected GHI and cap_e19_per_mpc, approximating the expected behind-the-meter solar suppression signal.

e19_cf_rolling_7d : 7-day rolling mean of the E19 capacity factor (see eq. (3.1)); captures recent fleet export behaviour.

e19_cf_lag168 : E19 capacity factor lagged by 168 hours (same hour, previous week).

cap_growth_yoy : Year-on-year growth in installed E19 capacity; flags regime changes in fleet size.

All five features are derived from data available at least 48 hours before the forecast horizon, satisfying the operational lag constraint (Section 3.4).

Evaluation protocol: Six quarterly test folds span July 2024 to December 2025 (Table 3.6). In each fold, the model is trained on all available data up to three months before the test window; the preceding three months serve as the validation set for early stopping; and forecasts are generated under a 24-hour rolling day-ahead regime (Section 3.4). Figure 3.2 illustrates the expanding-window structure.

Table 3.5: Stage 2 model ladder. M5_clean is evaluated only for the private category.

Model	Type	Features
M_naive	Persistence	kwh_lag_168h — same hour last week.
M0a	OLS	Calendar only (Tao vanilla baseline).
M0b	OLS	M0a + temperature recency lags ($T_{\text{lag}24}$, $T_{\text{lag}48}$, $\bar{T}_{1d}$, $\bar{T}_{2d}$).
M0c	OLS	M0b + irradiance recency lags ($G_{\text{lag}24}$, $G_{\text{lag}48}$, $\bar{G}_{1d}$, $\bar{G}_{2d}$).
LGB_M0c	LightGBM	Identical feature set to M0c.
M3	LightGBM	Calendar + temperature + solar + cloud cover + extended weather + AR lags (lag_48h, lag_168h). Full weather ceiling; forwarded to Stage 3.
M4_mpc	LightGBM	M3 + log metering-point count features.
M4_spot	LightGBM	M3 + day-ahead spot price features (spot price and intraday price rank).
M5_clean	LightGBM	M3 + prosumer bundle

Table 3.6: Stage 2 walk-forward folds.

Fold	Train end	Val (3 months)	Test quarter
1	Mar 2023 – Mar 2024	Apr – Jun 2024	Jul – Sep 2024
2	Mar 2023 – Jun 2024	Jul – Sep 2024	Oct – Dec 2024
3	Mar 2023 – Sep 2024	Oct – Dec 2024	Jan – Mar 2025
4	Mar 2023 – Dec 2024	Jan – Mar 2025	Apr – Jun 2025
5	Mar 2023 – Mar 2025	Apr – Jun 2025	Jul – Sep 2025
6	Mar 2023 – Jun 2025	Jul – Sep 2025	Oct – Dec 2025

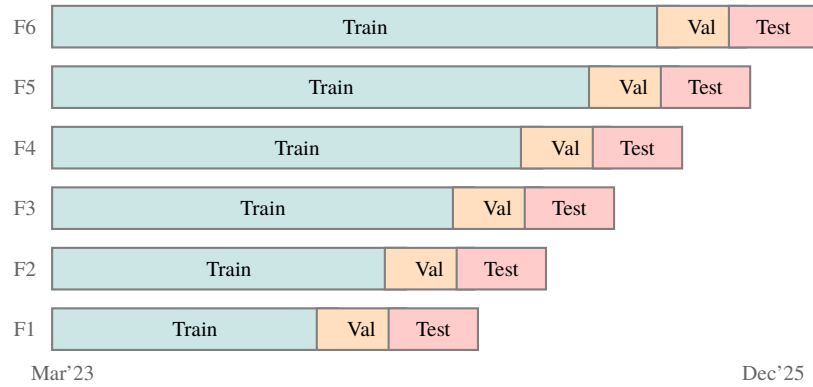


Figure 3.2: Expanding-window walk-forward protocol (Stage 2). Each fold extends the training set by one quarter. The validation window is used exclusively for early stopping.

The prosumer bundle (M5_clean) is evaluated only for the private category. M4_spot that includes features emerged from the spot prices is used only in the industry category. Stage 2 produces the model decision forwarded to Stage 3: M3 as the default for all categories, and M5_clean as a candidate for private municipalities with sufficient penetration evidence.

3.7 Stage 3: National Three-Fold Validation

Stage 3 scales the models promoted from Stage 2 to all 350 municipalities with full extended ERA5 coverage, providing national-scale evidence on model performance across all penetration tiers, consumer categories, and seasons. Each category is evaluated independently with a category-specific model scope, reflecting the findings of Stage 2.

Models evaluated per category:

- **Private:** M_naive, M0c, M3, M5_clean
- **Business:** M_naive, M0c, M3
- **Industry:** M_naive, M0c, M3, M4_spot

Stage 2 established that M3 is the default model across all categories. M5_clean, which augments M3 with the prosumer feature bundle, is evaluated only for private consumption. M4_spot, which augments M3 with day-ahead spot price features, is evaluated only for industry. Models showing

no consistent benefit in Stage 2 (M4_mpc) are excluded from Stage 3 (see Appendix B Table B.1 for the full feature set per model).

Fold structure: Three temporally stratified folds are used (Table 3.7). Fold A and Fold B together cover the full 2025 calendar year as two non-overlapping semesters, providing annual evidence. Fold C targets the peak solar season (May–August 2025) and serves as a dedicated robustness check for the solar signal.

Table 3.7: Stage 3 evaluation folds. Fold A + Fold B provide full-year evidence; Fold C is the solar-season robustness check.

Fold	Train	Val	Test	Role
A	Mar'23–Jun'24	Jul–Dec'24	Jan–Jun 2025	Annual (winter/spring)
B	Mar'23–Dec'24	Jan–Jun'25	Jul–Dec 2025	Annual (summer/autumn)
C	Mar'23–Dec'24	Jan–Apr'25	May–Aug 2025	Solar peak robustness

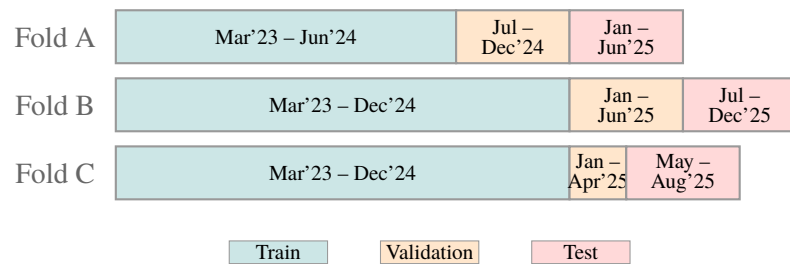


Figure 3.3: Temporal structure of the three Stage 3 evaluation folds.

Evaluation and assignment logic: All predictions are generated under a 24-hour rolling day-ahead regime and evaluated using MAPE (Section 3.8). Results are stratified by penetration tier (T0–T4) and consumer category.

For the private residential category, the primary comparison is M3 versus M5_clean. Two assignment criteria are applied to the two annual folds (Fold A and Fold B):

- The **conservative criterion** assigns M5_clean to a municipality only if M5_clean achieves strictly lower MAPE than M3 in *both* Fold A and Fold B. This ensures the recommendation is robust across seasons and not driven by a single anomalous fold.

- The **liberal criterion** assigns M5_clean if it outperforms M3 in at least one of the two annual folds.

Fold C (May–August 2025) is evaluated separately as a summer-peak robustness check. It does not affect the assignment decision but quantifies whether the portfolio gain holds during the period of highest solar activity.

For the business and industry categories, the comparison is M3 versus M_naive (and LGB_M0c for business), using the same conservative criterion: M3 is assigned to a municipality only if it outperforms the baseline in both Fold A and Fold B.

Penetration threshold analysis: For the private category, the conservative M5_clean win rate is examined across penetration tiers T0–T4, identifying whether a robust empirical threshold exists above which M5_clean provides consistent annual benefit.

3.8 Evaluation Framework

The primary evaluation framework is the three-stage ablation and cross-validation protocol described in Section 3.3. The primary metric is the Mean Absolute Percentage Error (MAPE), defined for a single municipality i over a test period of T_i hourly observations as

$$\text{MAPE}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \left| \frac{y_{i,t} - \hat{y}_{i,t}}{y_{i,t}} \right| \times 100, \quad (3.4)$$

where $y_{i,t}$ and $\hat{y}_{i,t}$ are the observed and predicted net load (kWh) at hour t . Hours with $y_{i,t} = 0$ are excluded to avoid division by zero.

National-level performance is summarised by two complementary aggregations. The **mean municipality MAPE** weights each municipality equally,

$$\overline{\text{MAPE}} = \frac{1}{N} \sum_{i=1}^N \text{MAPE}_i, \quad (3.5)$$

and serves as the primary deployment metric because each municipality constitutes an independent operational unit for a DSO regardless of its consumption volume. The **normalised (volume-weighted) MAPE**

aggregates errors across all municipalities proportionally to their consumption,

$$\text{nMAPE} = \frac{\sum_{i=1}^N \sum_{t=1}^{T_i} |y_{i,t} - \hat{y}_{i,t}|}{\sum_{i=1}^N \sum_{t=1}^{T_i} |y_{i,t}|} \times 100, \quad (3.6)$$

and serves as a national-aggregate robustness check; since larger municipalities carry proportionally more weight, a discrepancy between the two metrics signals that performance differs systematically between large and small municipalities.

Model comparisons in the same municipalities are made using two assignment criteria. The **conservative criterion** assigns a model to a municipality only if it achieves strictly lower MAPE than the comparative model in *both* annual folds (A and B). The **liberal criterion** assigns a model if it wins in at least one of the two annual folds. The conservative criterion is the primary decision rule, as it requires fold-consistent evidence of improvement and is robust to single-fold variance.

Chapter 4

Exploratory Data Analysis

This chapter presents the exploratory analysis and preprocessing results that form the empirical foundation for the forecasting experiments. The purpose is threefold: to characterise the quality and structure of the data prior to modelling, to establish empirical baselines for national demand patterns and solar penetration distribution, and to document the preprocessing decisions that shaped the pipeline.

Sections 4.1, 4.2, and 4.3 cover the consumption, production, and capacity datasets respectively; Section 4.4 reports the outcomes of the Weather Station Selection algorithm across the Norwegian municipalities; Section 4.5 describes the integration of day-ahead spot prices as an additional feature for the industrial consumer category.

The three primary datasets used in this thesis are all sourced from the Elhub open data portal. Although all three datasets originate from the same source, their acquisition methods differ due to the distinct API constraints that govern each endpoint, as summarised in Table 4.1.

Table 4.1: Elhub API endpoint constraints for the three datasets used in this thesis.

Dataset	Resolution	Max window	Acquisition method
Consumption	Hourly	1 day	37 monthly CSV files
Production (E19)	Hourly	1 day	37 monthly CSV files
Installed capacity	Daily	1 month	API calls per municipality

The consumption and production endpoints return aggregated hourly records but impose a maximum query window of one calendar day, making bulk historical retrieval impractical. Elhub makes the same data available as

monthly CSV files through the open data portal, which provide an efficient alternative for historical analysis. Accordingly, 37 monthly CSV files were downloaded for each dataset, covering March 2023 to February 2026 and saved as Parquet format, reducing storage from 3.6 GB to 320 MB for consumption and from 1.2 GB to 95 MB for production.

The installed capacity endpoint has a different access pattern: it returns daily snapshots of registered solar PV capacity (in kW) per municipality and metering point type, with a maximum query window of one month. More critically, capacity data for a specific municipality is only accessible through the per-municipality endpoint and not through any csv format, which requires iterating over all 357 Norwegian municipality identifiers individually.

4.1 Consumption Dataset

The consumption Parquet file contains 28,198,944 rows covering 358 municipalities across three consumer groups — private residential, commercial, and industrial — at hourly resolution. After excluding the *Ukjent* (unknown) municipality code 0000, which aggregates unattributed consumption, the working dataset covers 357 municipalities. Each row reports the total net electricity imported from the grid in kilowatt-hours for a given municipality, consumer group, and hour, together with the number of active metering points. No null values are present in any column.

A temporal continuity check was performed by computing the time difference between consecutive hourly records for each municipality and flagging any gap. No such gaps were found across the entire dataset. Every municipality reports a complete, uninterrupted hourly series for the full 36-month period, confirming that the Elhub infrastructure provides a reliable data stream with no systematic reporting outages. This absence of gaps is significant for the forecasting models, which rely on autoregressive lag features.

Table 4.2 summarises the key characteristics of the dataset after preprocessing.

Figure 4.1 shows national monthly electricity consumption disaggregated by consumer group over the full study period.

Aggregated over the full three-year period, the industrial consumer group accounts for the largest share of national electricity consumption at 48.5% (191.2 TWh), followed by private residential at 31.6% (124.7 TWh) and commercial at 19.9% (78.2 TWh). Figure 4.2 illustrates the national distribution of energy per category among all the study period.

Table 4.2: Summary characteristics of the consumption dataset after preprocessing.

Attribute	Value
Total rows	28,198,944
Municipalities (excl. Ukjent)	357
Consumer groups	Private residential, Commercial, Industrial
Temporal coverage	March 2023 – February 2026 (36 months)
Temporal resolution	Hourly (UTC)
Missing values	None
Temporal gaps > 25 h	None across all 357 municipalities
Raw csv source files	37 monthly files, $\approx$ 3.6 GB total
Parquet file size (Snappy)	$\approx$ 320 MB

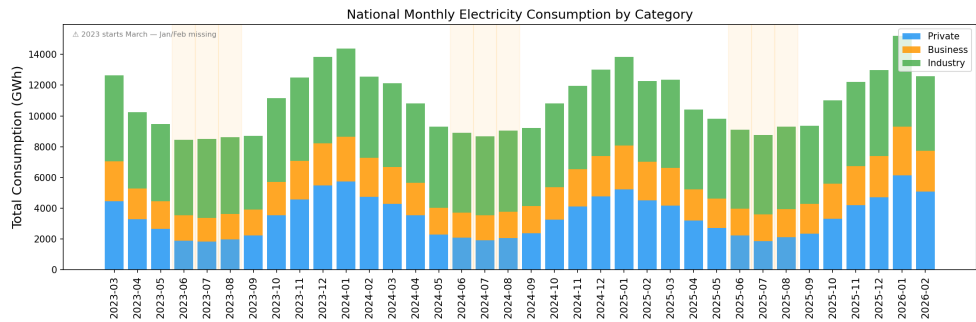


Figure 4.1: National monthly electricity consumption by consumer group (Norway, March 2023 – February 2026).

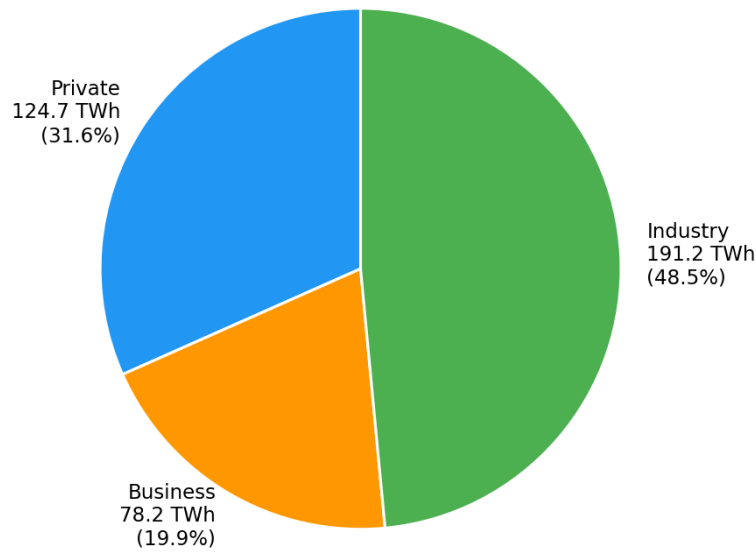


Figure 4.2: National electricity consumption by consumer group for the whole study period.

Table B.2 of Appendix B presents the annual breakdown using the March–December window common to all three years, to avoid the partial-year bias introduced by the missing January–February 2023 records.

Figure A.1 of Appendix A illustrates how the category shares remain stable year-over-year during the comparable March–December window, confirming that the relative composition of Norwegian electricity demand did not change dramatically over the study period.

This composition has a direct implication for the modelling approach. The three consumer groups exhibit structurally different demand dynamics: industrial load is driven by production schedules, commercial load follows business hours, and private residential load is shaped by household routines and behind-the-meter signals. Because each municipality has a distinct consumer mix and metering dynamic, a single aggregate model applied to total municipal load would dilute these structural signals. Training separate forecasters for each consumer group therefore enables the pipeline to isolate category-specific behaviour, attribute forecast errors to the relevant load segment, and assess the generalisability of the modelling approach across a heterogeneous set of municipalities. Figures A.2, A.3, A.4, and A.5 of

Appendix A illustrates this heterogeneity at the municipality level. Pie charts for Oslo, Bergen, Trondheim, and the municipality with the highest industry share (Sunndal) in the dataset reveal how markedly the consumer mix differs across cities, underscoring the need for a category-disaggregated modelling framework that does not require manual reconfiguration per municipality.

Figures A.6, A.8 and A.7 of Appendix A show the evolution of the active metering point count (MPC) for each consumer group across the five municipalities (Oslo, Vestby, Bergen, Trondheim, Stavanger). Each figure presents the growth index relative to the first observation (left panel) alongside the raw counts (right panel), enabling comparison of growth rates across municipalities of very different absolute sizes.

In Figure A.6, all five municipalities exhibit a slow but consistent upward trend over the three-year period, consistent with ongoing residential electrification and new prosumer registrations. Oslo holds approximately 350,000 private meters — roughly 35 times more than Vestby — yet the indexed growth rates are comparable across the group, suggesting that residential connection growth is a national phenomenon rather than a local one.

In Figure A.7, business metering point counts are broadly flat or slightly declining across most municipalities. Oslo and Bergen register a modest contraction, possibly reflecting business consolidation or metering point reclassification. Vestby displays an anomalous spike in mid-2024 followed by an immediate reversal, which is more consistent with a data registration artefact than a genuine structural change in the local commercial sector.

In Figure A.8, industry metering point counts show a pronounced decline across all focus municipalities. A near-simultaneous sharp drop is visible across all municipalities around mid-2025, consistent with a possible metering point reclassification event at the Elhub level rather than actual industrial contraction.

These observations have a direct implication for model design. Because the number of active metering points changes continuously as connections are added or removed, the raw kilowatt-hour aggregate contains a slow structural trend driven by fleet size changes in addition to weather and behavioural effects. The autoregressive lag features included in the forecasting models (`kwh_lag_48h`, `kwh_lag_168h`) are necessary to confront the level shift without requiring explicit MPC modelling.

Figure A.9 shows the ten largest municipalities by total electricity consumption for each consumer group over the study period. Oslo dominates the private residential and commercial rankings, reflecting its population

density. The industrial top-10 is concentrated in western and northern Norway, where energy-intensive manufacturing is located, and is largely non-overlapping with the population-driven rankings. This spatial separation between industrial and residential load centres reinforces the value of category-level disaggregation, as a national model aggregating all consumer types would be dominated by the geographic distribution of heavy industry.

4.2 Production Dataset

Of the 357 municipalities present in the consumption dataset, 276 (77.3%) also report E19 solar export data; the remaining 81 (22.7%) municipalities fall below Elhub’s privacy suppression threshold and are excluded from the production dataset as described in 3.2.2.

Table 4.3 reports national E19 export totals using a March–December window common to all three years. National solar export grew from 159.1 GWh in 2023 to 315.1 GWh in 2025 (+98% over two years), reflecting the accelerating deployment of pv installations in Norway.

Table 4.3: National E19 solar export — annual totals, March–December window

Year	E19 (GWh, Mar–Dec)	YoY growth
2023	159.1	—
2024	237.1	+49.0%
2025	315.1	+32.9%

Figure 4.3 shows the solar penetration metric 3.3 for all 276 municipalities, sorted in descending order. The distribution is strongly right-skewed: the large majority of municipalities record penetration below 1%, while only a handful exceed 5% or approach 10%. Dashed reference lines at mark tier boundaries used to stratify municipalities. The distribution is not static: both the consumption base and the installed solar capacity evolve continuously over the study period, meaning that penetration tier assignments reflect a snapshot of summer 2024 conditions rather than a fixed structural property of each municipality.

Table 4.4 shows the solar penetration metric 3.3 for June–August of each year across five focus municipalities. Penetration increases consistently year-over-year in all cases. Vestby stands out sharply, rising from 6.2% in 2023 to 44.8% in 2025 — meaning that in the summer of 2025, solar export

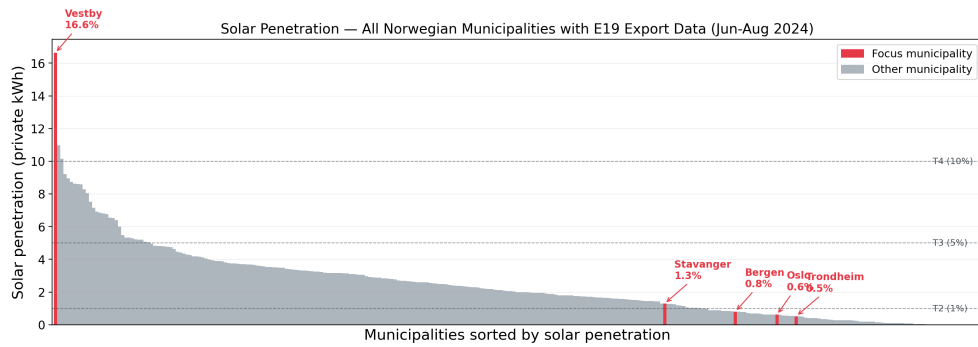


Figure 4.3: Solar penetration for all 276 municipalities with E19 data, June–August 2024. Focus municipalities are highlighted in red. Dashed lines indicate penetration tier thresholds.

from Vestby equaled nearly half of private residential electricity consumption, and this is a *lower bound* since behind-the-meter self-consumption is not observable in either dataset. The other remaining municipalities remain below 2% throughout the study period, illustrating the heterogeneity of the Norwegian solar penetration as well as consumption landscape.

Table 4.4: Solar penetration (%) — E19 export as a fraction of private residential summer consumption, June–August.

Municipality	Jun–Aug 2023	Jun–Aug 2024	Jun–Aug 2025
Vestby (3216)	6.17	16.63	44.76
Stavanger (1103)	1.12	1.30	1.59
Bergen (4601)	0.74	0.80	1.08
Oslo (0301)	0.43	0.62	0.82
Trondheim (5001)	0.39	0.53	0.61

The national summer-to-winter E19 ratio (June–August versus December–February, all years pooled) is $25.6\times$ (370.1 GWh versus 14.5 GWh), confirming the known information of the solar-driven signal, for solar fed to the grid. This extreme seasonality directly motivates the summer-scoring phase necessity of the Weather Station Selection algorithm described in Section 3.3.1.

4.3 Installed Solar Capacity Dataset

Installed solar capacity data was retrieved via the Elhub per-municipality API endpoint, as described in Section 3.2.3, yielding daily snapshots of registered E18 and E19 solar capacity for all 357 municipalities over the study period.

Of the 357 municipalities, 320 (89.6%) report a non-zero E19 installed capacity in the latest available snapshot (March 2026), while 37 (10.4%) record zero registered capacity. These 37 zero-capacity municipalities are a subset of the 81 municipalities that lack E19 export data in the production dataset (Section 4.2): the remaining 44 have some registered capacity but fall below the Elhub privacy threshold of metering points required for production data disclosure.

The capacity registry distinguishes between E19 meters and E18 meters. Over the study period, national E19 capacity grew from approximately 323 MW to 879 MW (+172%), while E18 capacity grew from 11 MW to 67 MW (+499%) and remained a small fraction of total solar capacity. The dominance of E19 is used as an assumption that this production signal is driven primarily by residential installations. It should be noted again, that E19 meters aggregate all net-metered connections at the municipal level, encompassing residential rooftop solar, small commercial rooftop systems, and other small-scale prosumer generators. The extent to which each installation type contributes to the observed export, and therefore to behind-the-meter load suppression is not observable from the available open data. For the purposes of this thesis, E19 export is treated as a residential-impact proxy: a signal whose magnitude probably can correlate with the degree of behind-the-meter solar activity, while acknowledging that its composition represents a limitation and threat of the open-data approach that is respecting deployability.

Table 4.5 shows the ten municipalities with the highest E19 installed capacity as of December 2025. Large cities such as Oslo, Bergen and Kristiansand dominate in absolute terms, while Vestby, despite its small size, ranks third nationally, reflecting its high solar adoption rate.

Table 4.6 shows the capacity evolution for five focus municipalities across the study period. Vestby records the highest relative growth at +179.8% (2023–2025), which directly explains the rapid increase in its E19 export and solar penetration documented in Section 4.2.

Table 4.5: Top 10 municipalities by E19 installed capacity, December 2025 snapshot.

Municipality	ID	E19 capacity (kW)
Oslo	0301	41,820
Kristiansand	4204	28,645
Vestby	3216	28,035
Bergen	4601	27,744
Stavanger	1103	20,015
Sandefjord	3907	19,360
Drammen	3301	19,019
Lillestrøm	3205	18,742
Sandnes	1108	17,616
Fredrikstad	3107	15,273

Table 4.6: E19 installed capacity for focus municipalities at end-of-year snapshots.

Municipality	End-2023 (kW)	End-2024 (kW)	End-2025 (kW)	Growth 2023–25
Oslo	25,342	35,369	41,820	+65.0%
Vestby	10,018	15,658	28,035	+179.8%
Bergen	19,704	23,825	27,744	+40.8%
Stavanger	11,636	16,293	20,015	+72.0%
Trondheim	7,522	9,965	11,351	+50.9%

4.4 Weather Station Selection Results

The WSS algorithm was applied successfully to 350 of 357 municipalities. Table 4.7 reports aggregate statistics across all processed municipalities.

The median $k_T^* = 2$ and $k_G^* = 1$ indicate that a single well-chosen ERA5 irradiance cell is sufficient for most municipalities, while temperature representation benefits from averaging two grid cells to reduce local noise.

Table 4.8 reports WSS outcomes for five representative municipalities used in the Stage 2 walk-forward evaluation.

Figure 4.4 illustrates the ERA5 grid cell selection for Oslo (area $\approx 929 \text{ km}^2$, box half-width 0.5° , 25 candidate grid cells). The WSS bounding box extends beyond the administrative municipality boundary, and grid cells are selected purely on the basis of minimising seasonal validation MAE regardless of whether they fall inside or outside the municipality polygon. The

Table 4.7: WSS aggregate results across municipalities

	Mean	Median	Min	Max
k_T^* (temperature grid cells)	4.6	2	1	25
k_G^* (irradiance grid cells)	2.7	1	1	25
Winter MAE (kWh/MPC/h)	0.145	0.136	0.060	1.470
Summer MAE (kWh/MPC/h)	0.059	0.055	0.031	0.303

Table 4.8: WSS results for the five Stage 2 representative municipalities.

Municipality	ID	k_T^*	k_G^*	Winter MAE	Summer MAE
Oslo	0301	2	1	0.186	0.047
Vestby	3216	1	1	0.180	0.052
Hå	1119	5	2	0.155	0.057
Gran	3446	2	1	0.130	0.052
Stavanger	1103	6	17	0.133	0.052

two selected temperature cells (T_22 and T_23) lie north of the city centre yielded the lowest winter validation MAE for Oslo’s private residential load. The first-ranked irradiance cell already minimises summer validation MAE, and adding further cells increases it monotonically.

Figure 4.5 and Figure 4.6 show the greedy MAE curves for Oslo. The temperature curve exhibits a clear minimum at $k_T^* = 2$ (MAE = 0.186 kWh/MPC/h), after which adding further grid cells increases the winter validation error. The irradiance curve is monotonically increasing from $k = 1$ (MAE = 0.047 kWh/MPC/h), so the greedy algorithm stops immediately at the first grid cell as the spatial averaging over additional cells only degrades the fit.

It is worth noting that the MAE differences across grid cell combinations are generally small, suggesting that the WSS selection is not highly sensitive to the exact choice of grid cells for most municipalities. The ERA5 reanalysis grid has a high spatial resolution, which may be too coarse to capture variability in temperature or irradiance. A higher-resolution meteorological dataset, such as a numerical weather prediction model, with finer spatial resolution could offer a more accurate representation and detection of the local conditions that actually can drive residential load, and would be a natural extension of the WSS framework developed here.

The WSS-selected grid cell coordinates (sel_T and sel_G) were reused

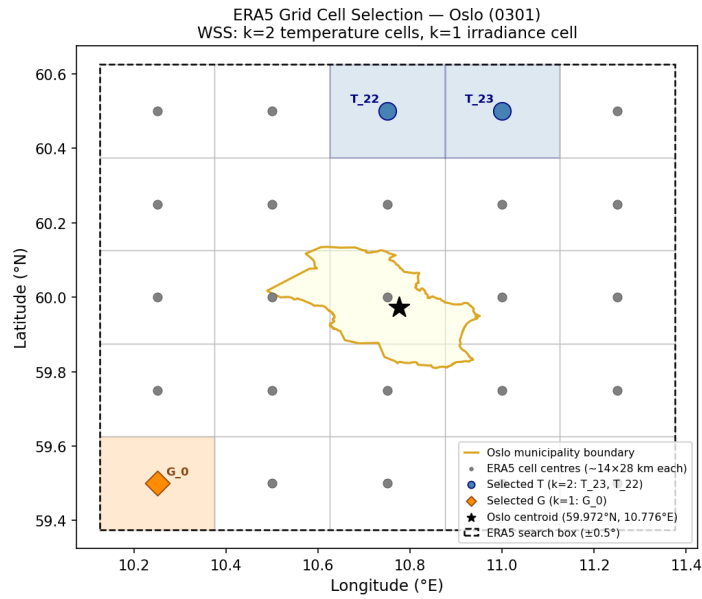


Figure 4.4: ERA5 grid cell selection for Oslo (0301).

directly to download five additional ERA5 variables via the Rebase Energy API: wind direction (WD, encoded as sine and cosine to preserve circularity), relative humidity (RH), total precipitation (TP), snow depth (SD), and cloud cover (CC). Wind direction, relative humidity, precipitation, and snow depth were averaged over the sel_T coordinates; cloud cover was averaged over the sel_G coordinates, as it is physically linked to irradiance variability. This targeted approach required fewer API calls per municipality rather than downloading the full bounding box grid. Extended features were successfully retrieved for 350 of 357 municipalities 7 remote Arctic municipalities failed and are excluded from forecasting modeling.

4.5 Day-Ahead Spot Prices

Large industrial consumers are known to exhibit price-responsive demand behaviour, adjusting consumption in response to hourly electricity price signals [43]. For the industrial forecasting models, day-ahead spot prices are therefore included as external future covariates. The ENTSO-E day-ahead market clears by 12:30 CET, making prices for the entire following day known at forecast time with no lag [44].

Hourly day-ahead prices for all five Norwegian bidding zones (NO1–NO5) were retrieved from the ENTSO-E Transparency Platform [45] via the

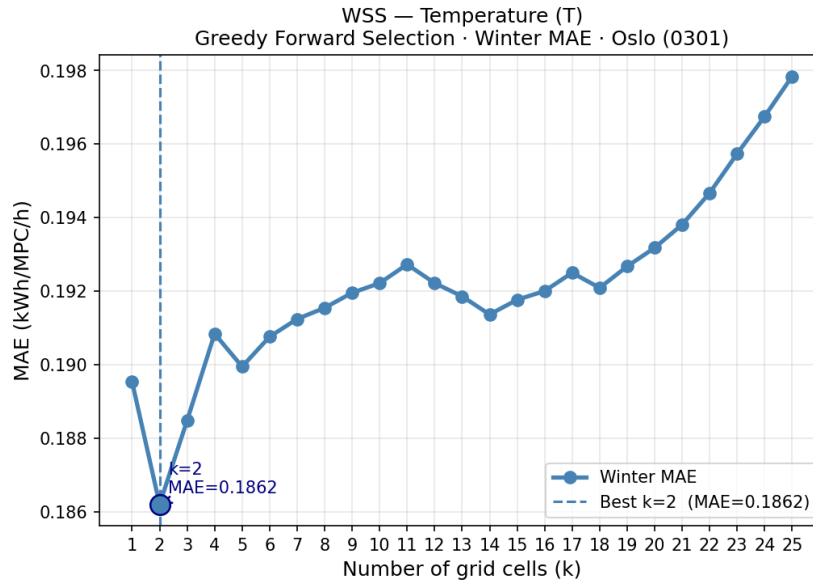


Figure 4.5: Greedy forward selection for temperature (T), winter validation MAE, Oslo (0301).

`entsoe-py` Python client [46], covering the study window. Where the API returns 15-minute resolution, mean resampling to a uniform UTC-aligned hourly index was applied.

Each municipality was assigned to its enclosing bidding zone via a spatial point-in-polygon join: the Kartverket centroid of each municipality was tested against the zone boundary polygons bundled with `entsoe-py`. All 357 municipalities were matched and Table 4.9 summarises the distribution.

Table 4.9: Distribution of the 357 Norwegian municipalities across ENTSO-E bidding zones.

Zone	Region	Reference city	Municipalities
NO1	East	Oslo	85
NO2	South	Stavanger	81
NO3	Central	Trondheim	78
NO4	North	Bodø	83
NO5	West	Bergen	30
Total			357

Figure 4.7 illustrates day-ahead prices for all five zones over a representative week in January 2024. Inter-zonal price divergence is

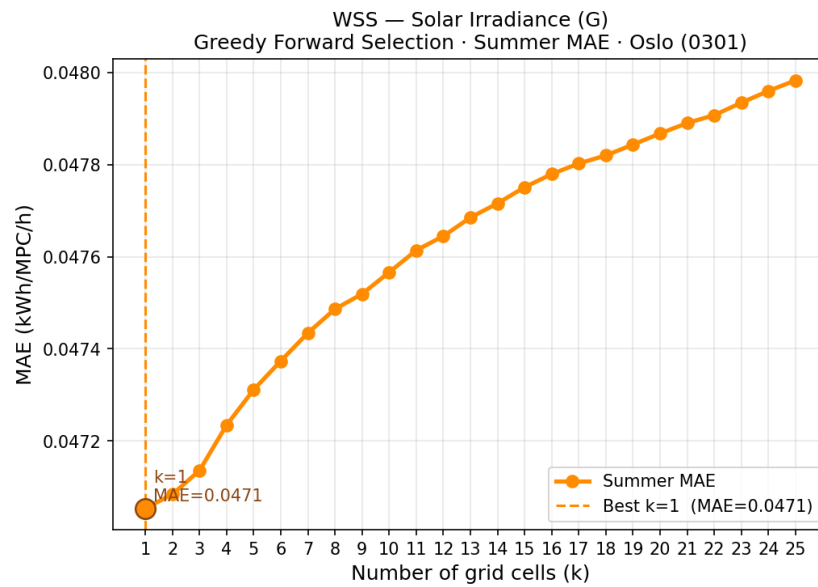


Figure 4.6: Greedy forward selection — irradiance (G), summer validation MAE, Oslo (0301).

substantial, confirming that a single national price series would be an inadequate representation.

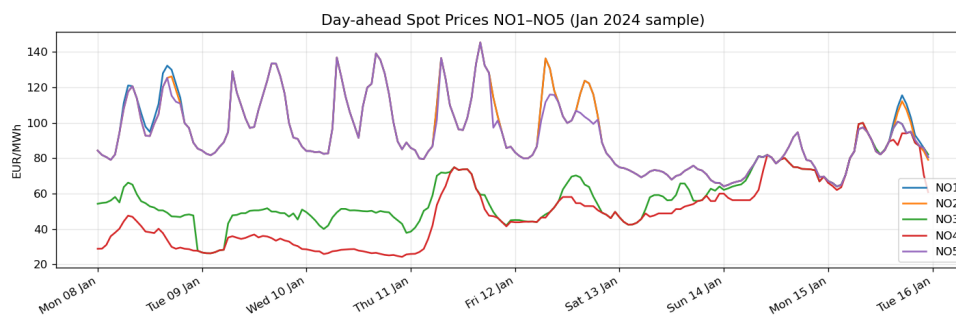


Figure 4.7: Day-ahead spot prices for the five Norwegian bidding zones (NO1–NO5), 8–14 January 2024.

Chapter 5

Results and Analysis

The results are organised according to the three-stage pipeline introduced in Chapter 3 and illustrated in Figure 3.1. Section 5.1 reports the weather feature ablation results across all municipalities (Stage 1). Section 5.2 presents the walk-forward evaluation on the eight representative municipalities (Stage 2). Section 5.3 reports the national three-fold validation across all 350 municipalities (Stage 3), followed by the synthetic BTM solar experiment in Section 5.4.

5.1 Stage 1: Weather Feature Ablation

Stage 1 quantifies the marginal forecasting value of each meteorological variable group by running a cumulative ablation across all available Norwegian municipalities and all three consumer categories. No autoregressive lags are included at this stage, so the results isolate the exogenous weather signal from the load-history signal, as described in Section 3.5. Table 5.1 reports the mean MAPE (%) across all municipalities, by consumer category and cumulative feature group. Positive Δ indicates improvement over the preceding model.

The single largest gain across the entire ablation ladder, after the calendar effect, is the addition of temperature features (M0_cal $\rightarrow$ M1_T): +5.74 pp for private residential, +4.21 pp for commercial, and +2.00 pp for industrial consumption. This reflects Norway's temperature-load relationship, creating a L-shaped temperature-load curve which is different from the usual U-shaped load curve as in Norway the cooling needs in summer are limited. This can be illustrated in Figure A.10 of Appendix A. Moreover, the seasonal spread visible in that figure whereby the same temperature corresponds to different load levels depending on the season provide empirical motivation for

Table 5.1: Weather feature ablation

Model	Features added	MAPE (%)			$\Delta 0$ (pp)		
		Priv.	Bus.	Ind.	Priv.	Bus.	Ind.
M0_cal	Calendar only	17.75	16.91	26.81	—	—	—
M1_T	+ Temperature	12.01	12.70	24.81	+5.74	+4.21	+2.00
M2_G	+ Solar irradiance	11.72	12.58	24.60	+0.29	+0.12	+0.21
M3_W	+ Wind speed and direction	11.23	12.44	24.40	+0.49	+0.14	+0.20
M4_RH	+ Relative humidity, cloud cover	11.13	12.35	24.04	+0.09	+0.09	+0.36
M5_P	+ Precipitation, snow depth	10.93	12.24	24.20	+0.20	+0.12	-0.16

including calendar interaction terms in the temperature feature specification. The comparatively modest gain for industry (+2.00 pp) is consistent with the expectation that large industrial consumers operate according to production schedules rather than ambient temperature conditions.

Adding solar irradiance features (M1_T $\rightarrow$ M2_G) yields small consistent gains across all three categories: +0.29 pp for private residential, +0.12 pp for commercial, and +0.21 pp for industrial consumption. The physical relationship between irradiance and private residential load is illustrated for two municipalities in Figures A.11 and A.12 (Appendix A). In Oslo (T1, $\sim 0.5\%$ penetration), increasing irradiance is associated with a modest load decline. In Vestby (T4, $\sim 10.7\%$ penetration), the suppression is more pronounced, observable in summer too, probably because of the higher solar capabilities and BTM generation. This motivates the inclusion of irradiance features including calendar interaction terms in the solar feature specification. Moreover, cloud cover, included in the subsequent M4_RH group, is physically related to solar availability and contributes an additional +0.09 to 0.36 pp depending on the category.

Wind speed and direction (M3_W) and humidity and cloud cover (M4_RH) each contribute further reductions for private and commercial consumption. For the industrial category, precipitation and snow depth (M5_P) produce a marginal *increase* in MAPE of 0.16 pp, indicating that these variables introduce noise rather than signal for process-driven industrial load. The full weather stack (M5_P) is nevertheless adopted as the confirmed feature ceiling for all categories in Stages 2 and 3. A key advantage of the ablation design is that it can replace arbitrary or empirical feature selection with an empirically validated decision: each feature group is retained or excluded based on measured MAPE evidence on a held-out validation set, providing a principled and reproducible basis for the downstream model specifications.

Figure A.13 of the Appendix A shows the LightGBM feature importance

by group for the private residential category under M5_P, across the Stage 2 representative municipalities. Calendar effects account for approximately 42–51% of total importance, confirming that human activity patterns are the primary determinant of residential load. Temperature contributes 15–21%, while humidity and cloud cover (13–16%) consistently exceed solar irradiance (4–7%) in all municipalities. This ordering is consistent with the ablation MAPE deltas. Full numerical values of the feature importances are reported in Table B.3 (Appendix B).

Norway’s pronounced climatic heterogeneity — ranging from arctic darkness and severe cold in the north to mild coastal conditions in the west and extended summer daylight at high latitudes — means that the marginal value of each meteorological variable is unlikely to be uniform across municipalities. The ablation ladder therefore serves not only as a feature selection procedure but also as an empirical test of which weather signals carry load-relevant information across this diverse setting.

5.2 Stage 2: Walk-Forward Evaluation on Representative Municipalities

Stage 2 evaluates a structured model ladder on eight representative municipalities spanning all four penetration tiers, using a six-fold expanding-window walk-forward protocol with 24-hour rolling prediction blocks and a three-month validation window for early stopping as described in Section 3.6. Each fold answers the same question under a progressively longer training horizon, covering test periods from July 2024 through December 2025. The model ladder is designed to isolate the contribution of each modeling component in sequence and aims to promote the most competitive baselines along with the most competitive proposed modeling approaches.

5.2.1 Private Residential

Table 5.2 reports mean MAPE across the six evaluation folds for the eight Stage 2 municipalities described in Table 3.4. **Red** indicates the model promoted to Stage 3 national evaluation; bold indicates the best model per municipality. M3 is promoted as the primary candidate for all categories, and M5_clean is additionally promoted for the private residential category to test whether its penetration-dependent gains generalise at national scale.

The three OLS baselines reveal a clear progression. The Tao vanilla model

Table 5.2: Stage 2 walk-forward results — private residential. Mean MAPE (%) across six folds per municipality. Bold = best per municipality; red = promoted to Stage 3.

Municipality	Tier	Baselines				Tested Models			
		M_naive	M0a	M0b	M0c	LGB_M0c	M3	M4_mpc	M5_clean
Bergen	T1	12.05	5.85	5.03	4.68	6.79	4.56	4.98	4.59
Oslo	T1	10.00	7.07	6.23	6.11	9.73	4.12	4.21	4.38
Stavanger	T2	11.48	6.90	6.24	5.56	8.32	5.24	5.87	5.42
Ås	T2	11.37	8.59	8.49	8.27	7.88	5.01	4.92	4.97
Gran	T3	12.33	6.74	7.38	6.93	8.64	5.48	5.48	5.68
Hå	T3	10.74	6.37	5.88	4.89	6.94	4.84	5.05	4.82
Vestby	T4	11.46	6.34	6.21	6.03	6.96	5.37	5.36	5.21
Hægebostad	T4	12.59	7.15	7.15	6.55	6.66	5.22	5.28	5.42
Mean		11.50	6.88	6.58	6.13	7.74	4.98	5.14	5.06

(M0a, mean 6.88%) recovers 4.62 pp over the naive baseline, confirming that calendar and temperature structure alone explains a substantial fraction of residential load variation. Adding temperature recency terms (M0b, +0.30 pp) captures the thermal inertia of buildings: the previous day’s temperature and its moving average explain additional load variation beyond the contemporaneous reading. Incorporating solar irradiance in the OLS framework (M0c, +0.45 pp) yields a further modest improvement, with the largest gains in municipalities where the solar signal is strongest relative to the overall load level. M0c (mean 6.13%) constitutes the best linear baseline and serves as the primary OLS comparator for the LightGBM models.

A key finding of Stage 2 is that LightGBM applied to the same feature set as M0c, but without autoregressive lags, performs *worse* than the OLS baseline: LGB_M0c achieves a mean MAPE of 7.74%, compared with 6.13% for M0c. This result demonstrates that the gradient-boosted model without AR lags overfits the weather and calendar signals during training and fails to generalise to out-of-sample periods, whereas the OLS model is regularised by its parametric structure. The result carries an important methodological implication: *algorithm choice is secondary to feature design*, and the inclusion of autoregressive lags is a prerequisite for competitive LightGBM performance in this setting.

Adding deployable autoregressive lags at 48 h and 168 h (one-week periodicity) together with the full extended weather feature set yields the largest single improvement in the ladder: M3 gains +2.76 pp over LGB_M0c, bringing the mean MAPE to 4.98%. M3 is the best or joint-best model in six of the eight municipalities and achieves its strongest

absolute performance in Oslo (4.12%), where the large metering point base provides a stable, predictable aggregate signal. The meter-count model (M4_mpc, mean MAPE 5.14%) fails to improve consistently over M3 across municipalities, confirming that metering point count information is already implicitly captured by the autoregressive lags and trend feature.

M5_clean adds the solar prosumer bundle — capacity-per-MPC, irradiance-capacity interaction, and E19 capacity-factor lags (Section 3.6) — on top of M3. Across all eight municipalities the mean gain is -0.08 pp, meaning M3 is marginally better on average and M5_clean provides no consistent benefit. However, a penetration-dependent pattern emerges when results are disaggregated by season. Table 5.3 reports the M3 $\rightarrow$ M5_clean gain broken down by season.

Table 5.3: M3 $\rightarrow$ M5_clean MAPE gain (pp) by season and municipality. Positive = M5_clean better. Summer folds: Jul–Sep 2024, Apr–Jun 2025, Jul–Sep 2025. Winter folds: Oct–Dec 2024, Jan–Mar 2025, Oct–Dec 2025.

Municipality	Tier	Summer	Winter
Bergen	T1	-0.25	$+0.21$
Oslo	T1	-0.28	-0.25
Stavanger	T2	-0.43	$+0.08$
Ås	T2	-0.10	$+0.18$
Gran	T3	-0.55	$+0.14$
Hå	T3	-0.25	$+0.30$
Vestby	T4	$+0.32$	$+0.00$
Hægebostad	T4	-0.15	-0.25

Vestby (T4) is the only municipality where M5_clean consistently outperforms M3, and the gain is concentrated in summer folds ($+0.32$ pp), when solar generation and the E19 export signal are at their peak. All other municipalities show zero or negative M5_clean gains in summer, and small mixed results in winter where solar features carry no physical meaning. This pattern provides the first quantitative evidence that solar prosumer features are probably penetration and season dependent: they add value only when both the penetration level and the seasonal irradiance are sufficient to produce a detectable E19 signal relative to the background load noise. The fold-level breakdown (Table B.4, Appendix B) confirms that M3 outperforms M_naive in every fold across all municipalities, and that the M5_clean gain in Vestby is driven by the three summer folds, with the largest single-fold improvement of $+1.59$ pp observed in Fold 1 (July–September 2024).

5.2.2 Commercial

Table 5.4 reports mean MAPE across the six evaluation folds for the commercial consumer category. No prosumer features are evaluated for commercial consumption, as behind-the-meter solar adoption is concentrated in the residential sector (Section 3.3.2); M3 therefore constitutes both the performance ceiling and the model promoted to Stage 3.

Table 5.4: Stage 2 walk-forward results — commercial. Mean MAPE (%) across six folds per municipality. Bold = best per municipality; red = promoted to Stage 3.

Municipality	Tier	Baselines				Tested Models		
		M_naive	M0a	M0b	M0c	LGB_M0c	M3	M4_mpc
Bergen	T1	7.52	4.91	4.71	4.61	4.37	3.67	4.08
Oslo	T1	5.93	6.54	6.29	6.27	4.26	3.35	3.65
Stavanger	T2	6.43	5.09	4.97	4.70	4.13	3.65	3.81
Ås	T2	7.20	7.35	7.08	7.10	6.12	4.33	4.37
Gran	T3	8.81	6.18	5.54	5.42	6.35	4.22	4.60
Hå	T3	8.26	6.92	6.65	5.73	5.97	4.66	4.87
Vestby	T4	10.22	9.78	9.60	8.64	8.80	6.83	6.64
Hægebostad	T4	10.38	9.50	9.10	8.75	10.01	7.02	7.20
Mean		8.09	7.04	6.74	6.40	6.25	4.72	4.90

Commercial load exhibits a structurally different pattern from private residential. The OLS baseline M0a recovers only 1.05 pp over M_naive (from 8.09% to 7.04%), a substantially smaller gain than for private residential (−4.62 pp), reflecting the weaker temperature sensitivity of commercial premises: office buildings, retail, and service facilities are primarily governed by occupancy schedules rather than heating demand. Temperature recency (M0b) provides a small additional gain (+0.30 pp on average), consistent with the commercial sector’s modest thermal inertia response.

Unlike the private category in which LGB_M0c *underperformed* OLS due to overfitting without AR lags, LGB_M0c (6.25%) modestly outperforms M0c (6.40%) for commercial load. This suggests that commercial consumption is more strongly driven by deterministic calendar rules (business hours, weekday patterns, public holidays) whose nonlinear structure LightGBM can capture from features alone, without requiring historical load lags. However, the advantage is inconsistent across municipalities — in Gran and Hå LGB_M0c is worse than M0c — and in Vestby and Hægebostad it deteriorates

dramatically ($LGB_M0c > M_naive$), indicating overfitting in municipalities with sparse and volatile commercial metering points.

M3 provides the decisive improvement, gaining +1.53 pp over LGB_M0c (from 6.25% to 4.72%) and is the best or joint-best model in seven of the eight municipalities. The exception is Vestby, where $M4_mpc$ (6.64%) marginally outperforms M3 (6.83%): in this small municipality the metering point count captures seasonal fluctuations in the number of active commercial connections that the AR lags alone do not fully resolve. The meter-count feature otherwise adds no consistent value (mean 4.90% vs M3 4.72%), and M3 remains the recommended model for commercial forecasting at all penetration tiers.

The per-fold breakdown (Table B.5, Appendix B) confirms that M3 outperforms M_naive in every fold for six of the eight municipalities. The two exceptions — Vestby and Hægebostad — are the two smallest municipalities in the evaluation set, where commercial load is dominated by a handful of large metering points and therefore exhibits higher fold-to-fold volatility.

5.2.3 Industrial

Table 5.5 reports Stage 2 results for the industrial consumer category, which exhibits qualitatively different behaviour from both private residential and commercial load.

Table 5.5: Stage 2 walk-forward results — industrial. Mean MAPE (%) across six folds per municipality. Bold = best per municipality; red = promoted to Stage 3. $M4_spot$ includes day-ahead spot prices (Section 4.5).

Municipality	Tier	Baselines				Tested Models			
		M_naive	$M0a$	$M0b$	$M0c$	LGB_M0c	M3	$M4_mpc$	$M4_spot$
Bergen	T1	15.50	16.83	16.34	16.36	15.67	13.03	14.27	12.98
Oslo	T1	25.46	28.98	29.17	29.49	31.31	28.12	32.54	25.36
Stavanger	T2	9.13	9.35	9.09	9.36	7.89	6.23	5.98	6.19
Ås	T2	13.13	21.47	21.38	20.61	20.73	14.79	14.87	15.03
Gran	T3	12.27	14.84	14.71	14.44	13.72	10.02	9.46	10.08
Hå	T3	8.88	8.91	8.92	8.24	7.50	6.62	6.48	6.44
Vestby	T4	12.98	17.84	17.79	17.69	15.76	11.88	12.28	11.70
Hægebostad	T4	19.41	30.15	29.95	29.82	20.13	16.22	18.11	17.00
Mean		14.60	18.55	18.42	18.25	16.59	13.36	14.25	13.10

The most striking result in Stage 2 is the systematic failure of all OLS baselines for industrial consumption: $M0c$ achieves a mean MAPE of 18.25%, substantially *worse* than the naive persistence baseline (14.60%). This reversal is consistent across six of the eight municipalities. So in this case the

naive model will be a more competitive baseline at the national evaluation. Industrial load is primarily driven by production schedules and shift patterns rather than weather or calendar effects. The Tao–Hong modelling approaches, which assume a smooth temperature and calendar-driven load function, is structurally misspecified for the industrial category: the dominant drivers of industrial consumption are production schedules, operational shifts, and price-responsive demand and those are not captured by temperature or calendar features, resulting in systematically high residuals that do not average out across seasons. These OLS approaches are producing predictions worse than simply repeating last week’s observation. LightGBM without AR lags (LGB_M0c, 16.59%) reduces this gap probably through its ability to capture nonlinear interactions, but still underperforms M_naive, confirming that the problem is not the model class but the absence of load-history information.

In general, the industrial category exhibits the largest cross-municipality variation in forecast difficulty of any category in this study: M3 MAPE ranges from 6.23% in Stavanger to 28.12% in Oslo. This disparity reflects the heterogeneity of industrial load composition at the municipal level: municipalities where industrial consumption is concentrated in a small number of facilities with regular operating patterns are substantially easier to forecast than municipalities where the industrial aggregate mixes different types of industrial loads with diverse and partially offsetting schedules. The per-fold results (Table B.6) further reveal substantial seasonal variation within municipalities: in Oslo, M_naive MAPE ranges from 7.02% in the summer fold (Jul–Sep 2025) to 43.46% in the winter fold (Jan–Mar 2025), suggesting that industrial consumption is considerably more volatile during winter months.

The addition of autoregressive lags (M3, 13.36%) is a step that is recovering +3.23 pp over LGB_M0c. M3 achieves its best results in municipalities where industrial load is again probably more homogeneous: Stavanger (6.23%), Hå (6.62%), and Gran (10.02%). Oslo remains the notable exception and structurally more difficult: M3 (28.12%) is still *worse* than M_naive (25.46%), and the meter-count feature (M4_mpc) is harmful (32.54%).

A particularly instructive case is Ås, where M_naive (13.13%) outperforms *all* proposed models including M3 (14.79%) and M4_spot (15.03%). This result suggests that Ås industrial load is dominated by a facility with a highly stable weekly production cycle that the 168-hour persistence captures optimally, while the additional features in M3 overfit to variation that does not repeat across folds.

M4_spot (M3 augmented with the zonal day-ahead spot price and its intraday rank) achieves the best overall result: mean MAPE 13.10%, a gain of +0.26 pp over M3. The improvement is consistent in six of the eight municipalities. Most notably, M4_spot is the *only* model to beat M_naive in Oslo (25.36% vs 25.46%), providing tentative evidence that price signals capture a component of large-facility demand flexibility. The gain remains modest and municipality-dependent; both M3 and M4_spot are promoted to Stage 3 for the industrial category.

The per-fold breakdown (Table B.6, Appendix B) confirms the high fold-to-fold volatility of industrial load, and highlights that no single model dominates consistently across all municipalities and seasons and structurally the industrial load category is a more difficult case for global approaching.

5.3 Stage 3: National Three-Fold Validation

5.3.1 Private Residential

Table 5.6 reports the mean municipality MAPE and normalized (volume-weighted) nMAPE across all 350 municipalities, evaluated on the two annual folds (Fold A: January–June 2025; Fold B: July–December 2025) and the summer robustness fold (Fold C: May–August 2025). The mean MAPE across municipalities is the primary deployment metric, as each municipality constitutes an independent forecasting task irrespective of its consumption volume. However, the pooled (normalized) metric serves as a national-aggregate robustness check, since volume-weighted averages can be distorted by a few large municipalities with atypically good or poor performance.

Table 5.6: Stage 3 national evaluation — private residential ($n = 350$ municipalities). Mean municipality MAPE and pooled MAPE (%). Bold = best model per column.

Model	Fold A		Fold B		Annual (A + B)		Fold C (summer)	
	Mean	Pooled	Mean	Pooled	Mean	Pooled	Mean	Pooled
M_naive	14.20	13.05	12.82	11.92	13.51	12.54	11.97	10.55
M0c	8.41	7.23	8.63	6.01	8.52	6.67	8.89	7.31
M3	6.45	5.42	6.93	5.13	6.69	5.29	6.99	5.52
M5_clean	6.48	5.45	6.96	5.25	6.72	5.36	7.04	5.58

M3 achieves the best performance under both metrics and across all folds. Annually, it attains a mean municipality MAPE of 6.69%, a gain of

+6.83 pp over M_naive (13.51%) and +1.83 pp over the best OLS baseline M0c (8.52%). M5_clean trails M3 by only 0.03 pp on the municipality-mean metric and 0.07 pp on the pooled metric, confirming no national benefit from the solar feature bundle at the population level. Both metrics lead to identical qualitative conclusions across all folds: pooled MAPE is uniformly lower in absolute terms because larger municipalities, which dominate volume weighting, tend to have more stable aggregate load. The model ordering is stable in both individual annual folds and in the summer robustness fold C (M3: 6.99%, M5_clean: 7.04%), ruling out any reversal during peak solar irradiance.

Table 5.7 quantifies model dominance at the municipality level for all pairwise comparisons. A model wins in a fold if its MAPE is strictly lower than the comparator for that municipality. The conservative criterion requires wins in *both* Folds A and B; the liberal criterion requires a win in *at least one*.

Table 5.7: Stage 3 private residential — pairwise municipality-level win counts ($n = 350$)

Candidate	Baseline	Fold A	Fold B	Fold C	Conservative	Liberal	Mean δ (pp)
M0c	M_naive	346	344	333	344 (98.3%)	346 (98.9%)	+4.99
M3	M_naive	349	347	340	347 (99.1%)	349 (99.7%)	+6.83
M5_clean	M_naive	349	347	343	347 (99.1%)	349 (99.7%)	+6.80
M3	M0c	258	332	309	246 (70.3%)	344 (98.3%)	+1.83
M5_clean	M0c	256	327	307	242 (69.1%)	341 (97.4%)	+1.80
M5_clean	M3	129	109	115	48 (13.7%)	190 (54.3%)	−0.03

M3 outperforms M_naive conservatively in 347 of 350 municipalities, establishing it as the near-universal default for private residential net load forecasting across Norway. It likewise outperforms M0c conservatively in 246 of 350 municipalities (70.3%), confirming that the modeling approach proposed and evaluated in Stage 2 generalises to national scale.

The M5_clean picture is more nuanced. In 190 of 350 municipalities (54.3%), M5_clean outperforms M3 in at least one annual fold. However, only 48 municipalities (13.7%) show consistent improvement across *both* annual folds, and the mean annual gain is −0.03 pp — effectively zero at the population level. The Fold C win count (115/350, 32.9%) is marginally higher than Fold B (109/350, 31.1%), consistent with a weak summer concentration of the M5_clean signal, but even in the dedicated summer fold a clear majority of municipalities are better served by M3. M5_clean therefore does not emerge as a viable deployment alternative at the national level.

The natural hypothesis that M5_clean can gain concentrate above a solar penetration threshold is also tested in Table 5.8.

Table 5.8: M5_clean vs M3 by penetration tier — Stage 3 private residential, annual Folds A+B. Conservative = wins in both A and B. $\delta > 0$ means M5_clean better.

Tier	<i>n</i>	Cons.	Cons. %	Lib.	Lib. %	Mean δ (pp)
T0 [†] (no E19)	80	1	1.3%	2	2.5%	+0.001
T1 (<1%)	69	9	13.0%	43	62.3%	−0.055
T2 (1–5%)	171	32	18.7%	128	74.9%	−0.028
T3 (5–10%)	27	6	22.2%	15	55.6%	−0.066
T4 ($\geq 10\%$)	3	0	0.0%	2	66.7%	−0.075

[†]T0: zero observed E19 export during Jun–Aug 2024 penetration window.

No robust penetration threshold emerges. The conservative win rate rises modestly from T1 (13.0%) to T3 (22.2%), but collapses to zero at T4 — the three highest-penetration municipalities in the study — where M5_clean is consistently worse than M3 in both annual folds. Mean gains remain negative across all tiers with E19 data, and a continuous threshold sweep confirms that no cut-off between 0% and 10% produces a positive mean annual gain. The T4 result warrants specific attention: Vestby, with 16.6% Jun–Aug penetration, showed concentrated M5_clean gains in the Stage 2 summer folds, yet here M5_clean fails the conservative criterion. This is explained by the structure of the Stage 3 folds: both Fold A (January–June) and Fold B (July–December) contain substantial non-solar months that dilute the weak possible summer signal.

Given the absence of a robust penetration rule, the operational question becomes an empirical selective assignment. M5_clean can be used only where it demonstrably outperforms M3 in both annual folds in order to test if it can improve the aggregate portfolio accuracy. Table 5.9 reports the result.

The conservative assignment achieves an annual mean municipality MAPE of 6.671%, a minimal gain of 0.018 pp over pure M3 resulting not a systematic actual benefit. Also in the summer fold there is an improvement. The liberal assignment also improves portfolio accuracy relative to pure M3, at the cost of including municipalities where this modeling choice won in only one fold.

Figure 5.1 maps the resulting assignment across Norway.

Three conclusions emerge from the Stage 3 national evaluation:

Table 5.9: Portfolio mean municipality MAPE and pooled MAPE (%) under alternative assignment strategies — Stage 3 private residential ($n = 350$).

Strategy	M5 munis	Annual (A + B)		Fold C
		Mean	Pooled	Mean
Pure M3	—	6.689	5.286	6.988
Pure M5_clean	—	6.719	5.360	7.036
Conservative assignment	48	6.671	5.277	6.952
Liberal assignment	190	6.679	5.285	6.975

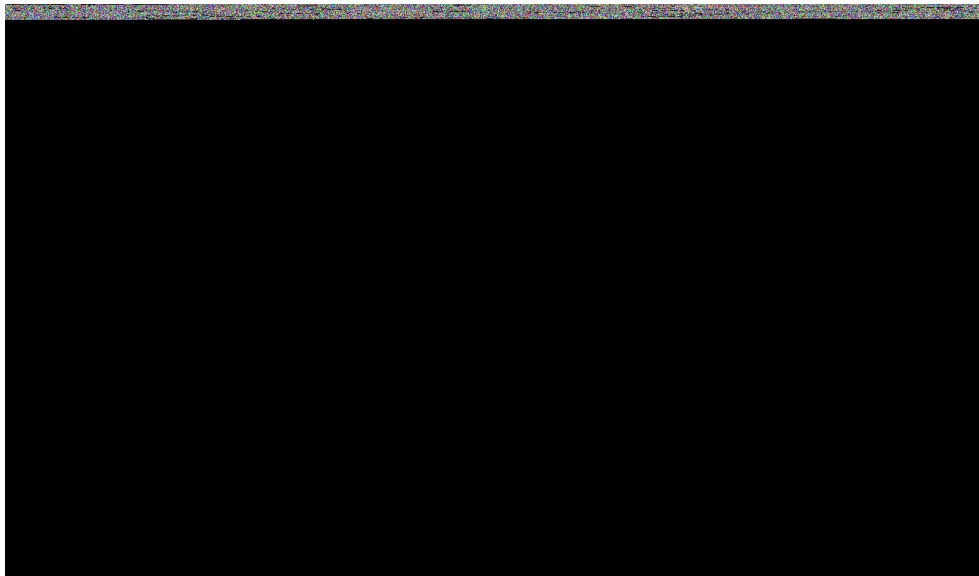


Figure 5.1: Conservative model assignment for private residential forecasting across 350 Norwegian municipalities. **Orange** = M5_clean (48 municipalities, 13.7%); **blue** = M3 default (302 municipalities, 86.3%). Criterion: M5_clean outperforms M3 in both annual Folds A and B.

1. **M3 is the universal default.** It outperforms M_naive conservatively in 347 of 350 municipalities (+6.83 pp mean annual gain) and M0c conservatively in 246 of 350 (70.3%), confirming national generalisation.
2. **M5_clean provides no systematic national benefit.** The mean annual gain over M3 is -0.03 pp, no penetration threshold yields a robust conservative win rate, and the T4 result shows M5_clean is still harmful at the highest penetration levels over annual evaluation windows.
3. **Selective deployment has limited but real value.** Conservative retrospective assignment of M5_clean to 48 municipalities reduces the portfolio mean MAPE by 0.018 pp, confirming that the solar feature selection probably can be informative where the E19 signal is more sufficient to the assumptions. The liberal signal in 190 municipalities (54.3%) motivates continued monitoring as penetration grows.

Whether the absence of a robust M5_clean signal reflects a genuine lack of predictive value in the E19 proxy, or rather the inherent limitations of this indirect solar measurement, cannot be resolved from observed data alone. This question motivates the controlled synthetic experiment of Section 5.4.

5.3.2 Commercial

Table 5.10 reports the mean municipality MAPE and pooled MAPE across all 350 municipalities for the two annual folds (Fold A: January–June 2025; Fold B: July–December 2025) and the summer robustness fold (Fold C: May–August 2025). The commercial sector evaluates three models only: M_naive, M0c, and M3.

Table 5.10: Stage 3 national evaluation — commercial ($n = 350$ municipalities). Mean municipality MAPE and pooled MAPE (%). Bold = best model per column.

Model	Fold A		Fold B		Annual (A + B)		Fold C (summer)	
	Mean	Pooled	Mean	Pooled	Mean	Pooled	Mean	Pooled
M_naive	12.074	9.654	13.290	7.438	12.682	8.568	9.209	6.433
M0c	11.665	10.462	14.748	11.381	13.207	10.912	11.804	10.759
M3	8.200	8.420	11.110	8.350	9.655	8.386	8.004	6.602

M3 achieves the best mean municipality MAPE across all folds, reaching 9.655% annually — a gain of +3.027 pp over M_naive (12.682%) and

+3.552 pp over M0c (13.207%). The OLS baseline M0c is the weakest model overall, finishing 0.525 pp *worse* than M_naive on an annual basis, confirming at national scale the finding first observed in Stage 2.

Examining individual folds reveals that the OLS failure is not uniform but strongly seasonal. In Fold A (January–June), M0c (11.665%) outperforms M_naive (12.074%) — OLS can capture the stable, near-linear winter and spring demand of commercial premises. In Fold B (July–December), M0c deteriorates sharply to 14.748% while M_naive reaches only 13.290%, a reversal of 1.458 pp. The summer and autumn months introduce non-linear occupancy and cooling patterns that the Tao–Hong polynomial specification cannot represent, causing systematic over- and under-estimation that a seasonal naive benchmark avoids entirely.

The pooled metric reveals an equally important structural finding. Across all folds, the gap between mean and pooled MAPE is consistently larger for M_naive than for M3, indicating that M3’s advantage is concentrated in small and medium municipalities while large municipalities are already well served by a seasonal naive forecast. This divergence is most pronounced in Fold B, where M_naive records a mean MAPE of 13.290% but a pooled MAPE of only 7.438% — a gap of 5.85 pp, the largest mean–pooled divergence observed across all models and folds in the commercial evaluation. Large commercial consumers, such as office complexes and retail centres, exhibit unusually stable aggregate demand in the summer–autumn period, which a seasonal naive forecast captures almost as effectively as a learned model. By contrast, M3 reduces this gap substantially (11.110% mean vs 8.350% pooled in Fold B, a difference of 2.76 pp), confirming that its primary benefit is in smaller municipalities where load heterogeneity and non-linearity are higher.

A further asymmetry appears in Fold A, where M3 is the only model for which the pooled MAPE (8.420%) exceeds the mean MAPE (8.200%). For M_naive and M0c, larger municipalities are easier to forecast than smaller ones in the winter–spring period; for M3, the relationship inverts slightly, suggesting that the learned model captures small and medium commercial patterns more effectively than the demand of the largest consumers in this season.

Table 5.11 quantifies model dominance at the municipality level.

M3 outperforms M_naive conservatively in 313 of 350 municipalities (89.4%) and M0c conservatively in 304 of 350 (86.9%), establishing it as the clear national default for commercial net load forecasting. The M3 vs M_naive conservative rate (89.4%) is notably lower than the private residential equivalent (99.1%), consistent with the greater variability and heterogeneity

Table 5.11: Stage 3 commercial — pairwise municipality-level win counts ($n = 350$). Conservative = wins in both Folds A and B; liberal = wins in at least one. $\delta > 0$ means candidate better.

Candidate	Baseline	Fold A	Fold B	Fold C	Conservative	Liberal	Mean δ (pp)
M3	M_naive	328	320	307	313 (89.4%)	335 (95.7%)	+3.027
M3	M0c	326	321	323	304 (86.9%)	343 (98.0%)	+3.552
M0c	M_naive	282	245	195	226 (64.6%)	301 (86.0%)	−0.525

of commercial load across Norwegian municipalities.

The M0c row is the most instructive result. Despite winning conservatively in 226 of 350 municipalities (64.6%), its mean annual delta relative to M_naive is −0.525 pp. This divergence between win count and mean gain indicates that M0c’s losses, where they occur, are large enough to outweigh its modest wins: OLS systematically collapses in the subset of municipalities with strongly non-linear commercial demand, and those collapses dominate the population average. The Fold C win count (195/350, 55.7%) further confirms the seasonal concentration of this failure — in the dedicated summer fold, M0c barely outperforms naive in a majority of municipalities, and never by a sufficient margin to recover its annual deficit.

Three conclusions emerge from the Stage 3 commercial evaluation:

1. **M3 is the national default for commercial load.** It outperforms M_naive conservatively in 313 of 350 municipalities (+3.027 pp mean annual gain) and M0c conservatively in 304 of 350 (+3.552 pp), confirming that the Stage 2 recommendation generalises to national scale.
2. **OLS fails nationally for commercial load, with a pronounced seasonal pattern.** M0c finishes 0.525 pp worse than M_naive on the annual mean despite a 64.6% conservative win rate, driven by systematic collapse in Fold B (14.748% vs 13.290% for M_naive). The failure is concentrated in summer and autumn, when non-linear commercial occupancy patterns exceed the representational capacity of the Tao–Hong polynomial.
3. **The mean-pooled divergence reveals a size-dependent structure in commercial forecasting.** Large commercial municipalities exhibit stable aggregate demand that naive forecasts capture effectively, particularly in summer — as evidenced by M_naive achieving the best pooled MAPE in Folds B and C. M3’s primary contribution is in

small and medium municipalities, where load heterogeneity is higher and a learned non-linear model provides the greatest marginal value. This finding motivates a municipality-size-aware deployment strategy as commercial solar penetration grows and load patterns become less predictable.

5.3.3 Industrial

Industrial net load forecasting presents a structurally different problem from both private residential and commercial categories. Private and commercial load are primarily driven by weather, calendar features and occupancy patterns — signals that are observable, learnable, and well represented by the feature set constructed in this thesis. Industrial load, by contrast, is driven by individual production schedules, contracts and operational decisions of individual large consumers. These drivers are neither observable in the available data nor proxied by any feature in the model. A municipality whose industrial consumption is dominated by a single industrial facility exhibits a load profile that reflects production ramp-ups and shutdowns rather than temperature or calendar effects and no model trained on weather and calendar features can forecast such a signal reliably.

The mean municipality MAPE is the natural metric for evaluating a global modelling approach, in which a single model is trained once and deployed across all municipalities without municipality-specific fine-tuning. Under this setting, each municipality constitutes an independent generalisation test, and mean MAPE measures how well the model’s learned representations transfer across diverse industrial load profiles. For private residential and commercial load, this generalisation was achievable because the primary drivers are shared across municipalities. For industrial load, each municipality’s dominant consumers operate under fundamentally different physical and economic processes, and the results below demonstrate the limits of global modelling in this context.

Table 5.12 reports the mean municipality MAPE and pooled MAPE for the full evaluation set of 350 municipalities.

Two qualitative cases illustrate the structural breadth of Norwegian industrial load. Sunndal (municipality 1563) hosts Europe’s largest aluminium smelter, accounting for approximately 9.6% of total national industrial consumption volume. Its load profile is almost perfectly flat at a constant operating level regardless of season, and a seasonal naive benchmark forecasts it with a MAPE of 0.62% in the winter fold — better than any learned

Table 5.12: Stage 3 national evaluation — industrial ($n = 350$ municipalities, full set). Mean municipality MAPE and pooled MAPE (%). Bold = lowest value per column. *Note: mean MAPE is distorted by 22 outlier municipalities see Appendix B Table B.7.*

Model	Fold A		Fold B		Annual (A + B)		Fold C (summer)	
	Mean	Pooled	Mean	Pooled	Mean	Pooled	Mean	Pooled
M_naive	26.759	11.212	43.113	10.614	34.936	10.915	42.405	11.514
M0c	68.081	18.981	58.579	17.983	63.330	18.485	67.837	19.712
M3	40.834	13.371	38.193	13.809	39.513	13.589	46.242	15.466
M4_spot	40.699	12.928	31.824	12.841	36.261	12.885	39.962	14.478

model. Oslo (municipality 0301), by contrast, aggregates a heterogeneous mix of urban industrial consumers including construction and manufacturing producing a highly volatile, spike-laden profile. In the same fold, M_naive reaches 30.2%, M3 achieves 18.7%, and M4_spot — which incorporates spot price features carrying partial production-decision signal — records the best result at 13.8%. These two municipalities represent the extremes of a spectrum that a single global model trained on weather and calendar features cannot span with uniform success, and they explain the two headline results in Table 5.12.

M_naive records the lowest annual mean MAPE (34.936%), but this apparent victory is an artefact of outlier distortion rather than genuine superiority: the mean MAPE here does not reflect successful generalisation of the proposed modelling approach but rather the inability of any global model to handle industrial municipalities whose load is governed by facility-level decisions. M_naive also achieves the best pooled MAPE across every fold (10.915% annually), a result driven by Sunndal's 9.6% volume share and sub-1% naive error — a single municipality's predictable continuous-process load compresses the volume-weighted metric for the entire sector. M0c fails catastrophically at 63.3% annual mean MAPE, consistent with the structural OLS failure observed in the commercial sector but further amplified by the extreme non-linearity of industrial demand.

Of the 350 industrial municipalities, 22 (6.3%) record an annual mean MAPE exceeding 100% for at least one model. These are not random forecast failures: they are structurally identifiable municipalities whose industrial consumption is based on different features. Modalen (3133% for M_naive, 3573% for M3), Sirdal (1053% and 845%), and Aurland (647% and 1229%) are representative examples of some small Norwegian municipalities where a single industrial installation can influence the total

consumption. When industrial load is effectively a step function governed by facility-level production decisions, neither weather features nor calendar effects provide forecasting signal, and MAPE values become measures of load variability rather than model quality. A sensitivity analysis excluding these 22 municipalities is provided in Appendix B in Table B.7 ; without them, annual mean MAPE falls to 16.379% (M3) and 16.868% (M_naive), and M3 is the marginal winner by 0.49 pp.

Table 5.13 reports pairwise municipality-level win counts for the full 350-municipality set.

Table 5.13: Stage 3 industrial — pairwise municipality-level win counts ($n = 350$). Conservative = wins in both Folds A and B; liberal = wins in at least one. $\delta > 0$ means candidate better.

Candidate	Baseline	Fold A	Fold B	Fold C	Conservative	Liberal	Mean δ (pp)
M3	M_naive	263	204	228	175 (50.0%)	292 (83.4%)	−4.577
M3	M0c	316	304	292	278 (79.4%)	342 (97.7%)	+23.817
M4_spot	M_naive	260	196	222	167 (47.7%)	289 (82.6%)	−1.326
M4_spot	M3	146	156	150	65 (18.6%)	237 (67.7%)	+3.252

The win count table delivers the clearest quantitative statement about global modelling limits for industrial load. M3 outperforms M_naive conservatively in exactly 175 of 350 municipalities (50.0%), with a negative mean annual delta of −4.577 pp. This is not a model failure: it reflects the structural absence of learnable signal in a substantial fraction of industrial municipalities, where the proposed global model cannot generalise successfully because each municipality’s dominant industrial process follows different drivers. The liberal win rate (83.4%) confirms that M3 captures useful signal intermittently — it is better in at least one fold in most municipalities — but cannot sustain this consistently across both annual folds. M4_spot tells a nearly identical story (47.7% conservative, −1.326 pp delta), confirming that additional spot price features do not resolve the underlying driver mismatch at the population level, though the Oslo case shows they can provide meaningful improvement where industrial demand is price-responsive. M3 outperforms M0c convincingly in 278 of 350 municipalities (79.4%), with a mean delta of +23.817 pp — the largest pairwise gap observed across all three sectors. The M4_spot versus M3 comparison is nuanced: M4_spot wins conservatively in only 65 of 350 municipalities (18.6%), yet its mean annual delta is +3.252 pp in favour of M4_spot, indicating that when M4_spot wins, it wins by large margins, while M3’s wins are smaller in magnitude.

Three conclusions emerge from the Stage 3 industrial evaluation:

1. **Industrial load forecasting is structurally a different problem from the other examined categories, and global models reach their generalisation limits here.** The near-50% conservative win rate of M3 against M_naive reflects the absence of learnable production-side features, not a deficiency of the LightGBM architecture. The same model achieves 99.1% conservative wins for private residential load with an identical training procedure, confirming that the performance gap is structural rather than methodological. Because each municipality's industrial load is governed by different physical and economic processes, a single global model trained on weather and calendar features cannot generalise with uniform success across the full evaluation set.
2. **22 outlier municipalities (6.3%) are forecastable only at the facility level.** These are municipalities where a single large industrial consumer dominates aggregate load. Their extreme MAPE values are measures of load variability rather than forecast quality, and they should be addressed through facility-level modelling or flagged for exclusion from population-level evaluation. The sensitivity analysis in Appendix B (Table B.7) confirms that their removal reduces the annual mean MAPE by more than half for every model.
3. **M3 is the recommended default, with calibrated expectations.** For the 328 non-outlier municipalities, M3 achieves the best annual mean MAPE (16.379%), marginally ahead of M_naive (16.868%). M0c must not be deployed for industrial load under any circumstances (+23.817 pp gap vs M3). The pooled MAPE advantage of M_naive is an artefact of Sunndal's volume dominance and does not constitute a case for naive deployment across the sector.

5.4 Synthetic BTM Solar Experiment

5.4.1 Motivation

The Stage 3 results in Section 5.3.1 reveal a persistent puzzle: the M5_clean model, which is designed to improve results and capture the effect of behind-the-meter (BTM) solar generation on net grid load,

fails to improve systematically over the alternative M3 model even in municipalities with the highest recorded penetration. The conservative criterion — requiring M5_clean to win in *both* Fold A and Fold B — is satisfied in only 48 of the 350 evaluated municipalities, and the mean MAPE advantage across the full portfolio is a marginal 0.018 percentage points.

Two competing explanations are consistent with this finding.

- (a) **The problem is empirically hard.** Under the data assumptions adopted in this thesis, the available signal may simply not be sufficient to improve over a well-specified model such as M3. BTM solar penetration remains low across the vast majority of Norwegian municipalities, and the E19 export signal aims to describe only the grid-visible fraction of an already small installed base. Furthermore, the assumption that Norway’s *Plusskunde* incentive renders the Elhub E19 data representative of residential solar exclusively may not hold uniformly across all municipalities. The combination of low penetration, unverifiable compositional assumptions, and structural BTM invisibility may render the forecasting gain too small to manifest consistently across folds.
- (b) **The proposed framework is fundamentally limited.** The M5_clean feature architecture, as deployed under the operational constraints of this thesis, may be incapable of extracting a reliable and useful solar signal. Under this explanation the framework is not merely data-limited but architecturally insufficient: the deployability constraints imposed to ensure real-world applicability may simultaneously prevent the model from capturing the signal it needs.

These two explanations cannot be distinguished from observed Elhub data alone. The synthetic experiment described in this section is designed to address explanation 2 directly: by constructing a fully controlled environment in which the model is given exact, noiseless capacity information and a known BTM fraction at multiple penetration levels, we test whether the M5_clean feature architecture is *capable* of improving forecasts when the informational incompleteness is removed. If M5_clean gains meaningfully over M3 in this idealised setting, the failure on real data cannot be attributed to a fundamental architectural flaw; explanation 1 — low penetration, BTM invisibility, and

unverifiable data assumptions — remains the more likely bottleneck. If it does not, the framework itself is insufficient regardless of data quality.

5.4.2 Experimental Setup

Bergen municipality (MUNICIPALITY_NUMBER 4601) is selected for two reasons. First, it belongs to penetration tier T1 in the real dataset, which means the private residential consumption series used as gross load is effectively solar-free as there is negligible BTM influence. Second, Bergen’s Elhub records place its real installed E19 capacity at approximately 12 MW as of 2023-03-01, providing a concrete and empirically grounded starting point for the growth scenarios.

Solar capacity factor derivation: The hourly global horizontal irradiance (GHI) series for Bergen is already available from the ERA5 weather pipeline described in Section 3.3.4; no additional data collection is required. The `pvl` Python library [47] is used to convert this series into a plane-of-array (POA) irradiance estimate for a representative south-facing rooftop panel at the Bergen location ($\phi = 60.39^\circ \text{ N}$, $\lambda = 5.33^\circ \text{ E}$, altitude 48 m, tilt 35° , azimuth 180°). The *panel capacity factor* $CF(t)$ — distinct from the E19 capacity factor defined in Section 3.3.3 — is the POA series normalised by its maximum observed value, yielding a dimensionless fraction in $[0, 1]$ that represents the instantaneous utilisation of installed capacity.

Synthetic net load construction: Let $L^{\text{gross}}(t)$ denote the actual Bergen private residential consumption as recorded in Elhub — the solar-free gross load. For a given installed capacity $c(t)$ (kW) and a fixed BTM fraction $\alpha = 0.5$ [48], the synthetic quantities are

$$G_{\text{BTM}}(t) = \alpha c(t) CF(t), \quad (5.1)$$

$$L^{\text{net}}(t) = \max(L^{\text{gross}}(t) - G_{\text{BTM}}(t), 0), \quad (5.2)$$

$$E_{19}^{\text{syn}}(t) = (1 - \alpha) c(t) CF(t). \quad (5.3)$$

The BTM fraction is fixed at $\alpha = 0.5$, consistent with the empirical mean self-consumption ratio of approximately 49% of PV generation [48] across over 1,300 fully-metered residential households in six countries. In practice, self-consumption varies with household size, occupancy, and season; fixing α at a constant value is a simplification of the synthetic setup. All M5_clean features are then derived from

E_{19}^{syn} and $c(t)$ in exactly the same way as in the main pipeline: $cap_e19_per_mpc$ is the E19 capacity per metering point, $G \times cap$ is the irradiance–capacity interaction, $e19_cf_rolling_7d$ and $e19_cf_lag168$ are the seven-day rolling and one-week lag of the export capacity factor, and cap_growth_yoy is the year-on-year fractional capacity change. Autoregressive lags (48-hour and 168-hour) are derived from L^{net} to respect the same Elhub publication delay.

Capacity growth scenarios: Four compound annual growth scenarios are evaluated, all starting from the real Bergen baseline of $c_0 = 12\,000$ kW:

$$c(t) = c_0 \times (1 + r)^{\Delta t}, \quad (5.4)$$

where r is the annual growth rate and Δt is years elapsed since 2023-03-01. Table 5.14 summarises the scenarios and the resulting installed capacity at the end of 2025.

Table 5.14: Capacity growth scenarios for the synthetic experiment.

Scenario	Annual rate	c_{final} (end-2025)
current	+30%/yr	25 MW
moderate	+60%/yr	46 MW
fast	+100%/yr	86 MW
ambitious	+130%/yr	128 MW

Pipeline alignment: To ensure the synthetic results are directly comparable to Stage 3, the experimental pipeline mirrors the main methodology exactly. The same three temporal folds are used (Fold A: test Jan–Jun 2025; Fold B: test Jul–Dec 2025; Fold C: test May–Aug 2025). The same LightGBM hyperparameters are applied (num_leaves=63, learning_rate=0.05, feature_fraction=0.8, min_child_samples=20, n_estimators=2000 and early stopping after 50 rounds of no improvement on the validation MAPE). Predictions are generated using 24-hour rolling blocks with the 48-hour lag floor enforced throughout. The two competing models are M3 (weather features + AR lags, no prosumer features) and M5_clean (M3 feature set augmented with the five prosumer features listed above).

5.4.3 Results

Table 5.15 reports mean MAPE averaged across the three folds for each scenario.

Table 5.15: Synthetic BTM experiment: mean MAPE (3 folds) for M3 and M5_clean. *BTM pen.* is the mean fraction of gross load suppressed by behind-the-meter generation.

Scenario	Growth (c_{final})	BTM pen.	M3 MAPE	M5_clean MAPE	Δ (pp)
current	+30%/yr (25 MW)	0.53%	4.663%	4.502%	+0.161
moderate	+60%/yr (46 MW)	0.74%	4.681%	4.565%	+0.117
fast	+100%/yr (86 MW)	1.08%	5.157%	4.948%	+0.209
ambitious	+130%/yr (128 MW)	1.39%	5.840%	5.475%	+0.365

The key result is that M5_clean improves over M3 in *all four scenarios* and across all three folds without exception. Under the most aggressive growth trajectory, the gain reaches +0.365 percentage points. The mean MAPE values increase monotonically with the growth rate, even though the gains of M5_clean over M3 do not follow a perfectly monotone pattern — the moderate scenario yields a gain of +0.117 pp, marginally below the +0.161 pp of the current scenario. This minor non-monotonicity likely reflects fold-composition variance at a single municipality rather than a structural reversal. The mean trend across the four points, however, is unambiguously positive, and the +0.365 pp gain at the ambitious scenario confirms that the relationship between penetration and M5_clean utility is real.

5.4.4 Discussion

The synthetic environment isolates the BTM invisibility problem.

The most important contrast between Table 5.15 and the Stage 3 findings (Table 5.7) is the *consistency* of M5_clean’s advantage. In Stage 3, M5_clean satisfies the conservative criterion (wins in both Fold A and Fold B) in only 48 of 350 municipalities; in the synthetic experiment it wins across every fold and scenario. The difference between the two settings is precisely the information content of the capacity proxy:

in the synthetic case the model is given the exact installed capacity and the exact BTM fraction, so the extra solar features carry true predictive signal; in the real case, the Elhub E19 metering points record only the exported fraction, leaving the self-consumed majority of BTM generation fluctuating among the different types of installations.

Why M5_clean fails on real Elhub data.

Three compounding mechanisms explain the near-zero net benefit in Stage 3.

First, *BTM invisibility* is structural. With a BTM fraction of 0.5, only half of total solar generation is visible through the E19 export signal. The capacity feature *cap_e19_per_mpc* therefore encodes, at best, half the true suppression effect.

Second, *penetration is low across the majority of the portfolio*. The mean BTM penetration in Stage 3 is well below 1% even for the highest prosumer tier (T4, $\geq 10\%$ E19 penetration as a share of metering points). The synthetic results show that even under ideal proxy conditions, a 0.53% actual BTM penetration yields only a minor +0.161 pp improvement. When the proxy is noisy and the penetration is simultaneously suppressed by the BTM fraction, the signal is in reality too weak to clear the threshold of fold-level consistency.

Third, *E19 pool heterogeneity* contaminates the signal. Real Elhub E19 metering points include commercial, agricultural, and mixed-use installations alongside residential rooftop systems. Each category has a different self-consumption ratio, a different diurnal profile, and a different seasonal trajectory, so the aggregate E19 capacity factor is a noisy average that partially decorrelates from residential BTM suppression.

Implications for the framework.

The synthetic experiment provides a constructive *proof of concept*: the M5_clean feature architecture is sufficient to capture BTM solar effects when the capacity signal is accurate. The practical barrier is measurement, not modelling. Three routes could close the gap in a production setting. A *metered gross load* pilot — where a sample of prosumers additionally meters total generation — would make the BTM

fraction directly observable. An *aerial or satellite estimation* of rooftop panel area, combined with a standard performance ratio, would possibly provide a municipality-level installed capacity estimate independent of export metering.

Until such data sources are available, the current E19 proxy remains useful as a soft signal in high-penetration regimes, but the expected gain is bounded by the BTM fraction and the heterogeneity of the metered population. The conservative portfolio strategy adopted in Stage 3 — applying M5_clean only when it demonstrates fold-consistent improvement — implicitly accounts for this limitation by reverting to the robust M3 baseline in the vast majority of municipalities.

Chapter 6

Discussion

The three-stage empirical programme reported in Chapter 5 produces a set of findings that, taken together, constitute a validated modelling proposal for municipality-level net load forecasting in Norway using exclusively open data. The discussion is organised around four themes: the national generalisation of the weather-based framework, the behaviour of the prosumer-aware solar feature bundle under real versus controlled conditions, the structural heterogeneity of industrial load, and the practical deployment implications that follow from these findings.

6.1 A Validated Forecasting Framework at National Scale

A core contribution of this thesis is the establishment of an end-to-end, open-data forecasting procedure that has been validated at two levels of granularity: a six-fold expanding-window walk-forward evaluation on eight representative municipalities in Stage 2, and a three-fold national evaluation across all 350 Norwegian municipalities in Stage 3. The two-level design addresses a gap common in the short-term load forecasting literature, where models are frequently evaluated on a single site or a single temporal fold, making it difficult to assess whether reported improvements reflect genuine generalisation or a favourable interaction between model and evaluation window. By requiring consistent improvement across two independent annual folds (Fold A: January–June; Fold B: July–December), the conservative criterion applied throughout Stage 3 provides a high-specificity filter against seasonal over-fitting.

For private residential and commercial load, the framework demonstrates robust national generalisation. Model M3 — a LightGBM model trained on the full ERA5 weather stack and 48-hour and 168-hour autoregressive lags — achieves a mean annual municipality MAPE of 6.69% for private residential load, winning conservatively in 347 of 350 municipalities (99.1%) against the seasonal naive benchmark and in 246 of 350 (70.3%) against the best OLS baseline. For commercial load, M3 reaches 9.655% annual mean MAPE, winning conservatively in 313 of 350 municipalities (89.4%) against the naive benchmark and in 304 of 350 (86.9%) against OLS. These results confirm that M3 is an appropriate national default for both categories and that the Stage 2 recommendations transfer to the full Norwegian portfolio without systematic degradation.

An important secondary finding is the consistent failure of the OLS baseline M0c across both consumer categories. For commercial load, M0c finishes 0.525 pp *worse* than the seasonal naive benchmark on an annual basis, despite winning conservatively in 226 of 350 municipalities. While calendar and temperature features constitute the standard foundation of short-term load forecasting models [14], this thesis demonstrates that domain understanding of the underlying consumption patterns is at least as important as feature selection. The failure of both the OLS baseline M0c and the LightGBM model M3 on industrial load — where M3 records a mean MAPE of 39.51%, worse than the naive benchmark at 34.94% — illustrates that neither statistical nor machine learning frameworks are immune to structural misspecification. Applying a weather-and-calendar feature set to a consumption category driven primarily by production schedules and price-responsive demand produces results worse than a naive benchmark, regardless of model complexity. Model selection should be driven by data understanding rather than methodological convention.

6.2 The Prosumer Feature Bundle: A Data Problem, Not a Modelling Problem

The prosumer-aware model M5_clean was designed to reduce net load forecasting error by incorporating the Elhub E19 export signal as an indirect proxy for behind-the-meter (BTM) solar capacity. The Stage 3 national results show that this approach does not produce systematic improvement under current Norwegian data conditions: M5_clean trails M3 by 0.03 pp on the annual mean MAPE across 350 municipalities, wins conservatively in only 48

municipalities (13.7%), and the conservative portfolio assignment achieves a gain of just 0.018 pp over a pure M3 deployment. No penetration tier produces a robust positive conservative win rate; notably, the conservative win rate collapses to zero at T4.

The synthetic experiment in Section 5.4 is designed to determine whether this failure reflects an architectural deficiency or a data limitation. By providing M5_clean with exact, noiseless capacity information and a known BTM fraction in a fully controlled setting, the experiment tests whether the feature architecture is *capable* of extracting predictive signal. The result is unambiguous: M5_clean improves over M3 consistently across all four capacity growth scenarios and all three evaluation folds, with gains ranging from +0.161 pp at current Bergen growth rates to +0.365 pp at the ambitious scenario. The architecture is sound; the problem lies in the observational data.

Three compounding mechanisms explain the near-zero real-data benefit. First, *BTM invisibility is structural but symmetric*: both the real and synthetic settings share the same fundamental invisibility of self-consumed generation. What differs is the *consistency* of the proxy: in the synthetic setting the BTM fraction α is fixed and known, making the exported signal a perfectly consistent proxy for BTM suppression. In the real setting, α varies with season, occupancy, and battery adoption, introducing noise that partially decorrelates the E19 export signal from the BTM suppression it is intended to capture. Second, *E19 pool heterogeneity* introduces noise: real E19 metering points aggregate residential, commercial, agricultural, and mixed-use installations, each with different self-consumption ratios and diurnal profiles, partially decorrelating the aggregate capacity factor from the residential BTM suppression it is intended to proxy. Third, *penetration is low*: even in the highest-penetration tier, mean BTM suppression of gross load is well below 1% during most hours, and the synthetic results show that gains of this magnitude require near-ideal proxy conditions to register consistently across evaluation folds.

The conclusion is that M5_clean represents a valid modelling approach for conditions that do not yet prevail in Norway at scale: higher BTM penetration, metered gross load for prosumer households, or disaggregation of the E19 pool by installation type. Under current conditions, the conservative portfolio strategy adopted in Stage 3 — deploying M5_clean selectively in the 48 municipalities where it demonstrates fold-consistent improvement — is the appropriate operational response, preserving the ability to expand deployment as penetration grows and data quality improves.

6.3 Industrial Load: Heterogeneity and the Case for a Hybrid Strategy

The industrial category results reveal a structural problem that is qualitatively different from the residential and commercial findings. M3's near-50% conservative win rate against the naive benchmark for industrial load (175 of 350 municipalities, -4.577 pp mean annual delta) does not indicate a model deficiency: the same LightGBM architecture and training procedure achieves a 99.1% conservative win rate for private residential load. The performance gap is structural. Norwegian industrial electricity consumption is concentrated in a small number of large, energy-intensive facilities, like the aluminium smelter in the municipality of Sunndal (1563), that operate according to production contracts and schedules that are not primarily affected by weather and calendar features.

This structural heterogeneity manifests across the full evaluation set as distribution of forecast difficulty. At one extreme, Sunndal (municipality 1563) hosts a single aluminium smelter that accounts for approximately 9.6% of total national industrial consumption volume. Its load is near-constant regardless of season, and the seasonal naive benchmark forecasts it with a MAPE of 0.62% — a level that no weather-based model can approach. At the other extreme, municipalities with heterogeneous urban industrial demand respond partially to price and calendar signals; M4_spot, which incorporates electricity spot price features, achieves meaningful improvement in Oslo (13.8% vs M3's 18.7% and M_naive's 30.2% in the winter fold). Between these extremes sit the 22 outlier municipalities where a single facility dominates load to such an extent that MAPE becomes a measure of load variability rather than forecast quality.

These findings motivate a hybrid deployment strategy for industrial load forecasting. Rather than applying a single global model to all 350 municipalities, a multiple-category disaggregation based on load profile characteristics is warranted. For municipalities dominated by a single large continuous-process installation — identifiable by low inter-annual load variance and near-constant load profiles — the seasonal naive forecast is the appropriate recommendation: it is computationally trivial, requires no weather data, and achieves sub-1% MAPE scores. For the remaining municipalities, where industrial load is driven by a heterogeneous mix of consumers with partial weather and price responsiveness, M3 combined with spot price features (M4_spot) provides the best available global model.

Critically, the OLS baseline must not be deployed for industrial load under any circumstances: its mean annual MAPE of 63.3%, driven by systematic collapse on non-linear industrial demand patterns.

The boundary between the two categories can be operationalised using a simple variance-based criterion applied to the historical load series — a profiling step that requires no additional data beyond what is already available in Elhub. Implementing this disaggregation would convert the near-50% conservative win rate of M3 against naive into a sharper, context-appropriate deployment protocol that acknowledges the limits of global modelling while preserving its benefits in the municipalities where it genuinely adds value.

6.4 Open-Data Forecasting: Potential and Boundaries

This thesis is, among other things, an empirical test of what is achievable using exclusively open, publicly available data sources in a Nordic electricity system context. For private residential and commercial net load forecasting, the ERA5 weather dataset and Elhub AMI metering records provide sufficient signal for a competitive national forecasting pipeline that generalises robustly across a heterogeneous set of 350 municipalities. For BTM solar forecasting, the Elhub E19 export records provide a useful indirect proxy that the synthetic experiment confirms would be capable of driving meaningful accuracy improvements if the underlying capacity signal were complete. The gap between the synthetic and real-data results thus identifies a specific, targeted improvement to the existing data infrastructure — not a fundamental barrier — that would unlock the modelling value already embedded in the M5_clean feature architecture.

Two interventions would have a disproportionately high impact relative to their cost. A *generation metering pilot* for a representative sample of prosumer households would make the BTM fraction directly observable, providing the ground truth needed to validate and calibrate the E19 proxy. An *installation-type flag* in the Elhub E19 registration database — distinguishing residential from commercial and industrial prosumers — would allow construction of a homogeneous residential capacity signal without requiring any new physical metering infrastructure. Both improvements are administrative rather than engineering interventions, and both build on the universal smart-metering programme that Norway completed in 2019. Even assuming that all E19 installations are residential, a reasonable choice based on

plusskundeordningen framework, the results of this thesis show that this assumption alone is a limitation and insufficient to extract a robust result from the available data.

Chapter 7

Conclusions and Future Work

This chapter summarizes the main findings of the thesis, identifies the limitations of the work, and outlines directions for future research. Section 7.1 returns to the overarching research question and answers it on the basis of the empirical evidence. Section 7.2 describes the boundaries within which those answers are valid. Section 7.3 proposes possible next steps. Section 7.4 situates the contribution within a broader sustainability and societal context.

7.1 Conclusions

This thesis set out to answer one overarching question: to what extent can a modular forecasting framework that explicitly incorporates prosumer characteristics and consumer-category disaggregation improve short-term net load forecasting accuracy at the Norwegian municipality level, and how does this improvement vary with local distributed solar PV penetration across a heterogeneous national set of municipalities?

The three-stage empirical programme — a national weather feature ablation (Stage 1), a six-fold walk-forward evaluation on eight representative municipalities (Stage 2), and a three-fold national evaluation across 350 municipalities (Stage 3) — produces a clear and internally consistent answer. Taken together, the results establish the framework as a validated proposal for forecasting many structurally different loads from a single open-data pipeline, while being explicit about the conditions under which each modelling component is and is not beneficial.

The weather-based model M3 is the universal national default for private residential and commercial net load forecasting. For private residential load, M3 achieves a mean annual municipality MAPE of 6.69%, winning

conservatively in 347 of 350 municipalities (99.1%) against the seasonal naive benchmark and in 246 of 350 (70.3%) against the best OLS baseline. For commercial load, M3 achieves 9.655% annual mean MAPE, winning conservatively in 313 of 350 municipalities (89.4%) against the naive benchmark and in 304 of 350 (86.9%) against OLS. The Stage 1 ablation establishes temperature as the dominant explanatory variable, contributing +5.74 pp of MAPE reduction for private residential, +4.21 pp for commercial, and +2.00 pp for industrial consumption. The OLS baseline M0c fails systematically for commercial load (−0.525 pp relative to naive) and collapses for industrial load (63.3% annual mean MAPE), establishing that the non-linear LightGBM architecture is a structural necessity in the Norwegian context.

The prosumer-aware model M5_clean does not produce systematic improvement at the national level under current data conditions: its mean annual MAPE gain over M3 is −0.03 pp across 350 municipalities, and no penetration tier produces a robust positive conservative win rate. The synthetic experiment in Section 5.4 isolates the cause: when M5_clean is provided with exact, noiseless capacity information, it improves over M3 consistently across all growth scenarios and evaluation folds (gains of +0.161 to +0.365 pp), confirming that the architecture is sound and the failure on real data is attributable to structural BTM invisibility, low current penetration, and E19 pool contamination rather than to a modelling deficiency. The conservative portfolio strategy — deploying M5_clean selectively in the 48 municipalities (13.7%) where it demonstrates fold-consistent improvement — is the appropriate operational response under current conditions.

For industrial net load, the near-50% conservative win rate of M3 against the naive benchmark reflects the structural absence of learnable production-side features in the available open data. Norwegian industrial consumption is concentrated in large energy-intensive facilities governed by production schedules uncorrelated with weather and calendar effects. A two-category hybrid strategy — seasonal naive for continuous-process municipalities, M3 with spot price features for heterogeneous urban industrial municipalities — is the recommended approach for improving industrial portfolio accuracy within the constraints of global open-data modelling.

The complete pipeline is built exclusively on open data sources and open-source software, designed to forecast many structurally different municipal loads from a single framework, while remaining deployable under realistic operational data constraints.

7.2 Limitations

- ERA5 as a performance ceiling and spatial resolution constraint.** Weather features are derived from ERA5 reanalysis, which provides historically observed conditions rather than forward-looking forecasts. Reported MAPE values therefore represent an *upper bound* on achievable operational accuracy: in a production setting, NWP forecast uncertainty would degrade performance relative to the reanalysis baseline. Additionally, ERA5 operates on an approximately 31-kilometre grid, which cannot resolve sub-municipal variability in load concentration or solar resource. The weather station selection (WSS) procedure assigns the most representative ERA5 cell to each municipality, but municipalities where dominant load centres or solar installations are geographically offset from the optimal cell — due to fjord microclimates, elevation differences, or spatially concentrated industrial facilities — will experience a systematic mismatch between the assigned weather signal and the actual conditions driving their load.
- The plusskunde assumption and E19 pool contamination.** The M5_clean feature bundle rests on the assumption that Elhub E19 metering points represent residential solar prosumers under the Norwegian *plusskunde* scheme. In practice, E19 metering aggregates residential, commercial, agricultural, and large-scale industrial generation units, each with different self-consumption ratios, diurnal profiles, and seasonal behaviour. Treating the aggregate E19 capacity as a residential solar proxy introduces contamination that is structurally irreducible without installation-type disaggregation of the E19 registry, and this represents an independent source of signal degradation beyond the BTM invisibility problem.
- Forecasting horizon and spatial granularity.** The study focuses on the day-ahead horizon with 24-hour rolling prediction blocks at hourly resolution. Intra-day, real-time, and medium-term horizons are outside the scope. The municipality-level aggregation means that feeder-level or household-level patterns that are relevant for congestion management and flexibility dispatch, cannot be addressed with the available data.
- Model class.** The empirical analysis is limited to LightGBM gradient boosting and an OLS baseline arised from literature. Deep learning architectures like temporal convolutional networks, transformers and

LSTM networks as well as foundation model approaches like Chronos, TimeGPT and TimesFM have demonstrated competitive performance in forecasting literature and are not evaluated here. Whether gradient boosting remains competitive against deep learning under the data volumes and municipality heterogeneity of the Norwegian setting is an open empirical question.

- **Geographic and market scope.** The analysis is conducted exclusively on Norwegian data for the period 2023–2025. Norway’s near-total electrification of heating, predominantly hydropower generation mix, and open-access Elhub infrastructure may limit transferability to markets with different structural characteristics.
- **Single site for the synthetic experiment.** The synthetic BTM experiment is conducted for Bergen municipality only. It demonstrates architectural capability but cannot predict M5_clean performance in other municipalities, where different local load profiles, climate conditions, and metering-point density would affect the magnitude of BTM suppression and feature extraction quality. Furthermore, the BTM fraction α is held constant at 0.5 throughout the experiment, whereas in reality self-consumption varies with season, occupancy patterns, and the presence of battery storage. This additional simplification means the synthetic setting is more favourable than real deployment conditions, and the observed gains should be interpreted as an upper bound on what the M5_clean architecture can achieve under ideal proxy conditions.

7.3 Future Work

- **Operational validation with NWP forecast inputs.** Replacing ERA5 reanalysis inputs with operational NWP forecast inputs would test the framework under realistic production conditions. This validation could span multiple NWP providers to identify the configuration that minimises the reanalysis-to-forecast accuracy gap and to assess whether the WSS spatial matching procedure remains appropriate at NWP resolutions.
- **Higher-resolution and geographically explicit weather inputs.** Higher-resolution gridded weather products would allow the weather station selection algorithm to identify even more optimal combinations

of nearby grid points. This could additionally improve the weather feature selection in municipalities.

- **Metered gross load and E19 installation-type disaggregation.** A generation metering pilot for a representative sample of prosumer households would make the BTM fraction more observable and closer to the synthetic environment. An installation-type flag in the Elhub E19 registration database — distinguishing residential from commercial, agricultural, and industrial prosumers, would allow construction of a homogeneous residential capacity signal without new physical metering infrastructure.
- **Multi-site and variable- α synthetic experiments.** The synthetic BTM experiment conducted in this thesis fixes both the study site (Bergen) and the BTM fraction ($\alpha = 0.5$). A natural extension is to repeat the experiment across municipalities that differ in geographic location, climate regime and baseline load profile. For instance, a high-latitude municipality in northern Norway with extreme photoperiod variation, a coastal west-facing municipality with persistent cloud cover and a high-penetration municipality from a different penetration tier than T1. Simultaneously varying α across a plausible range (e.g. 0.3 to 0.7, reflecting different battery adoption and household sizes) would quantify the sensitivity of the M5_clean gain to the self-consumption assumption and reveal whether the architecture’s predictive value is robust across the Norwegian geographic and climatic spectrum or concentrated in specific regional conditions. Such a study would directly strengthen the generalisability claim of the synthetic proof of concept and provide calibration guidance for future deployments.
- **Hybrid industrial forecasting protocol.** A variance-based or cluster-based classification of industrial municipalities — using historical load series to distinguish continuous-process facilities from heterogeneous urban industrial demand — would operationalise the two-category deployment strategy recommended in Section 6.3. This would allow the framework to apply the seasonal naive model where it excels and a feature-rich model where learnable signal exists, without requiring any additional data collection.
- **Per-municipality ensemble modelling across architectures.** A per-municipality ensemble combining LightGBM, temporal convolutional networks, and linear models — weighted by municipality-specific

validation performance — would allow the framework to adapt its effective model complexity to the local data environment. Linear models may generalise better in data-sparse municipalities; deep learning architectures may capture long-range temporal dependencies where load histories are rich and stable. A dynamic weighting or stacking scheme would exploit this complementarity and potentially recover accuracy in heterogeneous settings where a single global model underperforms.

- **Probabilistic forecasting.** Extending the framework to produce prediction intervals using quantile regression — natively supported by LightGBM — would increase its operational applicability for reserve sizing and balancing decisions. Probabilistic evaluation is particularly valuable for the industrial category and for municipalities experiencing rapid solar growth, where point forecast uncertainty is highest.
- **Extension to other Nordic markets.** Sweden, Denmark, and Finland share structural characteristics with Norway but differ in generation mix, grid architecture, and metering data availability. A systematic comparison across Nordic markets using nationally representative municipality-level datasets would test the generalisability of the framework and identify market-specific factors that condition the value of weather-based and prosumer-aware forecasting.

7.4 Reflections

Accurate short-term net load forecasting is a prerequisite for the efficient operation of electricity systems undergoing a fundamental transition toward distributed and variable renewable generation. At the municipality level, improved forecast accuracy translates into reduced balancing costs, more efficient use of flexible loads and storage, and a reduced need for fossil-fuel reserve capacity dispatched to compensate for forecast errors.

This work contributes to the United Nations Sustainable Development Goals in three specific ways. It supports **SDG 7** (Affordable and Clean Energy) by developing forecasting tools that facilitate the integration of distributed solar PV. It supports **SDG 11** (Sustainable Cities and Communities) by providing distribution system operators with municipality-level capabilities that can inform their planning for grid, electric infrastructure and distributed storage. It supports **SDG 13** (Climate Action) by demonstrating that the open

AMI data infrastructure built through national smart-metering investments can serve as the foundation for data-driven grid intelligence that aims to improve the imbalance of the power sector.

The framework proposed in this thesis is deliberately respectful of the constraints of real available data. Rather than assuming access to metered gross load, installation-level solar records, or proprietary operational data, it is designed to extract maximum forecasting value from what open AMI infrastructure already provides — while being explicit about where those data boundaries limit achievable accuracy. This transparency is itself a contribution: it maps the frontier between what is currently achievable with open data and what would become achievable with targeted data infrastructure improvements.

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Appendix A

Supplementary Figures

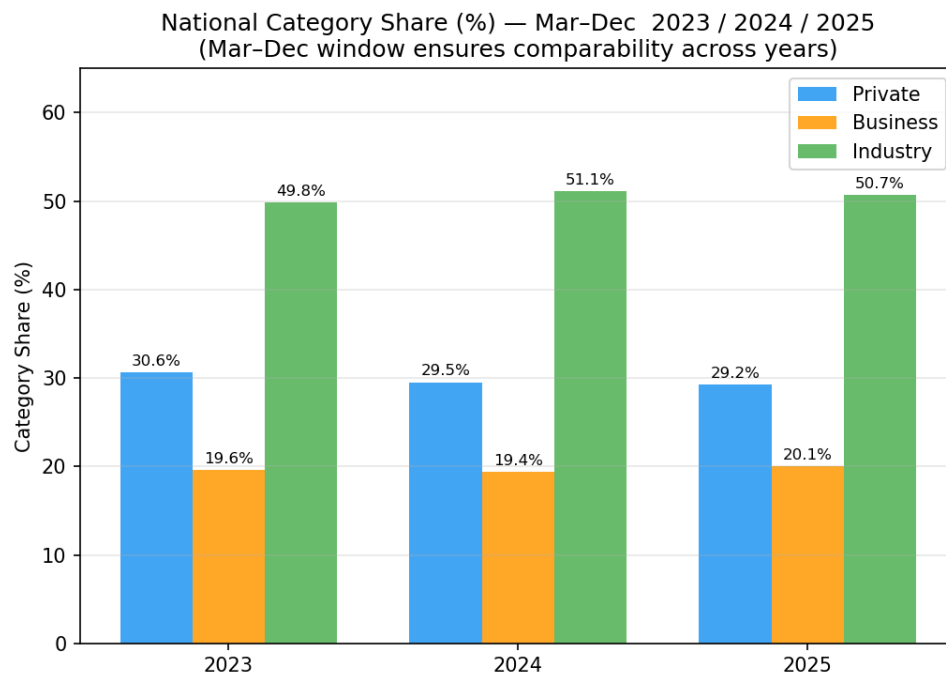


Figure A.1: National electricity consumption by consumer group for the March–December comparable window, 2023–2025.

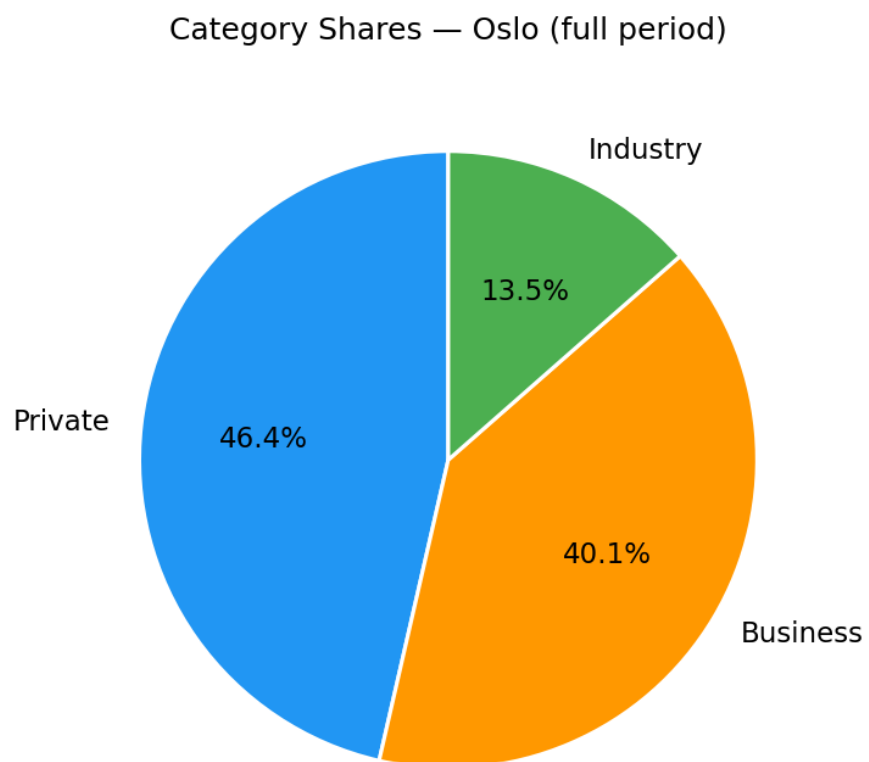


Figure A.2: Consumer group shares Oslo over the full study period.

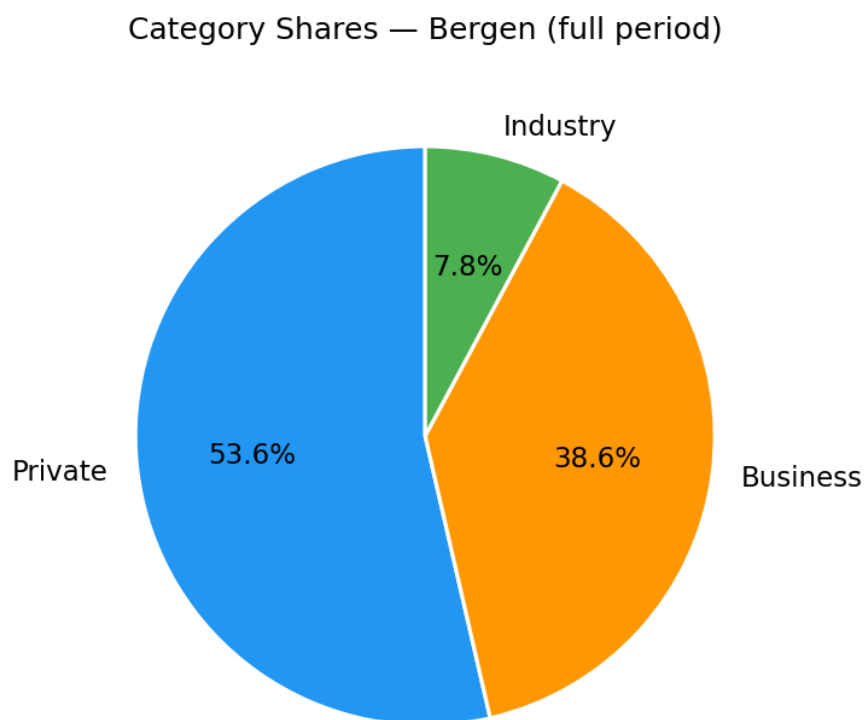


Figure A.3: Consumer group shares for Bergen over the full study period.

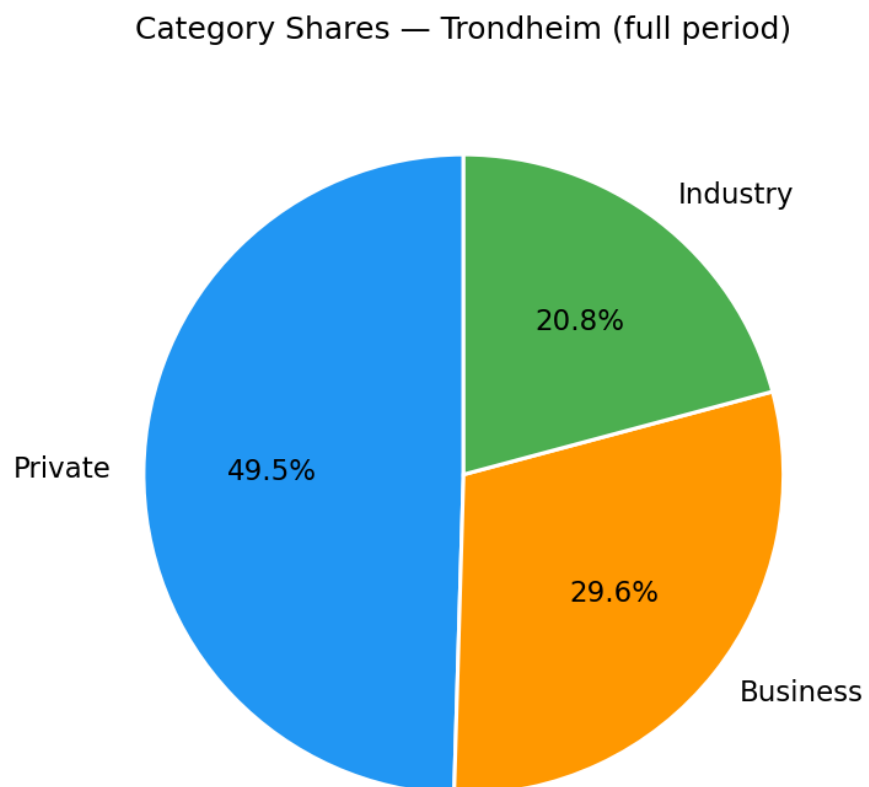


Figure A.4: Consumer group shares for Trondheim over the full study period.

Category Shares — Sunndal (full period)

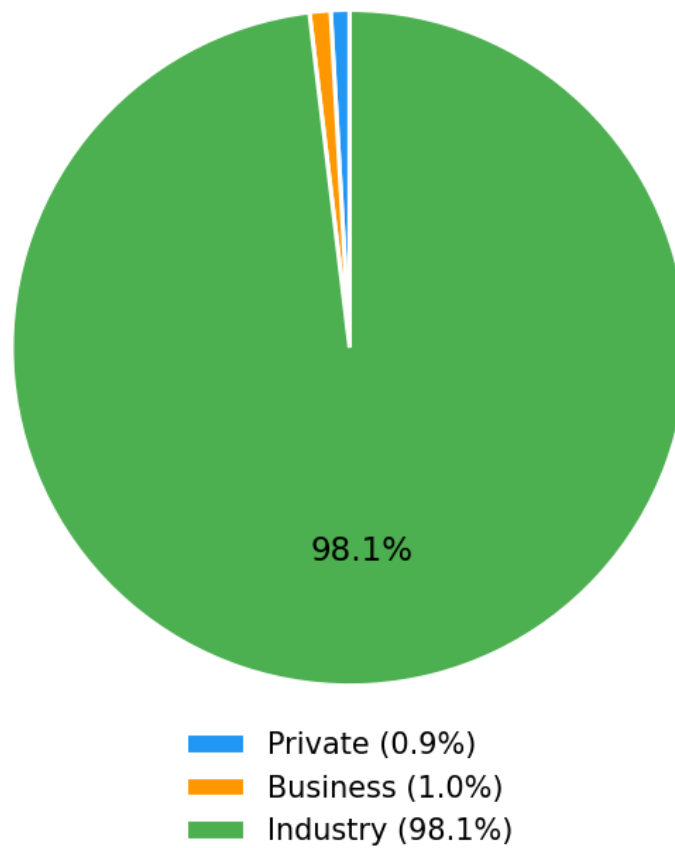


Figure A.5: Consumer group shares for Sunndal over the full study period.



Figure A.6: Private residential metering point count trends across five municipalities. Left: growth index (base = 100 at first observation). Right: raw MPC counts. All municipalities show a gradual upward trend over the study period.

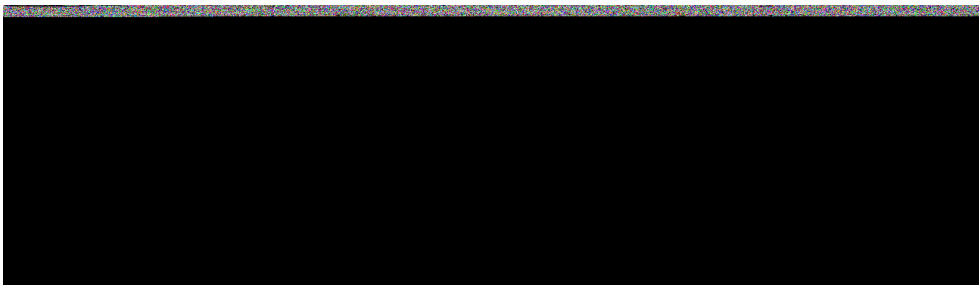


Figure A.7: Commercial metering point count trends across five focus municipalities.

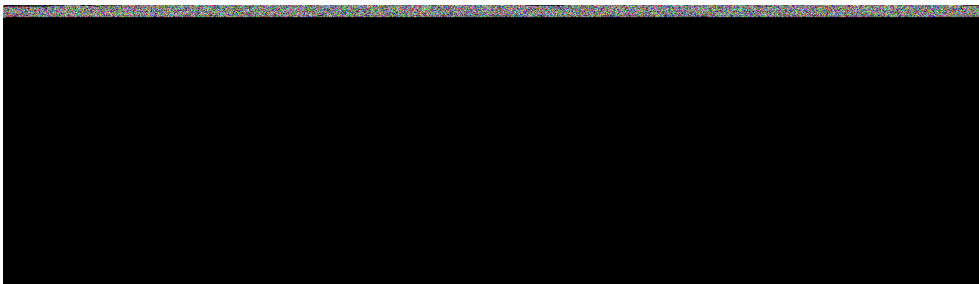


Figure A.8: Industrial metering point count trends across five focus municipalities



Figure A.9: Top-10 municipalities by total electricity consumption (TWh) for each consumer group, full study period (March 2023 – February 2026)

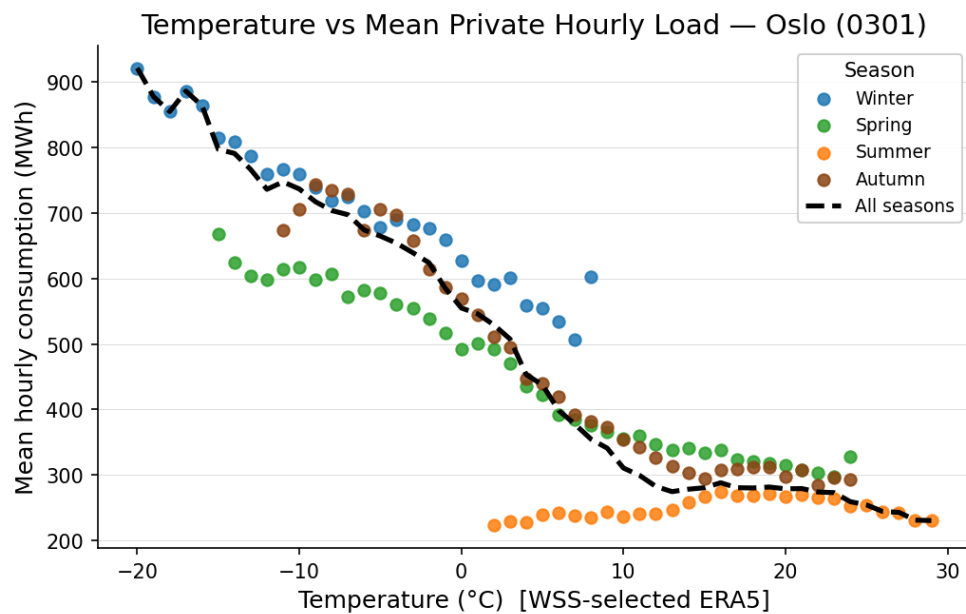


Figure A.10: Temperature versus mean hourly private residential load for Oslo (0301), coloured by season. The L-shaped profile confirms heating-dominated demand with negligible cooling load. The seasonal spread at identical temperatures motivates the inclusion of calendar interaction terms in the temperature feature specification.

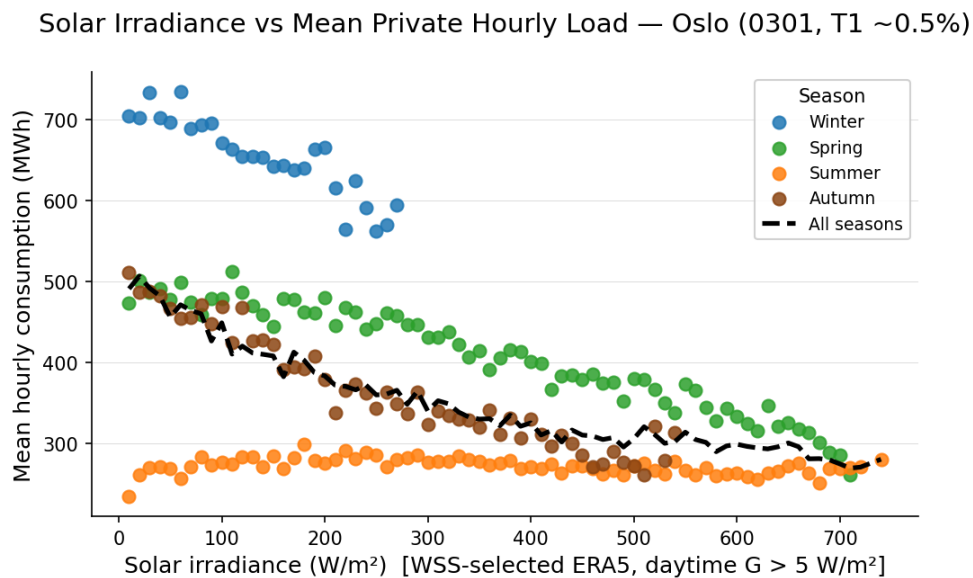


Figure A.11: Solar irradiance versus mean hourly private residential load for Oslo (0301, T1, ~0.5% penetration). Daytime hours only ($G > 5 \text{ W/m}^2$).

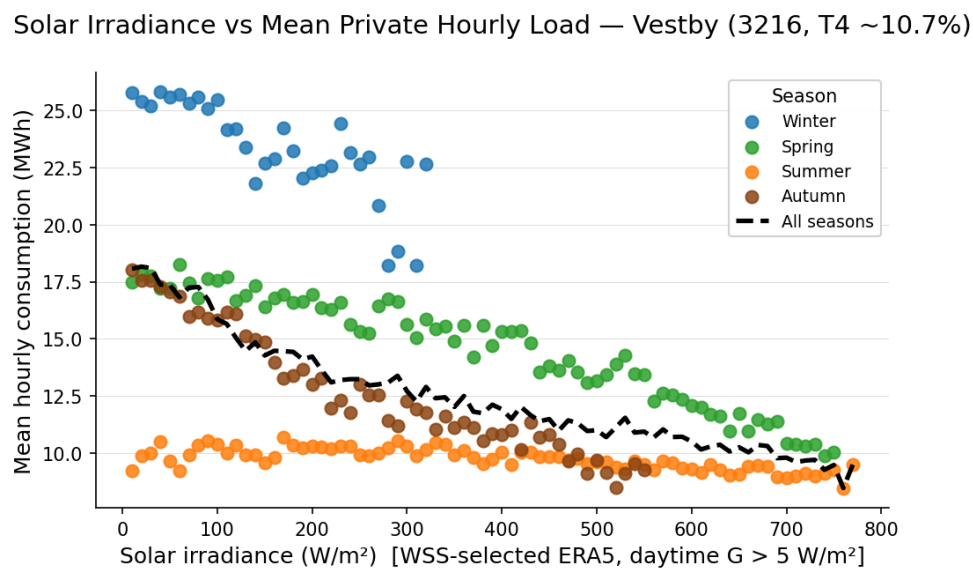


Figure A.12: Solar irradiance versus mean hourly private residential load for Vestby (3216, T4, ~10.7% penetration). Daytime hours only ($G > 5 \text{ W/m}^2$).

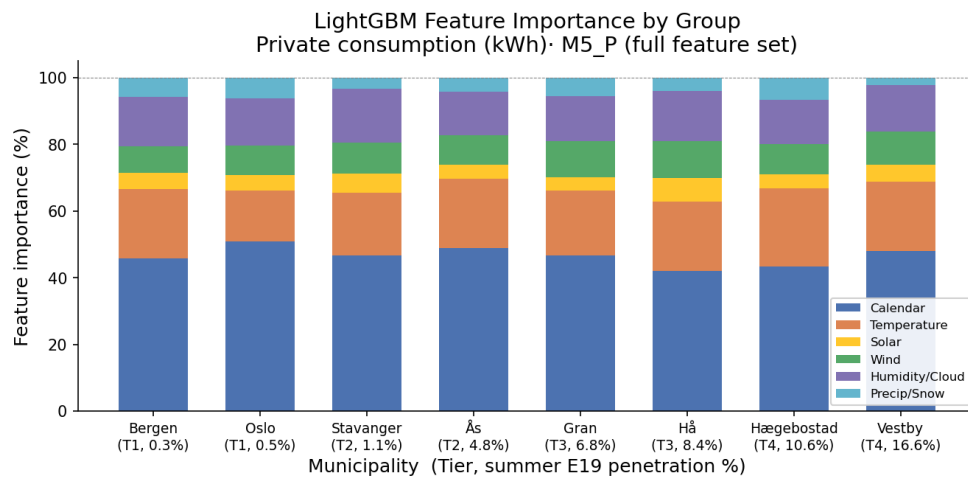


Figure A.13: LightGBM feature importance by group (private residential, model M5_P) for all eight Stage 2 representative municipalities, ordered by solar penetration tier from T1 (lowest) to T4 (highest).

Appendix B

Supplementary Tables

Table B.1: Complete feature set per model. A dot (•) indicates inclusion; a dash (—) indicates exclusion. M_naive uses no features (persistence of kwh_lag_168h only).

Feature group	Features	M0c (OLS)	M3	M5_clean (private)	M4_spot (industry)
<i>Calendar</i>					
	hour, dayofweek, month, Trend	•	•	•	•
	is_holiday, is_holiday_tomorrow	•	•	•	•
	hour_x_dow	•	•	•	•
<i>Temperature (WSS-selected ERA5)</i>					
	T_wss, T2_wss, T3_wss	•	•	•	•
	TxH, T2xH, T3xH	•	•	•	•
	TxM, T2xM, T3xM	•	•	•	•
	T_lag_24h, T_lag_24h_sq, T_lag_48h	•	•	•	•
	T_daily_ma_1d, T_daily_ma_2d	•	•	•	•
<i>Solar irradiance (WSS-selected ERA5)</i>					
	G_wss, GxH, GxM	•	•	•	•
	G_lag_24h, G_lag_48h, G_lag_168h	•	•	•	•
	G_daily_ma_1d, G_daily_ma_2d	•	•	•	•
<i>Extended weather (ERA5)</i>					
	CC_wss, CCxH, CCxM	—	•	•	•
	WS_wss, WD_sin, WD_cos, WSxH, WSxM	—	•	•	•
	RH_wss, RHxH, RHxM	—	•	•	•
	TP_wss, SD_wss	—	•	•	•
<i>Autoregressive lags</i>					
	kwh_lag_48h, kwh_lag_168h	—	•	•	•
<i>Prosumer bundle (E19 — private only)</i>					
	cap_e19_per_mpc, G_x_cap	—	—	•	—
	e19_cf_rolling_7d, e19_cf_lag168	—	—	•	—
	cap_growth_yoy	—	—	•	—
<i>Spot price (industry only)</i>					
	spot_price_eur, price_rank_24h	—	—	—	•
Total features		29	44	49	46

Table B.2: Annual national electricity consumption share by consumer group (%), computed over the March–December across 2023–2025.

Consumer Group	2023	2024	2025	Mean (2023–2025)
Private residential	30.6%	29.5%	29.2%	29.8%
Commercial	19.6%	20.4%	20.1%	20%
Industrial	49.8%	51.1%	50.7%	50.5%

Table B.3: LightGBM feature importance by group (%), private residential (M5_P), all eight Stage 2 municipalities.

Municipality	Calendar	Temp.	Solar	Wind	Hum./Cloud	Precip./Snow
Bergen (T1)	45.9	20.8	4.8	7.9	14.8	5.8
Oslo (T1)	51.0	15.2	4.6	9.0	14.1	6.1
Stavanger (T2)	46.6	18.9	5.7	9.4	16.0	3.4
Ås (T2)	48.8	20.8	4.2	8.9	13.1	4.2
Gran (T3)	46.6	19.5	4.0	10.9	13.5	5.5
Hå(T3)	42.0	20.8	7.0	11.2	15.1	3.9
Vestby (T4)	48.0	20.9	5.0	9.8	14.1	2.1
H(T4)	43.3	23.5	4.2	9.1	13.2	6.7

Table B.4: Private residential — MAPE (%) per fold for key models.
F1,F4,F5 = summer; F2,F3,F6 = winter.

Muni	Model	Mean	Summer folds			Winter folds		
			F1	F4	F5	F2	F3	F6
Bergen (T1)	M_naive	12.05	12.30	12.20	10.85	12.77	12.70	11.50
	M0c	4.68	5.74	4.65	5.60	4.43	4.26	3.43
	M3	4.56	5.39	4.58	4.04	5.22	3.90	4.24
	M5_clean	4.59	5.93	4.59	4.24	4.38	3.89	4.48
Oslo (T1)	M_naive	10.00	6.82	11.60	6.41	10.91	12.15	12.11
	M0c	6.13	6.89	5.93	6.78	7.15	6.39	3.51
	M3	4.12	4.83	5.21	2.66	4.64	3.76	3.59
	M5_clean	4.38	5.54	5.09	2.92	5.09	4.02	3.61
Stavanger (T2)	M_naive	11.48	12.43	11.62	9.93	11.95	11.73	11.23
	M0c	5.56	5.96	6.72	5.17	4.91	5.35	5.25
	M3	5.24	6.04	5.64	3.81	5.89	4.51	5.56
	M5_clean	5.42	6.93	5.69	4.17	5.05	4.80	5.87
Ås (T2)	M_naive	11.37	8.39	13.15	7.42	12.70	13.04	13.54
	M0c	8.27	20.30	7.29	6.96	6.36	5.27	3.45
	M3	5.01	7.26	5.28	3.38	5.46	4.19	4.49
	M5_clean	4.97	6.15	5.91	4.16	5.38	4.17	4.05
Gran (T3)	M_naive	12.33	10.08	13.02	8.12	12.11	15.85	14.76
	M0c	6.93	12.54	6.42	6.19	5.68	6.35	4.39
	M3	5.48	8.69	6.05	3.54	6.12	3.84	4.62
	M5_clean	5.68	10.19	6.13	3.61	6.03	3.63	4.50
Hå(T3)	M_naive	10.74	11.34	9.92	9.39	11.61	11.11	11.08
	M0c	4.89	5.94	5.58	5.10	4.44	4.40	3.86
	M3	4.84	5.87	4.74	3.80	5.47	3.72	5.44
	M5_clean	4.82	6.46	4.91	3.80	5.06	3.81	4.85
Vestby (T4)	M_naive	11.46	7.99	11.96	6.72	13.51	14.19	14.41
	M0c	6.03	11.34	5.48	5.38	5.41	4.85	3.71
	M3	5.37	7.90	5.04	3.43	5.85	4.62	5.39
	M5_clean	5.21	6.31	5.57	3.55	5.54	4.50	5.82
Haegbostad (T4)	M_naive	12.59	12.12	13.04	9.48	13.02	13.89	13.97
	M0c	6.55	12.55	6.81	5.75	5.50	4.25	4.43
	M3	5.22	5.13	5.24	4.17	6.56	4.44	5.77
	M5_clean	5.42	5.32	5.54	4.14	6.82	4.41	6.29

Table B.5: Commercial — MAPE (%) per fold for key models. F1,F4,F5 = summer; F2,F3,F6 = winter.

Muni	Model	Mean	Summer folds			Winter folds		
			F1	F4	F5	F2	F3	F6
Bergen (T1)	M_naive	7.52	5.29	8.47	5.02	9.20	9.13	8.00
	M0c	4.61	6.43	4.20	3.82	5.29	4.13	3.79
	M3	3.67	3.37	3.39	2.89	4.59	3.92	3.88
Oslo (T1)	M_naive	5.93	2.98	7.09	3.22	7.44	8.19	6.66
	M0c	6.27	8.54	4.84	3.20	10.99	5.75	4.30
	M3	3.35	2.00	2.95	1.85	5.72	4.20	3.38
Stavanger (T2)	M_naive	6.43	4.54	8.48	3.90	7.76	7.87	6.03
	M0c	4.70	5.04	5.14	4.11	5.98	4.10	3.85
	M3	3.65	3.52	3.83	2.37	5.24	3.53	3.40
Ås (T2)	M_naive	7.20	3.88	7.51	4.56	9.09	9.96	8.23
	M0c	7.10	7.24	5.53	7.17	6.00	5.06	11.58
	M3	4.33	3.53	3.57	3.69	4.36	4.34	6.52
Gran (T3)	M_naive	8.81	4.97	9.57	4.48	9.94	12.59	11.29
	M0c	5.42	4.83	7.47	4.46	5.15	5.58	5.02
	M3	4.22	3.53	4.46	3.24	5.16	4.09	4.87
Hå(T3)	M_naive	8.26	6.28	9.41	6.03	9.62	10.28	7.92
	M0c	5.73	6.27	5.84	5.81	6.35	5.77	4.31
	M3	4.66	4.28	4.87	3.56	6.19	4.88	4.15
Vestby (T4)	M_naive	10.22	7.83	15.15	7.20	10.07	11.85	9.20
	M0c	8.64	9.45	12.10	7.24	8.15	6.56	8.30
	M3	6.83	6.63	7.96	6.28	6.55	6.47	7.12
Hægebostad (T4)	M_naive	10.38	7.23	11.29	8.79	11.06	13.18	10.71
	M0c	8.75	8.66	10.05	9.77	7.77	8.93	7.33
	M3	7.02	6.83	10.34	6.17	6.18	6.81	5.76

Table B.6: Industrial — MAPE (%) per fold, key models. F1,F4,F5 = summer; F2,F3,F6 = winter.

Muni	Model	Mean	Summer folds			Winter folds		
			F1	F4	F5	F2	F3	F6
Bergen (T1)	M_naive	15.50	8.27	15.94	8.62	20.87	24.65	14.63
	M3	13.03	7.07	11.53	8.59	20.17	15.53	15.28
	M4_spot	12.98	6.70	11.56	8.47	19.61	15.54	16.02
Oslo (T1)	M_naive	25.46	17.92	17.32	7.02	36.97	43.46	30.06
	M3	28.12	28.36	26.08	32.33	21.18	29.96	30.82
	M4_spot	25.36	22.89	22.59	31.76	20.74	27.85	26.33
Stavanger (T2)	M_naive	9.13	8.49	13.79	9.87	7.60	8.10	6.95
	M3	6.23	5.84	8.37	6.52	5.49	5.66	5.51
	M4_spot	6.19	5.69	7.64	6.89	5.49	5.72	5.71
Ås (T2)	M_naive	13.13	9.79	17.85	16.71	11.39	12.69	10.35
	M3	14.79	7.70	14.66	29.85	7.56	7.77	21.21
	M4_spot	15.03	7.72	15.08	29.95	8.02	7.79	21.62
Gran (T3)	M_naive	12.27	11.88	17.39	12.59	10.86	10.37	10.54
	M3	10.02	10.93	11.90	12.09	10.56	6.73	7.87
	M4_spot	10.08	11.41	11.95	12.44	10.34	6.64	7.70
Hå(T3)	M_naive	8.88	10.18	10.34	10.62	7.37	7.60	7.14
	M3	6.62	7.64	6.79	8.43	5.36	6.29	5.22
	M4_spot	6.44	7.57	6.40	8.00	5.41	6.08	5.17
Vestby (T4)	M_naive	12.98	8.53	16.31	14.92	13.19	11.77	13.18
	M3	11.88	8.93	14.23	14.86	8.59	7.96	16.67
	M4_spot	11.70	8.67	13.75	14.68	8.56	7.93	17.11
Hægebostad (T4)	M_naive	19.41	19.88	26.93	24.54	15.89	12.47	16.77
	M3	16.22	16.34	18.40	23.00	15.21	10.08	14.30
	M4_spot	17.00	17.29	17.39	21.99	15.92	10.09	19.32

Table B.7: Stage 3 industrial — sensitivity analysis with 22 outlier municipalities excluded ($n = 328$). Mean municipality MAPE and pooled MAPE (%). Bold = lowest value per column.

Model	Fold A		Fold B		Annual (A + B)		Fold C (summer)	
	Mean	Pooled	Mean	Pooled	Mean	Pooled	Mean	Pooled
M_naive	17.027	9.305	16.708	8.251	16.868	8.781	17.516	8.907
M0c	22.270	15.548	24.834	14.854	23.552	15.202	24.985	15.691
M3	14.919	10.627	17.839	11.066	16.379	10.845	18.875	11.693
M4_spot	14.991	10.492	17.870	10.844	16.430	10.667	19.078	11.545

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Accurate short-term net load forecasting is essential for distribution system operators, energy traders, and flexibility service providers, particularly as distributed rooftop solar photovoltaic installations transform the electricity demand observed at the grid boundary. This thesis develops and validates a multi-stage municipality-level net load forecasting framework for Norway, where the `\textit{Plusskunde}` net-metering scheme --- allowing households with installations up to 100\,kW to export surplus generation --- has driven rapid residential prosumer adoption, using exclusively open data sources and machine learning methods.

The framework processes Elhub open AMI metering data --- comprising hourly consumption disaggregated by consumer category (private residential, commercial, and industrial), solar export records, and daily installed solar capacity snapshots --- across 350 Norwegian municipalities, combined with ERA5 reanalysis weather data. The empirical programme consists of three stages: a national weather feature ablation across all municipalities, a six-fold expanding-window walk-forward evaluation on eight representative municipalities spanning the full solar penetration spectrum and multiple modelling approaches, and a three-fold national validation covering the unseen 2025 calendar year.

For private residential net load, a conservative municipality-level model assignment strategy --- deploying a prosumer-aware LightGBM model in the 48 municipalities where fold-consistent improvement is demonstrated and a weather-based model in the remaining 302 --- achieves a mean annual municipality MAPE of 6.671\%, *winning against the seasonal naive benchmark* in 347 of 350 municipalities (99.1\%), *against the best linear regression baseline* in 246 of 350 (70.3\%), and never underperforming the weather-based model alone across the entire portfolio. For commercial load, the framework achieves 9.655\% *annual mean MAPE, winning conservatively against the naive benchmark* in 313 of 350 municipalities (89.4\%) *and against the linear regression baseline* in 304 of 350 (86.9\%). *Industrial net load forecasting is shown to be a structurally different problem* in which weather-and-calendar features reach their generalisation limits due to the heterogeneity of facility-level production patterns across municipalities, motivating a hybrid deployment strategy. A controlled synthetic experiment reveals that the near-zero benefit of solar-enriched modelling for the residential category under current data conditions is attributable to structural behind-the-meter invisibility, low solar penetration, and prosumer registry heterogeneity rather than an architectural deficiency: when provided with exact, noiseless capacity information, the prosumer-aware model improves consistently over its weather-based counterpart with gains of \$+\$0.16 to \$+\$0.37 percentage points across multiple solar growth scenarios.

The complete pipeline is built on open data and open-source software under realistic operational constraints, providing a reproducible benchmark for municipality-scale net load forecasting in Nordic open-data environments.

"Abstract[swe]": €€€€

Noggrann korttidsprognostisering av nettolast är avgörande för distributionsnätoperatörer, energihandlare och flexibilitetstjänsteleverantörer, särskilt i takt med att distribuerade solcellsinstallationer omvandlar den elefterfrågan som observeras vid nätgränsen. Denna avhandling utvecklar och validerar ett ramverk för nettolastprognostisering på kommunnivå för Norge, där \textit{Plusskundeordningen} --- ett nätmätningssystem som ger hushåll med installationer upp till 100\,kW rätt att exportera överskottsenergi --- har drivit en snabb tillväxt av bostadsprosumenter, med enbart öppna datakällor och maskininlärningsmetoder.

Ramverket bearbetar öppna AMI-mätardata från Elhub --- bestående av timvis förbrukning uppdelad per konsumentkategori (privathushåll, handel och industri), solexportdata samt dagliga ögonblicksbilder av installerad solkapacitet --- för 350 norska kommuner, kombinerat med ERA5-reanalysdata för väder. Det empiriska programmet består av tre steg: en nationell ablation av väderprediktorer, en sexfaldig expanderande walk-forward-validering på åtta representativa kommuner och en tre-faldig nationell validering för det hittills osedda kalenderåret 2025.

För privata hushålls nettolast uppnår en konservativ modelltilldelningsstrategi ett genomsnittligt kommunalt MAPE på 6.671\%, *vinner mot det säsongснаiva referensvärdet* i 347 av 350 kommuner (99.1\%) *och mot den bästa linjära regressionsbaslinjen* i 246 av 350 (70.3\%), *och underpresterar aldrig den vädersbaserade modellen ensam*. För kommersiell last uppnår ramverket 9.655\% *MAPE med en konservativ*

vinstkvot på 89.4\% mot naiv och 86.9\% mot regressionsbaslinje.
Industriell nettolastprognostisering identifieras som ett
strukturellt annorlunda problem som kräver en hybridstrategi.
Ett kontrollerat syntetiskt experiment visar att solmodellens
nära-noll-fördel under rådande dataförhållanden beror på
strukturell osynlighet bakom mätaren och låg
solpenetration snarare än en arkitektonisk brist.

Det fullständiga systemet är byggt på öppna data och
öppen källkod under realistiska driftsförhållanden, och
utgör ett reproducerbart riktmärke för nettolastprognostisering
i nordiska öppna datamiljöer.

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"Keywords[eng]": €€€€

Net load forecasting, Behind-the-meter solar, Distributed solar PV, Smart metering, LightGBM, Gradient-boosted trees, Norwegian electricity
system, Elhub, AML data, Municipality-level forecasting, Ensemble methods, Prosumer €€€€,

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\newacronym{KTH}{KTH}{KTH Royal Institute of Technology}
\newacronym{DiVA}{DiVA}{Digitala Vetenskapliga Arkivet}
\newacronym{UN}{UN}{United Nations}
\newacronym{SDG}{SDG}{Sustainable Development Goal}
\newacronym{API}{API}{Application Programming Interface}

% --- Power systems ---
\newacronym{DSO}{DSO}{Distribution System Operator}
\newacronym{TSO}{TSO}{Transmission System Operator}
\newacronym{AMI}{AMI}{Advanced Metering Infrastructure}
\newacronym{DER}{DER}{Distributed Energy Resource}
\newacronym{PV}{PV}{Photovoltaic}
\newacronym{BTM}{BTM}{Behind-The-Meter}

% --- Forecasting ---
\newacronym{NLF}{NLF}{Net Load Forecasting}
\newacronym{STLF}{STLF}{Short-Term Load Forecasting}
\newacronym{NWP}{NWP}{Numerical Weather Prediction}
\newacronym{WSS}{WSS}{Weather Station Selection}
\newacronym{AR}{AR}{Autoregressive}

% --- Models ---
\newacronym{MLR}{MLR}{Multiple Linear Regression}
\newacronym{OLS}{OLS}{Ordinary Least Squares}
\newacronym{RF}{RF}{Random Forest}
\newacronym{GBDT}{GBDT}{Gradient-Boosted Decision Trees}
\newacronym{XGBoost}{XGBoost}{Extreme Gradient Boosting}
\newacronym{LightGBM}{LightGBM}{Light Gradient-Boosting Machine}
\newacronym{CatBoost}{CatBoost}{Categorical Boosting}

% --- Metrics ---
\newacronym{MAE}{MAE}{Mean Absolute Error}
\newacronym{RMSE}{RMSE}{Root Mean Square Error}
\newacronym{MAPE}{MAPE}{Mean Absolute Percentage Error}
\newacronym{nRMSE}{nRMSE}{Normalised Root Mean Square Error}
\newacronym{nMAPE}{nMAPE}{Normalised Mean Absolute Percentage Error}

% --- Data ---
\newacronym{ERA5}{ERA5}{ECMWF Reanalysis v5}
\newacronym{ECMWF}{ECMWF}{European Centre for Medium-Range Weather Forecasts}
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